Lecture Notes

Dynamical Systems

(MA17029, MA1516)

PG Notes

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Preface

These lecture notes have been prepared for the course **Nonlinear Dynamics**. The notes are intended to supplement classroom lectures and provide students with a concise mathematical treatment of nonlinear dynamical systems, bifurcation theory, chaos, Hamiltonian systems, Lyapunov exponents, and related topics.

The material is compiled primarily for educational purposes. Students are encouraged to consult the prescribed textbooks and research articles for additional details and proofs.

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Course Information

Dynamical Systems

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Course Outcomes

CO-1	Identify Autonomous and Nonautonomous Systems.
CO-2	Understand Limit Cycle Motion and Periodic Attractor.
CO-3	Differentiate Dissipative and Conservative Systems.
CO-4	Understand Different Types of Bifurcations.
CO-5	Apply Poincaré-Bendixson Theory.

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Learning Resources

Text Books

1. M. Lakshmanan and S. Rajasekar, Nonlinear Dynamics: Integrability, Chaos and Patterns, Springer, First Edition, 2010.

2. George F. Simmons, Differential Equations with Applications and Historical Notes, McGraw-Hill, Second Edition, 2003.

Reference Books

1. Morris W. Hirsch, Stephen Smale and Robert L. Devaney, Differential Equations, Dynamical Systems, and an Introduction to Chaos, Elsevier Academic Press, 2004.

2. Lawrence Perko, Differential Equations and Dynamical Systems, Third Edition, Springer, 2010.

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1. Introduction

Nature is inherently dynamic. Every physical, biological, chemical and engineering system undergoes continuous evolution with time. The motion of planets, oscillations of a pendulum, propagation of electrical signals, chemical reactions, population growth, and the spread of infectious diseases are all examples of systems whose states change continuously with time. Understanding and predicting such changes form one of the principal goals of mathematical modelling.

The mathematical study of systems whose states evolve with time is known as **Dynamical Systems Theory**. It provides a unified framework for describing the evolution of a wide variety of natural and artificial systems. Depending on the application, the governing equations may be ordinary differential equations, partial differential equations, difference equations, delay differential equations, stochastic differential equations, or integro-differential equations.

Historically, the foundations of dynamical systems were laid by **Isaac Newton** through his laws of motion, which explained how the motion of an object is determined by the forces acting upon it. During the nineteenth century, the works of **Joseph-Louis Lagrange** and **William Rowan Hamilton** provided elegant reformulations of mechanics that remain fundamental in modern mathematics and theoretical physics.

Although many mathematical models appear similar, their qualitative behaviour depends strongly on whether the governing equations are *linear* or *nonlinear*. Linear systems possess several remarkable mathematical properties that make them relatively easy to analyze. Nonlinear systems, however, often exhibit complicated behaviours such as multiple equilibrium points, limit cycles, bifurcations and chaos. Consequently, nonlinear dynamical systems have become one of the most active areas of modern applied mathematics.

This course is devoted to understanding the mathematical theory underlying nonlinear dynamical systems.

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1.1 What is a Dynamical System?

One of the fundamental questions in science is

"How does a system evolve with time?"

A mathematical description of this evolution is called a **dynamical system**.

Definition 1. *A **dynamical system** is a mathematical model that describes the evolution of one or more state variables with respect to time according to a prescribed law.*

If the state vector is denoted by

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix},$$

then a general continuous-time dynamical system can be written as

$$\dot{x} = f(x, t),$$

where

- $x(t) \in \mathbb{R}^n$ is called the **state vector**,
- t denotes time,
- $f : \mathbb{R}^n \to \mathbb{R}^n$ describes the governing law.

The function f determines how the state evolves with time.

Examples of Dynamical Systems: Dynamical systems arise naturally in almost every branch of science and engineering.

System	State Variables
Simple Pendulum	Angular displacement and angular velocity
Planetary Motion	Position and velocity
Electrical Circuit	Voltage and current
Population Model	Population size
Neural Network	Activations of neurons
Robot Manipulator	Joint positions and velocities
Financial Market	Asset prices
Weather Model	Temperature, pressure and humidity

Although these systems appear unrelated, they can all be represented mathematically using differential equations (or difference equations).

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1.2 Continuous and Discrete Dynamical Systems

Dynamical systems are broadly classified into two categories.

Continuous-Time Systems A continuous-time system evolves continuously with respect to time and is generally described by an ordinary differential equation

$$\dot{x} = f(x, t).$$

Examples include

- motion of planets,
- electrical circuits,
- mechanical vibrations,
- fluid flow.

Discrete-Time Systems A discrete-time system evolves only at discrete instants

$$x_{k+1} = F(x_k).$$

Such systems naturally arise in

- digital control,

- numerical algorithms,
- economic models,
- population models with seasonal breeding.

Throughout these notes we shall mainly consider continuous-time systems but also discuss some concepts in discrete systems.

State Variables The smallest collection of variables that completely describes the present condition of a system is called the **state**.

Definition 2. *A set of variables is called a* ***state vector*** *if knowledge of these variables at an initial time together with the governing equations uniquely determines the future evolution of the system.*

Example For a simple pendulum the angular positions, $\theta(t)$ alone is insufficient because two pendulums may have the same angular position but different angular velocities. Hence the state vector is

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}.$$

Thus the pendulum is a two-dimensional dynamical system.

Why Nonlinear Dynamical Systems? Most physical systems encountered in practice are inherently nonlinear. Linear models are usually obtained by making simplifying assumptions valid only near an equilibrium point. While these approximations are mathematically convenient, they often fail to describe the complete behavior of the underlying physical system. Examples include large-angle pendulum motion, spacecraft attitude dynamics, fluid turbulence, ecological systems, climate dynamics, neural activity, robotics, and power systems. Nonlinear systems can exhibit remarkable phenomena that never occur in linear systems, including multiple equilibrium points, bifurcations, chaos, and sensitive dependence on initial conditions. The study of these behaviors forms the subject of nonlinear dynamics.

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2. Qualitative Features of Dynamical Systems

For many **nonlinear dynamical systems**, obtaining an exact analytical solution is either extremely difficult or impossible. Consequently, instead of searching for explicit solutions, mathematicians and scientists often investigate the **qualitative behaviour** of solutions. Its aim is to answer questions such as

- Does the system eventually settle down to an equilibrium?
- Is the equilibrium stable or unstable?
- Does the system exhibit periodic oscillations?
- Can the system display complicated or chaotic motion?

Rather than computing the exact trajectory, qualitative analysis studies the overall behavior of solutions.

Why Qualitative Analysis? Most nonlinear differential equations cannot be solved exactly. However, valuable information can still be obtained by examining certain special solutions and their neighborhood. The simplest objects to study are Equilibrium (fixed) points, Periodic solutions, Limit cycles, and More complicated invariant sets such as attractors. These objects determine the long-term behaviour of many dynamical systems.

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2.1 Periodic Motion

Not every dynamical system approaches an equilibrium. Many systems instead exhibit repeating motions. A solution satisfying

$$x(t+T) = x(t)$$

for some constant $T > 0$ is called a **periodic solution**.

Examples

- Oscillation of a pendulum.
- Planetary motion around the Sun.
- Vibrations of springs.
- Electrical oscillators.

Periodic solutions play an important role because they describe sustained oscillatory behavior.

Beyond Periodic Motion Some nonlinear systems exhibit behavior much more complicated than simple equilibria or periodic oscillations. Such systems may exhibit quasiperiodic, chaotic, or strange attractors. Although these motions are difficult to predict exactly, their overall geometric structure can still be studied.

2.2 Orders of Differential Equations

Physical systems may be described by differential equations of different orders.

Second-Order Equations Many mechanical systems obey Newton's second law and are naturally described by second-order differential equations.

Example:

$$m\ddot{x} + c\dot{x} + kx = 0.$$

Higher-Order Equations Some engineering systems contain third-order or higher-order equations, especially in control theory and beam dynamics.

First-Order Systems Many models in biology, chemistry, economics, and population dynamics are naturally expressed as systems of first-order differential equations.

Example:

$$\begin{aligned} \dot{x} &= f(x, y), \\ \dot{y} &= g(x, y). \end{aligned}$$

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2.3 Autonomous and Nonautonomous Systems

Dynamical systems are broadly classified into two categories.

2.3.1 Autonomous Systems

An autonomous system has no explicit dependence on time. Its general form is

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}).$$

The governing equations remain unchanged with time.

Examples

- Exponential oscillator
$$\ddot{x} + e^x = 0.$$
- Lorenz equations
$$\begin{aligned} \dot{x} &= \sigma(y - x), \\ \dot{y} &= rx - y - xz, \\ \dot{z} &= xy - bz. \end{aligned}$$

2.3.2 Nonautonomous Systems

A nonautonomous system explicitly depends on time. Its general form is

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t).$$

The external forcing or time-dependent parameters make the dynamics change with time.

Examples

- Forced harmonic oscillator

$$\ddot{x} + a\dot{x} + \omega_0^2 x = f\cos(\omega t).$$

- Forced Van der Pol oscillator

$$\ddot{x} - b(1 - x^2)\dot{x} + \omega_0^2 x = f\cos(\omega t).$$

- Forced pendulum

$$\ddot{x} + a\dot{x} + \sin x = f\sin(\omega t).$$

Difference Between Autonomous and Nonautonomous Systems

Autonomous	Nonautonomous
No explicit time dependence	Explicit time dependence
Form: $\dot{x} = f(x)$	Form: $\dot{x} = f(x, t)$
System parameters remain constant	External forcing present
Phase portrait remains unchanged	Phase portrait changes with time
Examples: Lorenz system	Examples: Forced pendulum

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2.4 Dynamical Systems as Coupled First-Order Differential Equations

Most dynamical systems arising in science and engineering are described by ordinary differential equations (ODEs). These equations may be linear or nonlinear, autonomous or nonautonomous, and of first or higher order. Fortunately, every finite-dimensional dynamical system can be represented as a system of coupled first-order differential equations. This representation is known as the **state-space representation** and forms the basis of modern dynamical systems theory, control theory, and numerical simulation.

Why Convert to First-Order Systems? The conversion of higher-order differential equations into first-order systems offers several advantages like that it provides a unified mathematical framework for studying all dynamical systems, the theory of phase space is naturally developed using first-order equations, stability, equilibrium points, and long-term behavior are easier to analyze and most numerical algorithms are designed for first-order systems.

Procedure for conversion of higher-order equations The general procedure is straightforward. Consider an n^{th}-order differential equation

$$x^{(n)} = f\left(x, \dot{x}, \ddot{x}, \ldots, x^{(n-1)}, t\right).$$

Introduce new variables

$$x_1 = x, \qquad x_2 = \dot{x}, \qquad x_3 = \ddot{x}, \qquad \cdots, \qquad x_n = x^{(n-1)}.$$

Then

$$\begin{aligned}\dot{x}_1 &= x_2,\\ \dot{x}_2 &= x_3,\\ &\vdots\\ \dot{x}_{n-1} &= x_n,\\ \dot{x}_n &= f(x_1, x_2, \ldots, x_n, t).\end{aligned}$$

Thus every higher-order equation becomes a system of first-order equations.

Example: Linear Harmonic Oscillator The equation of motion is

$$\ddot{x} + \omega_0^2 x = 0.$$

Introduce

$$y = \dot{x}.$$

The equivalent first-order system is

$$\begin{aligned}\dot{x} &= y,\\ \dot{y} &= -\omega_0^2 x.\end{aligned}$$

This system has two state variables,

$$X = (x, y)^T.$$

Example: Duffing Oscillator The Duffing oscillator is governed by

$$\ddot{x} + a\dot{x} + \omega_0^2 x + \beta x^3 = f\cos(\omega t).$$

Let

$$y = \dot{x}.$$

Then

$$\begin{aligned}\dot{x} &= y,\\ \dot{y} &= -ay - \omega_0^2 x - \beta x^3 + f\cos(\omega t).\end{aligned}$$

Since time appears explicitly, this is a **nonautonomous system**. To convert it into an autonomous system, introduce

$$z = \omega t.$$

Since

$$\dot{z} = \omega,$$

the autonomous system becomes

$$\begin{aligned}\dot{x} &= y,\\ \dot{y} &= -ay - \omega_0^2 x - \beta x^3 + f\cos z,\\ \dot{z} &= \omega.\end{aligned}$$

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Example: Rigid Body Equations The rotational motion of a rigid body is described by Euler's equations

$$I_1\dot{\omega}_1 - (I_2 - I_3)\omega_2\omega_3 = N_1,$$
$$I_2\dot{\omega}_2 - (I_3 - I_1)\omega_3\omega_1 = N_2,$$
$$I_3\dot{\omega}_3 - (I_1 - I_2)\omega_1\omega_2 = N_3.$$

These equations are already first-order differential equations in the angular velocities

$$(\omega_1, \omega_2, \omega_3).$$

Hence no further conversion is required.

Example: Lotka-Volterra Model The predator-prey model is

$$\dot{x} = ax - xy,$$

$$\dot{y} = xy - by.$$

These equations are naturally expressed as a coupled first-order system. Here

- $x(t)$ denotes the prey population,
- $y(t)$ denotes the predator population.

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General State-Space Representation Suppose a dynamical system contains n state variables

$$X = (x_1, x_2, \ldots, x_n)^T.$$

Its evolution is represented by

$$\dot{X} = F(X),$$

where

$$F(X) = \begin{bmatrix} F_1(X) \\ F_2(X) \\ \vdots \\ F_n(X) \end{bmatrix}.$$

Equivalently,

$$\dot{x}_1 = F_1(x_1, x_2, \ldots, x_n),$$
$$\dot{x}_2 = F_2(x_1, x_2, \ldots, x_n),$$
$$\vdots$$
$$\dot{x}_n = F_n(x_1, x_2, \ldots, x_n).$$

The vector X is called the **state vector**, and the space containing all possible states is called the **state space** or **phase space**.

State Variables The choice of state variables depends on the physical system.

Examples Position and velocity, Position and momentum, Angular velocity, Energy, Population densities, and Chemical concentrations. Once the state variables are selected, the complete future behavior of the system is determined by the current state.

2.5 Equilibrium (Fixed) Points

An equilibrium point is a state at which the system does not evolve with time. For

$$\dot{X} = F(X),$$

an equilibrium point satisfies

$$F(X^*) = 0.$$

Equivalently,

$$\dot{X} = 0.$$

If the system starts at an equilibrium point, it remains there forever. Therefore, equilibrium points are also called stationary points, fixed points, or steady states.

Procedure for Finding Equilibrium Points To determine the equilibrium points,

1. Write the system in first-order form.
2. Set every derivative equal to zero.
3. Solve the resulting algebraic equations simultaneously.

Example Consider the dynamical system

$$\dot{x} = x(1 - x).$$

The equilibrium points are obtained by solving

$$f(x_e) = x_e(1 - x_e) = 0.$$

Hence,

$$x_e = 0 \quad \text{or} \quad x_e = 1.$$

Therefore, the system has two equilibrium points:

$$x_e = 0, \qquad x_e = 1.$$

For example, if the initial condition is $x(0) = 1$, then

$$\dot{x} = 1(1 - 1) = 0.$$

Thus, the state remains at $x(t) = 1$ for all $t \geq 0$. Similarly, if $x(0) = 0$, the state remains at $x(t) = 0$ for all $t \geq 0$.

If the system starts exactly at an equilibrium point, it remains there forever.

Physical Interpretation An equilibrium represents a steady state of the system. Examples include

- A pendulum hanging vertically downward.
- A chemical reaction that has reached equilibrium.
- Constant population in an ecological model.
- Constant temperature in a thermal system.

The important question is not only whether an equilibrium exists, but also

Will the system remain close to the equilibrium if it is slightly disturbed?

This leads to the concept of **stability**.

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Example: A simple pendulum

$$\ddot{\theta} + \frac{g}{L}\sin\theta = 0,$$

where θ is the angular displacement, g is the acceleration due to gravity, and L is the length of the pendulum. This can be expressed as

$$\dot{x}_1 = x_2,$$
$$\dot{x}_2 = -\frac{g}{L}\sin x_1,$$

where $x_1 = \theta$ and $x_2 = \dot{\theta}$. The simple pendulum possesses two distinct equilibrium positions in the interval $0 \leq \theta < 2\pi$.

1. Pendulum hanging downward

   $$\theta = 0,$$

   which is a **stable equilibrium**. Small disturbances produce oscillations around this position.

2. Pendulum balanced vertically upward

   $$\theta = \pi,$$

   which is an **unstable equilibrium**. Even a tiny disturbance causes the pendulum to fall away from this position.

Hence, two equilibrium points may exist in the same system with completely different stability properties.

Example: Linear Harmonic Oscillator The system is

$$\dot{x} = y,$$

$$\dot{y} = -\omega_0^2 x.$$

Setting

$$\dot{x} = 0, \qquad \dot{y} = 0,$$

gives

$$y = 0,$$

$$x = 0.$$

Hence

$$X^* = (0, 0).$$

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Example The Lorenz equations are

$$\dot{x} = \sigma(y - x),$$

$$\dot{y} = rx - y - xz,$$

$$\dot{z} = xy - bz.$$

Setting all derivatives equal to zero,

$$\sigma(y - x) = 0,$$

$$rx - y - xz = 0,$$

$$xy - bz = 0.$$

The equilibrium points are

$$(0, 0, 0),$$

and

$$\left(\sqrt{b(r-1)}, \sqrt{b(r-1)}, r - 1\right),$$

$$\left(-\sqrt{b(r-1)}, -\sqrt{b(r-1)}, r - 1\right).$$

Thus the Lorenz system possesses three equilibrium points whenever $r > 1$.

2.6 Phase Space, Phase Plane, and Phase Trajectories

The qualitative study of a dynamical system is greatly simplified by examining its motion in a geometric space called the **phase space**. Rather than studying the variation of each state variable separately with time, the complete state of the system is represented by a single point whose motion describes the evolution of the system. Phase space analysis provides valuable information about the long-term behavior of dynamical systems, including equilibrium points, periodic solutions, attractors, and stability.

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Phase Space Consider an n-dimensional dynamical system

$$\dot{\mathbf{X}} = \mathbf{F}(\mathbf{X}),$$

where

$$\mathbf{X} = (x_1, x_2, \ldots, x_n)^T$$

is the state vector. The n variables

$$(x_1, x_2, \ldots, x_n)$$

serve as the coordinate axes of an **n-dimensional phase space**. Each point in this space uniquely represents the instantaneous state of the system. As time evolves, the point moves continuously through the phase space, tracing out a curve called the **phase trajectory**.

Example Consider the system

$$\dot{x} = -x,$$
$$\dot{y} = -2y,$$
$$\dot{z} = -3z.$$

The state vector is

$$X = (x, y, z)^T.$$

Therefore, the phase space is three-dimensional with coordinate axes x, y, and z. For the initial condition

$$X(0) = (1, 1, 1)^T,$$

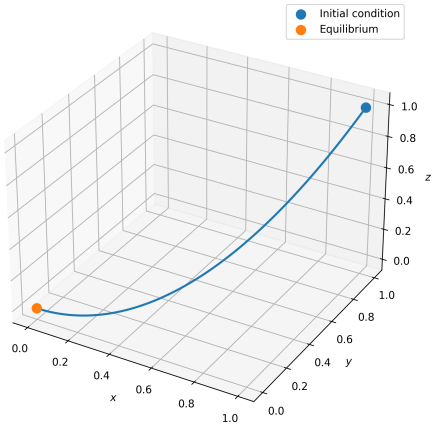
the solution is

$$x(t) = e^{-t}, \qquad y(t) = e^{-2t}, \qquad z(t) = e^{-3t}.$$

Hence, the phase point

$$X(t) = \left(e^{-t}, e^{-2t}, e^{-3t}\right)^T$$

moves through the three-dimensional phase space and approaches the origin as $t \to \infty$.



Phase Plane If the system contains only two state variables,

$$\mathbf{X} = (x, y),$$

the phase space becomes two-dimensional and is called the **phase plane**. Typical examples include

$$\dot{x} = f(x, y),$$
$$\dot{y} = g(x, y).$$

For second-order differential equations,

$$\ddot{x} = f(x, \dot{x}),$$

we usually define

$$y = \dot{x},$$

so that the phase plane consists of the variables

$$(x, y).$$

Each point on the phase plane represents the position and velocity of the system at a given instant.

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Example: Simple Harmonic Oscillator Consider the second-order differential equation

$$\ddot{x} + x = 0.$$

Introducing the state variable

$$y = \dot{x},$$

the equation can be written as the first-order system

$$\dot{x} = y,$$
$$\dot{y} = -x.$$

Thus, the phase plane is the (x, y)-plane, where x represents the position and y represents the velocity. For the initial condition

$$x(0) = 1, \qquad y(0) = 0,$$

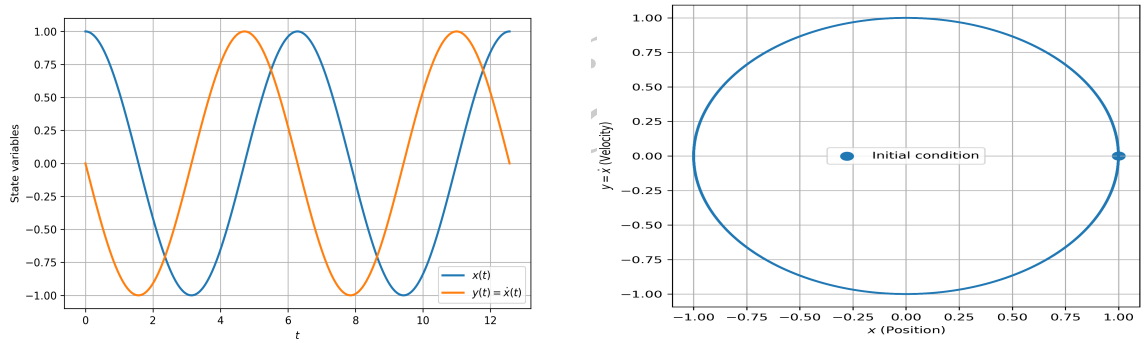
the solution is

$$x(t) = \cos t, \qquad y(t) = -\sin t.$$

Eliminating t, we obtain

$$x^2 + y^2 = 1.$$

Therefore, the phase trajectory is a circle in the phase plane.



Phase Trajectories The curve traced by the moving phase point in the phase space is called the **phase trajectory** (or phase orbit). Mathematically,

$$\mathbf{X}(t) = (x_1(t), x_2(t), \ldots, x_n(t))$$

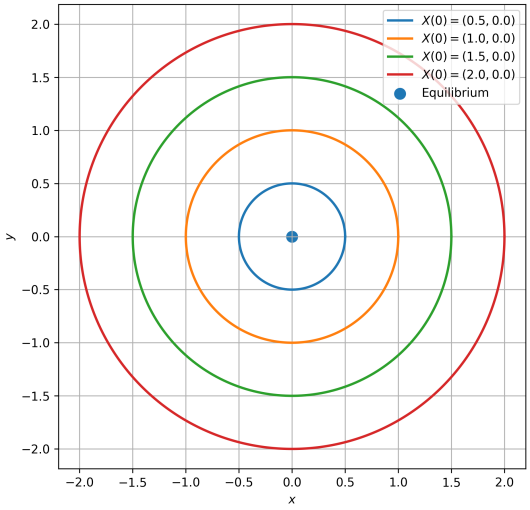
defines a trajectory in phase space. Different initial conditions generally generate different trajectories. Unlike ordinary graphs of $x(t)$ versus t, a phase trajectory directly illustrates the geometric behavior of the dynamical system. Now the quantity

$$x^2 + y^2$$

remains constant along every trajectory. Therefore,

$$x^2 + y^2 = C,$$

where the value of C depends on the initial condition. Hence, different initial conditions generate different circular trajectories in the phase plane. Each circle represents a different motion of the system.



2.7 Stability, Attractors and Repellers

Equilibrium Points in Phase Space Consider the dynamical system

$$\dot{X} = F(X).$$

An equilibrium point X^* satisfies

$$F(X^*) = 0.$$

In phase space, X^* appears as a fixed point because its coordinates do not change with time. If the system starts exactly at the equilibrium point,

$$X(0) = X^*,$$

then

$$X(t) = X^* \qquad \forall t \geq 0.$$

Example Consider the system

$$\dot{X} = -X.$$

The equilibrium point is obtained from

$$-X^* = 0,$$

which gives

$$X^* = 0.$$

If

$$X(0) = 0,$$

then the solution is

$$X(t) = 0 \qquad \forall t \geq 0.$$

Thus, the origin appears as a fixed point in phase space.

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Behavior Near Equilibrium Points Knowing the location of an equilibrium point alone is not sufficient to understand the behavior of a dynamical system. We must also examine what happens to trajectories that start close to the equilibrium.

Three typical behaviors are illustrated below.

Case 1: Asymptotically Stable Equilibrium If trajectories starting sufficiently close to an equilibrium remain close and eventually approach it as $t \to \infty$, the equilibrium is called *asymptotically stable*.

Example Consider

$$\dot{X} = -X,$$

where $X = (x, y)$. The solution corresponding to the initial condition $X(0) = X_0$ is

$$X(t) = X_0 e^{-t}.$$

Therefore,

$$\lim_{t\to\infty} X(t) = 0.$$

Hence, the equilibrium

$$X^* = 0$$

is asymptotically stable. Nearby trajectories are attracted toward the origin.

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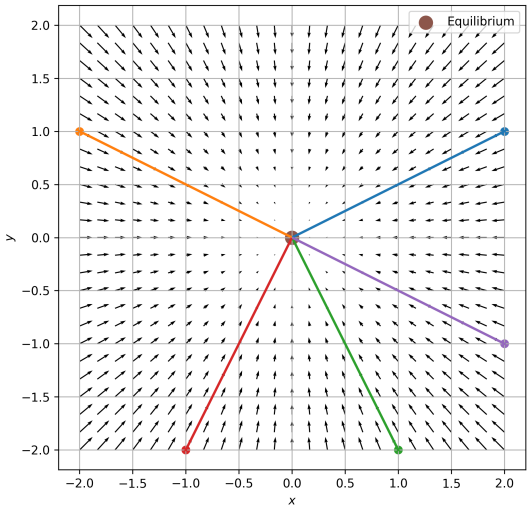


Case 2: Unstable Equilibrium If trajectories starting arbitrarily close to an equilibrium move away from it, the equilibrium is called *unstable*.

Example Consider

$$\dot{X} = X,$$

where $X = (x, y)$. Its solution is

$$X(t) = X_0 e^{t}.$$

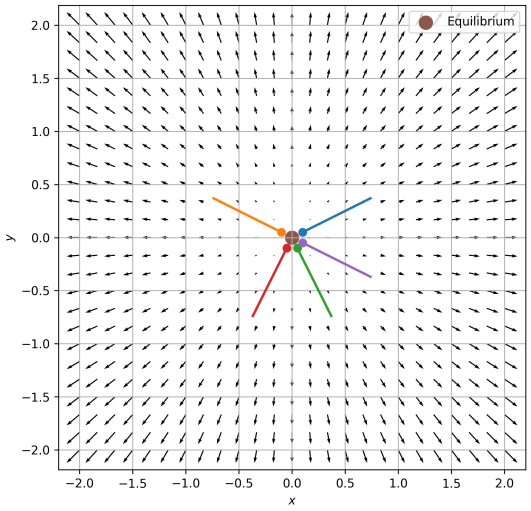
For every nonzero initial condition X_0,

$$|X(t)| \to \infty \qquad \text{as } t \to \infty.$$

Thus, the equilibrium

$$X^* = 0$$

is unstable. Nearby trajectories are repelled from the origin.



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Case 3: Neutrally Stable Equilibrium An equilibrium may be stable without attracting nearby trajectories. In this case, nearby trajectories remain close to the equilibrium but do not converge to it.

Example Consider the harmonic oscillator

$$\dot{x} = y,$$
$$\dot{y} = -x.$$

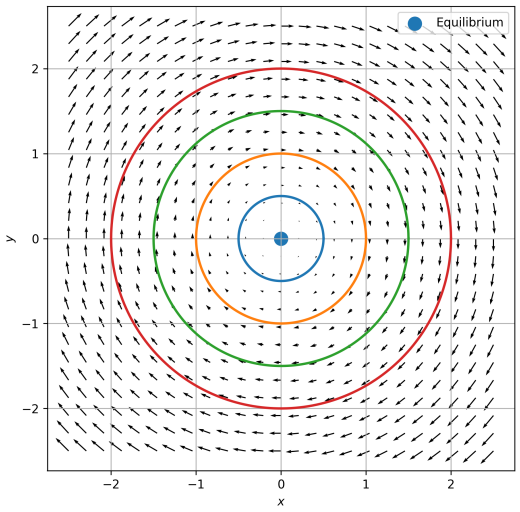
The quantity

$$x^2 + y^2$$

remains constant along every trajectory. Hence,

$$x^2 + y^2 = C.$$

The phase trajectories are circles centered at the origin. Trajectories neither approach nor move away from the equilibrium. Therefore, the origin is stable but not asymptotically stable. Such behavior is often referred to as neutral stability.



Definition 3. *An equilibrium point X^* is called* asymptotically stable *if it is stable and trajectories starting sufficiently close to X^* satisfy*

$$\lim_{t\to\infty} \|X(t) - X^*\| = 0.$$

Thus, the distance between the trajectory and the equilibrium tends to zero as time increases.

2.7.1 Attractors

An attractor is a set in phase space toward which nearby trajectories converge as time increases.

Definition 4. *A set A in phase space is called an* attractor *if trajectories starting from a suitable neighborhood of A approach the set as $t \to \infty$. Symbolically,*

$$\lim_{t\to\infty} \operatorname{dist}(X(t), A) = 0.$$

Examples of attractors include asymptotically stable equilibrium points, and stable limit cycles.

Example Consider a damped simple pendulum governed by

$$\dot{\theta} = \omega,$$
$$\dot{\omega} = -c\omega - \frac{g}{L}\sin\theta.$$

The downward vertical position

$$(\theta, \omega) = (0, 0)$$

is an equilibrium point. If the pendulum is slightly displaced, it oscillates about the downward position. Because of damping, energy is continuously dissipated and the amplitude of oscillation gradually decreases. In the phase plane, the trajectory spirals inward toward

$$(0, 0).$$

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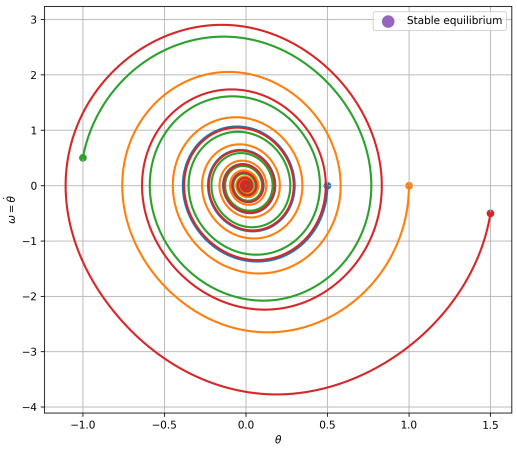
Hence, the downward equilibrium is an attractor.



2.7.2 Repellers

A repeller has behavior opposite to that of an attractor.

Definition 5. *A* repeller *is a point or invariant set from which nearby trajectories move away as time increases.*

Example Consider

$$\dot{X} = X.$$

The origin

$$X^* = 0$$

is an equilibrium point. However, for $X_0 \neq 0$,

$$X(t) = X_0 e^t.$$

Thus, even a small perturbation from the origin grows exponentially. The origin therefore acts as a repeller.

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2.7.3 Limit Cycles as Attractors

An attractor need not be an equilibrium point. A closed periodic trajectory may also attract nearby trajectories. Such an isolated closed trajectory is called a *limit cycle*.

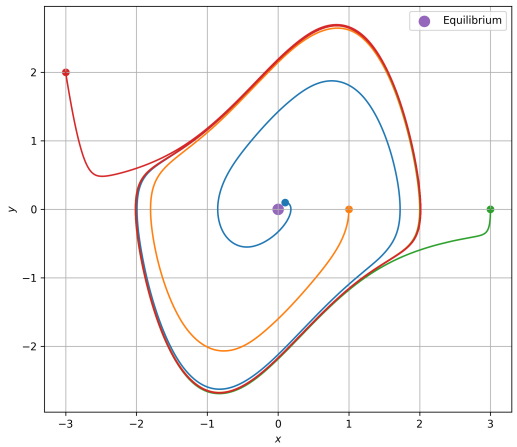
Example The Van der Pol oscillator

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \qquad \mu > 0.$$

This can be rewritten as follows:

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= \mu(1 - x^2)y - x. \end{aligned}$$

For many initial conditions, the trajectories approach a closed periodic orbit. This closed orbit is a stable limit cycle and acts as an attractor.



How do trajectories approach a limit cycle? Consider two trajectories with different initial conditions: one starting inside the closed orbit and another starting outside the closed orbit. The trajectory starting inside the closed orbit gradually moves outward, whereas the trajectory starting outside the closed orbit gradually moves inward. Despite having different initial conditions, both trajectories eventually approach the same closed periodic orbit. Thus, the long-term behavior of the system becomes independent of these initial conditions. The attracting set is not an equilibrium point but a closed periodic trajectory called a *stable limit cycle*. This illustrates an important principle: An attractor need not be a point. An attractor may be an equilibrium point, a periodic orbit such as a stable limit cycle, or a more complicated invariant set.

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2.8 Classification of Equilibrium Points: Two-Dimensional Systems

In the previous section, we introduced the concept of equilibrium (fixed) points and discussed their qualitative behavior. In this section, we develop a mathematical criterion to determine whether an equilibrium point is **stable**, **unstable**, or **neutrally stable**. The method is known as **linear stability analysis**. The basic idea is to introduce a very small perturbation around an equilibrium point and study the evolution of the perturbation with time. If the perturbation decays, the equilibrium is stable; if it grows, the equilibrium is unstable.

General Two-Dimensional Dynamical System Consider the autonomous system

$$\dot{x} = P(x, y),$$
$$\dot{y} = Q(x, y),$$

where $P(x, y)$ and $Q(x, y)$ are continuously differentiable functions. Let

$$X^* = (x_0, y_0)$$

be an equilibrium point. Then

$$P(x_0, y_0) = 0, \qquad Q(x_0, y_0) = 0.$$

Since both derivatives vanish at this point, the state of the system does not change with time. To study stability, we slightly disturb the equilibrium point by introducing small perturbations

$$x = x_0 + \xi(t), \qquad y = y_0 + \eta(t),$$

where

$$|\xi|,\ |\eta| \ll 1.$$

The functions $\xi(t)$ and $\eta(t)$ represent small deviations from the equilibrium. Our objective is to determine whether these perturbations decrease or increase with time. Expand the functions $P(x, y)$ and $Q(x, y)$ about the equilibrium point. Using the first-order Taylor expansion,

$$P(x, y) \approx P(x_0, y_0) + \left.\frac{\partial P}{\partial x}\right|_{X^*} \xi + \left.\frac{\partial P}{\partial y}\right|_{X^*} \eta,$$

$$Q(x, y) \approx Q(x_0, y_0) + \left.\frac{\partial Q}{\partial x}\right|_{X^*} \xi + \left.\frac{\partial Q}{\partial y}\right|_{X^*} \eta.$$

Since

$$P(x_0, y_0) = Q(x_0, y_0) = 0,$$

the constant terms vanish. Ignoring quadratic and higher-order terms gives the linear approximation

$$\dot{\xi} = a\xi + b\eta,$$
$$\dot{\eta} = c\xi + d\eta,$$

where

$$a = \left.\frac{\partial P}{\partial x}\right|_{X^*}, \quad b = \left.\frac{\partial P}{\partial y}\right|_{X^*}, \quad c = \left.\frac{\partial Q}{\partial x}\right|_{X^*}, \quad d = \left.\frac{\partial Q}{\partial y}\right|_{X^*}.$$

Thus, the nonlinear system is approximated by a linear system in a small neighborhood of the equilibrium. The coefficients of the linearized system form the **Jacobian matrix**

$$J(X^*) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The Jacobian contains all first-order partial derivatives evaluated at the equilibrium point. The local behavior of the nonlinear system depends entirely on this matrix. Assume a solution of the form

$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \mathbf{v}e^{\lambda t},$$

where $\mathbf{v}$ is a constant vector. Substituting into the linearized equations gives

$$(J - \lambda I)\mathbf{v} = 0.$$

For a nontrivial solution,

$$\det(J - \lambda I) = 0.$$

Expanding,

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0.$$

Hence the eigenvalues are

$$\lambda_{1,2} = \frac{(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}}{2}.$$

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Define

$$T = a + d,$$

called the **trace** of the Jacobian, and

$$D = ad - bc,$$

called the **determinant** of the Jacobian. The characteristic equation becomes

$$\lambda^2 - T\lambda + D = 0.$$

The eigenvalues are therefore

$$\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4D}}{2}.$$

Role of the Trace, Determinant, and Discriminant The classification of an equilibrium point can be understood from three quantities:

$$T = \mathrm{tr}(J), \qquad D = \det(J), \qquad \Delta = T^2 - 4D.$$

The trace determines the sum of the eigenvalues, $\lambda_1 + \lambda_2 = T$, while the determinant determines their product, $\lambda_1\lambda_2 = D$. The discriminant $\Delta = T^2 - 4D$ determines whether the eigenvalues are real or complex.

- If $D < 0$, the eigenvalues have opposite signs. Therefore, the equilibrium is a **saddle point** and is always unstable.
- If $D > 0, \quad \Delta > 0$, the eigenvalues are real and have the same sign. The equilibrium is a **node**. The sign of T determines whether the node is stable or unstable.
- If $D > 0, \quad \Delta < 0$, the eigenvalues are complex conjugates. The equilibrium is a **focus**. The sign of T determines whether trajectories spiral inward or outward.
- If $D > 0, \quad T = 0$, the eigenvalues are purely imaginary. For a linear system, the equilibrium is a **center**.
- If $D = 0$, at least one eigenvalue is zero. The equilibrium is non-hyperbolic, and linearization may be inconclusive.

Thus, the quantities T, D, and Δ provide a convenient method for classifying equilibrium points of two-dimensional systems.

Condition on Eigenvalues	**Local Behavior**
$\Re(\lambda_1) < 0,\ \Re(\lambda_2) < 0$	Asymptotically stable
$\Re(\lambda_1) > 0$ or $\Re(\lambda_2) > 0$ or $\Re(\lambda_1), \Re(\lambda_2) > 0$	Unstable
Eigenvalues have opposite signs	Saddle point; unstable
At least one eigenvalue has zero real part	Linearization may be inconclusive

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Interpretation If $\Re(\lambda_1) < 0, \quad \Re(\lambda_2) < 0$, all sufficiently small perturbations decay exponentially. Hence, nearby trajectories approach the equilibrium point. If at least one eigenvalue satisfies $\Re(\lambda) > 0$, there exists a perturbation that grows exponentially. Therefore, the equilibrium is unstable. If one or more eigenvalues have zero real part, the equilibrium is *non-hyperbolic*. In this case, linearization may not determine the stability of the original nonlinear system, and higher-order nonlinear terms must be examined. For a purely linear system, purely imaginary eigenvalues produce a center and closed trajectories. For a nonlinear system, however, purely imaginary eigenvalues alone do not guarantee neutral stability.

Linearization Linearization provides an accurate approximation of the nonlinear system only in a sufficiently small neighborhood of the equilibrium point. Far from the equilibrium, neglected nonlinear terms become significant and may alter the behavior of the system. Therefore, linear stability analysis is a **local** stability analysis.

Classification of Equilibrium Points The eigenvalues of the Jacobian determine the geometry of nearby phase trajectories. The principal types of equilibrium points are described below.

1. Stable Node (Sink) Suppose

$$\lambda_1 < 0, \qquad \lambda_2 < 0, \qquad \lambda_1 \neq \lambda_2.$$

Both eigenvalues are real and negative. Therefore, all nearby trajectories approach the equilibrium point as $t \to \infty$. The two eigenvalues generally correspond to different rates of exponential decay. Consequently, trajectories approach the equilibrium preferentially along the direction associated with the eigenvalue having the smaller magnitude. The equilibrium is asymptotically stable and acts as a sink.

2. Unstable Node (Source) Suppose

$$\lambda_1 > 0, \qquad \lambda_2 > 0, \qquad \lambda_1 \neq \lambda_2.$$

Both eigenvalues are real and positive. Perturbations grow exponentially and nearby trajectories move away from the equilibrium. The equilibrium is unstable and acts as a source.

3. Stable Focus (Spiral Sink) Suppose

$$\lambda_{1,2} = \alpha \pm i\beta, \qquad \alpha < 0, \qquad \beta \neq 0.$$

The imaginary part produces oscillatory motion, while the negative real part causes the amplitude of oscillation to decay exponentially. Hence, trajectories spiral inward toward the equilibrium. The equilibrium is asymptotically stable and is called a spiral sink or stable focus.

4. Unstable Focus (Spiral Source) Suppose

$$\lambda_{1,2} = \alpha \pm i\beta, \qquad \alpha > 0, \qquad \beta \neq 0.$$

The imaginary part produces oscillations, while the positive real part causes their amplitude to increase. Thus, trajectories spiral outward from the equilibrium. The equilibrium is unstable and is called a spiral source or unstable focus.

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5. Center For a linear system, suppose

$$\lambda_{1,2} = \pm i\beta, \qquad \beta \neq 0.$$

There is no exponential growth or decay. The phase trajectories are closed curves surrounding the equilibrium point. The equilibrium is stable but not asymptotically stable. For nonlinear systems, purely imaginary eigenvalues make the equilibrium non-hyperbolic. Linearization alone may not determine whether the equilibrium is a center, stable, or unstable.

6. Saddle Point Suppose

$$\lambda_1 < 0, \qquad \lambda_2 > 0.$$

The eigenvalues are real and have opposite signs. Perturbations decay along the eigendirection corresponding to the negative eigenvalue and grow along the eigendirection corresponding to the positive eigenvalue. Thus, trajectories approach the equilibrium along one direction and move away along another direction. A saddle point is always unstable.

7. Repeated Eigenvalues Suppose

$$\lambda_1 = \lambda_2 = \lambda.$$

The nature of the equilibrium depends not only on the repeated eigenvalue but also on the number of linearly independent eigenvectors. If the Jacobian has two linearly independent eigenvectors, the equilibrium is called a *star node*. Trajectories move along radial directions. If the Jacobian has only one linearly independent eigenvector, the equilibrium is called an *improper node*. Trajectories become tangent to the single eigendirection near the equilibrium. For

$$\lambda < 0,$$

the equilibrium is asymptotically stable, whereas for

$$\lambda > 0,$$

the equilibrium is unstable.

8. Degenerate or Non-Hyperbolic Equilibrium An equilibrium is called non-hyperbolic if at least one eigenvalue has zero real part. An important case occurs when

$$\det(J) = 0,$$

so that at least one eigenvalue is zero. In such cases, linearization may be insufficient to determine the local behavior. Higher-order nonlinear terms may significantly influence the dynamics. Techniques such as nonlinear stability analysis, Lyapunov methods, or center manifold theory may be required.

Example Consider the linear system

$$\dot{X} = AX.$$

The following examples illustrate the principal types of equilibrium points.

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1. **Stable Node**

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_1 = -1, \qquad \lambda_2 = -2.$$

Hence, the origin is a stable node.

2. **Unstable Node**

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_1 = 1, \qquad \lambda_2 = 2.$$

Hence, the origin is an unstable node.

3. **Stable Focus**

$$A = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_{1,2} = -1 \pm i.$$

Hence, the origin is a stable focus.

4. **Unstable Focus**

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_{1,2} = 1 \pm i.$$

Hence, the origin is an unstable focus.

5. **Center**

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_{1,2} = \pm i.$$

Hence, the origin is a center.

6. **Saddle Point**

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_1 = 1, \qquad \lambda_2 = -1.$$

Hence, the origin is a saddle point.

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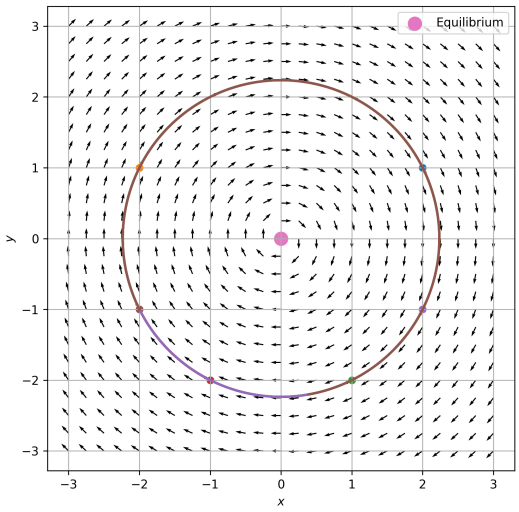
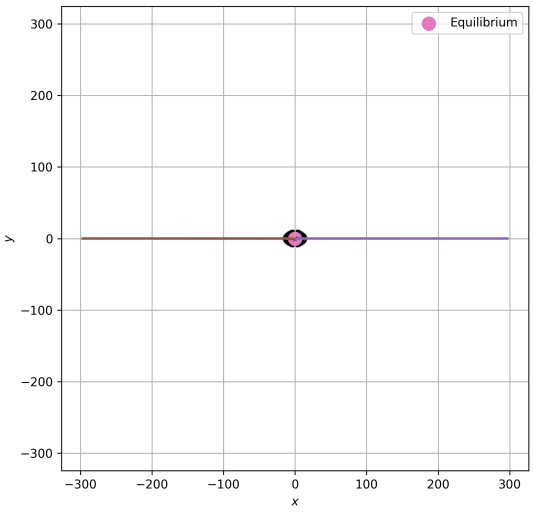

Equilibrium
x
y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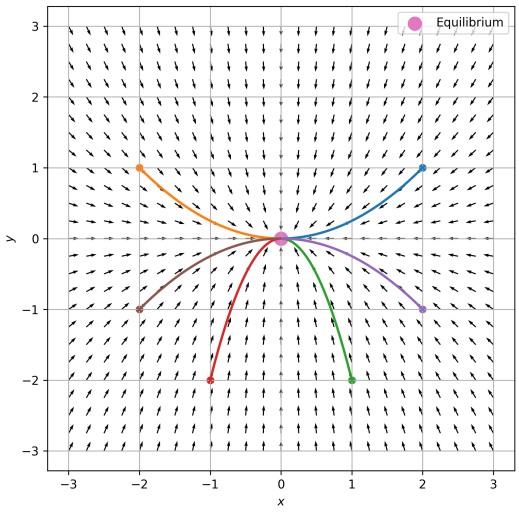
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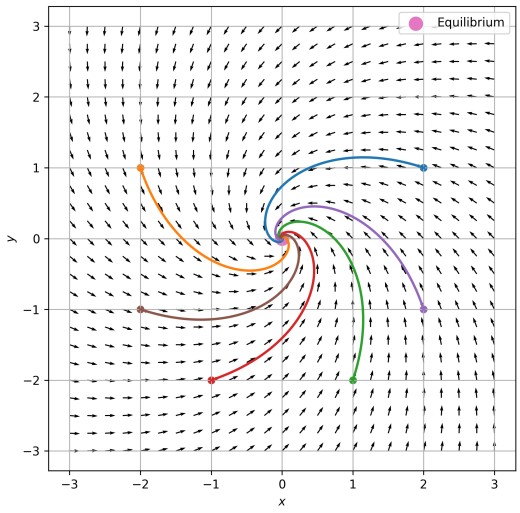
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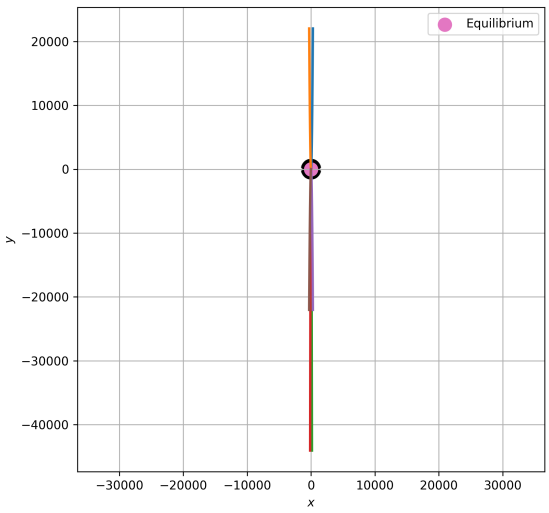
Equilibrium
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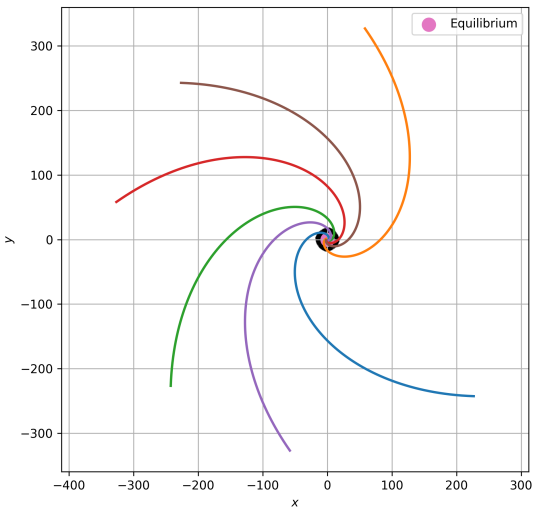
Equilibrium
x
y

Geometric Interpretation of Eigenvalues The eigenvalues determine whether perturbations grow or decay, while the eigenvectors determine the preferred directions of motion in the phase plane. For real eigenvalues, trajectories tend to align with the eigendirections. A negative eigenvalue produces decay along its eigendirection, whereas a positive eigenvalue produces growth. Complex eigenvalues produce rotational behavior. If

$$\lambda = \alpha \pm i\beta,$$

the real part α determines the change in amplitude, while the imaginary part β produces rotation or oscillation. Thus,

$\Re(\lambda)$ determines growth or decay,
$\Im(\lambda)$ determines oscillation or rotation.

This provides a direct connection between the algebraic properties of the Jacobian matrix and the geometric structure of trajectories in the phase plane.

Trace-Determinant Plane The classification of a two-dimensional equilibrium point can be represented geometrically in the trace-determinant plane. The curve

$$D = \frac{T^2}{4}$$

corresponds to

$$\Delta = T^2 - 4D = 0$$

and separates regions with real eigenvalues from regions with complex eigenvalues. Below this curve, when

$$D > 0,$$

the eigenvalues are real, and the equilibrium is a node. Above the curve, the eigenvalues are complex conjugates and the equilibrium is a focus. The line

$$D = 0$$

separates saddle points from nodes and foci. If

$$D < 0,$$

the eigenvalues have opposite signs, and the equilibrium is always a saddle point. Finally, the sign of the trace determines attraction or repulsion:

$$T < 0 \implies \text{attracting behavior},$$

whereas

$$T > 0 \implies \text{repelling behavior}.$$

Thus, the location of the point

$$(T, D)$$

in the trace-determinant plane provides a rapid geometric classification of the equilibrium point.

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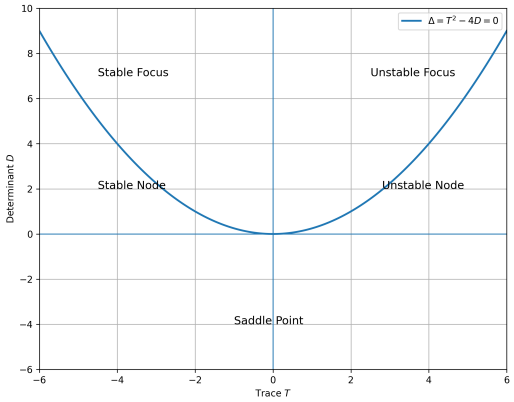

Example: Damped Cubic Oscillator Consider

$$\ddot{x} + d\dot{x} - ax + bx^3 = 0,$$

where

$$a > 0, \qquad b > 0, \qquad d > 0.$$

Thus

$$\dot{x} = y,$$
$$\dot{y} = -dy + ax - bx^3.$$

The equilibrium points are

$$(0,0), \qquad \left(\sqrt{\frac{a}{b}}, 0\right), \qquad \left(-\sqrt{\frac{a}{b}}, 0\right).$$

The Jacobian matrix is

$$J = \begin{bmatrix} 0 & 1 \\ a - 3bx^2 & -d \end{bmatrix}.$$

From the eigenvalues we have

At the origin

$$J(0,0) = \begin{bmatrix} 0 & 1 \\ a & -d \end{bmatrix}.$$

The determinant is

$$D = -a < 0.$$

Therefore, the eigenvalues have opposite signs, and the origin is a **saddle point**. Hence, the equilibrium

$$(0,0)$$

is unstable.

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At the equilibria

$$\left(\pm\sqrt{\frac{a}{b}}, 0\right),$$

we have

$$a - 3bx^2 = a - 3a = -2a.$$

Thus, the Jacobian becomes

$$J = \begin{bmatrix} 0 & 1 \\ -2a & -d \end{bmatrix}.$$

The trace and determinant are

$$T = -d < 0, \qquad D = 2a > 0.$$

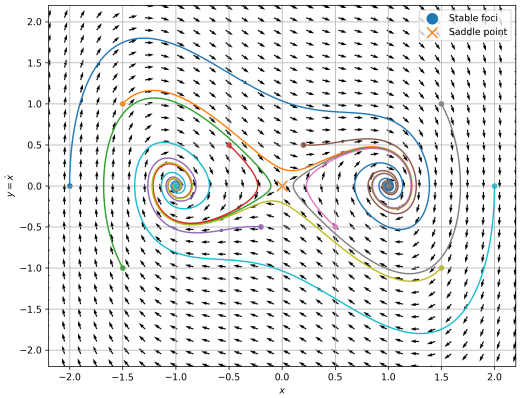
The discriminant is

$$\Delta = T^2 - 4D = d^2 - 8a.$$

Therefore, three cases arise:

$$\begin{aligned} d^2 > 8a &\implies \text{stable node,} \\ d^2 = 8a &\implies \text{repeated negative eigenvalue,} \\ d^2 < 8a &\implies \text{stable focus.} \end{aligned}$$

Thus, the same nonlinear dynamical system may contain equilibrium points of different types. The origin is always a saddle point, whereas the two outer equilibria are asymptotically stable. Their precise geometric classification depends on the damping parameter d.



Interpretation of the Phase Portrait For

$$a = 1, \qquad b = 1, \qquad d = 0.5,$$

the equilibrium points are

$$(-1, 0), \qquad (0, 0), \qquad (1, 0).$$

At the origin,

$$D = -a = -1 < 0.$$

Hence, the origin is a saddle point. Trajectories approach the origin along one direction and move away along another direction. At the two outer equilibria,

$$(-1, 0) \qquad \text{and} \qquad (1, 0),$$

the discriminant is

$$\Delta = d^2 - 8a = (0.5)^2 - 8 = -7.75 < 0.$$

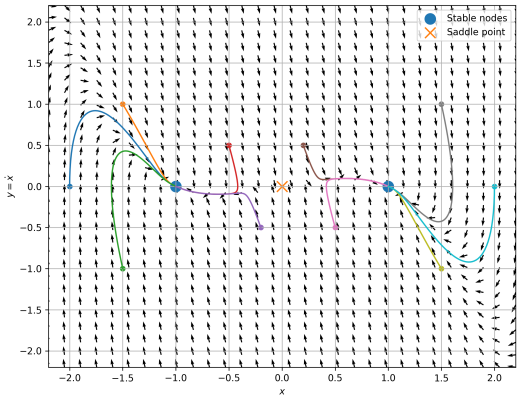
Therefore, the eigenvalues are complex conjugates with negative real parts. Consequently, the two outer equilibria are stable foci. The phase portrait shows trajectories spiraling toward one of the two stable equilibria. The long-term state of the system depends on the initial condition. Some trajectories approach

$$(-1, 0),$$

whereas others approach

$$(1, 0).$$

Thus, the system possesses two attracting equilibrium points separated by the saddle structure associated with the origin.



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2.9 Limit Cycle Motion: Periodic Attractors

In the previous sections, we studied equilibrium points as possible long-term states of dynamical systems. For an asymptotically stable equilibrium, nearby trajectories eventually approach a fixed point in phase space. However, many dynamical systems do not settle to a state of rest. Instead, they approach a sustained periodic motion. In the phase plane, such periodic motion may appear as an isolated closed trajectory called a **limit cycle**. A stable limit cycle represents a **periodic attractor**.

Closed Orbits and Limit Cycles Consider a two-dimensional autonomous system

$$\dot{x} = P(x, y),$$
$$\dot{y} = Q(x, y).$$

Suppose a solution satisfies

$$X(t + T) = X(t)$$

for some

$$T > 0.$$

The motion is periodic with period T. In the phase plane, the corresponding trajectory forms a closed curve. However, not every closed trajectory is a limit cycle.

Definition 6. *A **limit cycle** is an isolated closed trajectory in the phase plane.*

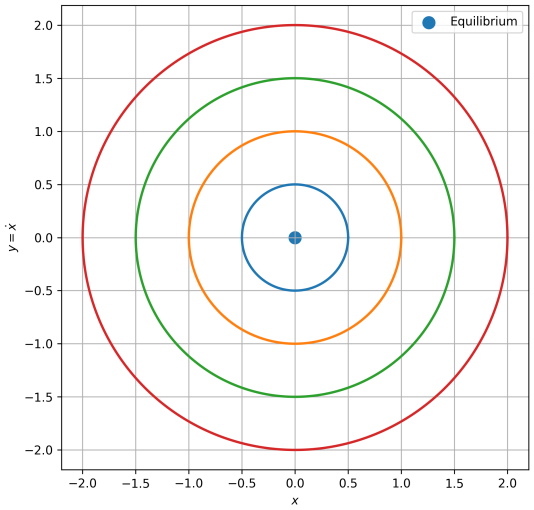
The word *isolated* means that sufficiently close to the closed trajectory, there are no other closed trajectories arbitrarily near it. For example, the ideal harmonic oscillator

$$\ddot{x} + x = 0$$

has the phase trajectories

$$x^2 + y^2 = C.$$

These trajectories form a continuous family of closed curves. Therefore, an individual circular trajectory is **not** a limit cycle. A limit cycle, in contrast, is an isolated periodic orbit.



The phase portrait of the harmonic oscillator contains a continuous family of closed trajectories. For every closed orbit, another closed orbit can be found arbitrarily close to it. Therefore, no individual periodic orbit is isolated. Hence, Closed orbit $\not\Rightarrow$ Limit cycle. A limit cycle must be an isolated closed trajectory.

Stable and Unstable Limit Cycles The behavior of nearby trajectories determines the stability of a limit cycle.

Stable Limit Cycle If trajectories starting sufficiently close to a closed orbit approach it as $t \to \infty$, the closed orbit is called a **stable limit cycle**. Thus, Stable Limit Cycle $\Longrightarrow$ Periodic Attractor.

Unstable Limit Cycle If nearby trajectories move away from the closed orbit, the orbit is called an **unstable limit cycle**. An unstable limit cycle acts as a periodic repeller.

Example: Stable Limit Cycle Consider the system in polar coordinates

$$\dot{r} = r(1 - r^2),$$
$$\dot{\theta} = 1.$$

The equilibrium points are $r = 0$ and $r = 1$. The angular equation

$$\dot{\theta} = 1$$

shows that every trajectory rotates around the origin with constant angular velocity. To determine the nature of the circle $r = 1$, consider three cases.

- If
$$0 < r < 1,$$
then
$$\dot{r} = r(1 - r^2) > 0,$$
so the radius increases with time. The trajectory spirals outward toward the unit circle.

- If
$$r > 1,$$
then
$$\dot{r} = r(1 - r^2) < 0,$$
so the radius decreases with time. The trajectory spirals inward toward the unit circle.

- If
$$r = 1,$$
then
$$\dot{r} = 0,$$
and the trajectory remains on the circle of radius one.

Hence, trajectories starting both inside and outside the unit circle approach the same closed orbit. Therefore,

$$r = 1$$

is a **stable limit cycle**. The unit circle acts as a **periodic attractor**. The origin,

$$r = 0,$$

is an unstable equilibrium point, since a small perturbation causes the trajectory to move away from the origin and eventually approach the stable limit cycle.

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Example: Unstable Limit Cycle Consider the system in polar coordinates

$$\dot{r} = r(r-1),$$
$$\dot{\theta} = 1.$$

The equilibrium solutions of the radial equation satisfy $r(r-1) = 0$, giving $r = 0$ and $r = 1$. The angular equation

$$\dot{\theta} = 1$$

shows that every trajectory rotates around the origin with constant angular velocity. To determine the nature of the circle $r = 1$, consider three cases.

- If $0 < r < 1$,
$$\dot{r} = r(r-1) < 0,$$
so the radius decreases, and trajectories spiral inward toward the origin.
- If $r > 1$,
$$\dot{r} = r(r-1) > 0,$$
so the radius increases, and trajectories spiral outward away from the unit circle.
- If $r = 1$,
$$\dot{r} = 0,$$
and the trajectory remains on the circle of radius one.

Hence, trajectories starting on either side of the circle move away from it. Therefore,

$$r = 1$$

is an **unstable limit cycle**. The origin

$$r = 0$$

is a stable equilibrium point.

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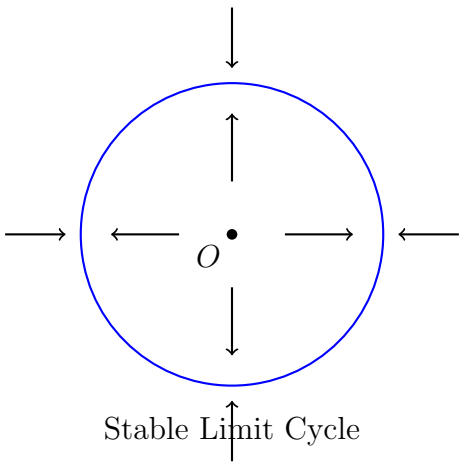

Figure 3: Trajectories starting inside the unit circle spiral outward, whereas those starting outside spiral inward. Both approach the same periodic orbit, making the unit circle a stable limit cycle (periodic attractor).

Physical Example: Wall-Clock Pendulum In a wall-clock pendulum. Two competing effects occur:

1. air resistance and friction continuously remove energy from the pendulum
2. the escapement mechanism periodically supplies energy to the system.

If the energy supplied by the escapement balances the energy dissipated by friction, the pendulum eventually exhibits oscillations of approximately constant amplitude. The long-term motion is periodic and is qualitatively similar to an attracting limit cycle. This suggests an important mechanism for sustained oscillations and can produce a stable periodic motion.

The Van der Pol Oscillator One of the classical mathematical models exhibiting stable limit-cycle behavior is the Van der Pol oscillator. Its equation can be written as the first-order system

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= b(1-x^2)y - x. \end{aligned}$$

The only equilibrium point is $(0,0)$. The Jacobian matrix is

$$J(x,y) = \begin{bmatrix} 0 & 1 \\ -2bxy - 1 & b(1-x^2) \end{bmatrix}.$$

At the origin,

$$J(0,0) = \begin{bmatrix} 0 & 1 \\ -1 & b \end{bmatrix}.$$

The characteristic equation is

$$\det(J - \lambda I) = 0,$$

which gives

$$\lambda^2 - b\lambda + 1 = 0.$$

Hence,

$$\lambda_{1,2} = \frac{b \pm \sqrt{b^2 - 4}}{2}.$$

For $b > 0$, the trace is $T = b > 0$, and the determinant is $D = 1 > 0$. Therefore, the equilibrium at the origin is unstable. For

$$0 < b$$

the eigenvalues are complex conjugates with positive real parts, and the origin is an unstable focus. Thus, trajectories starting close to the origin initially move away from the equilibrium.

Phase-Plane Interpretation The phase portrait of the Van der Pol oscillator reveals the mechanism of the periodic attractor. Consider two trajectories:

1. one starting inside the stable limit cycle
2. another starting outside the stable limit cycle.

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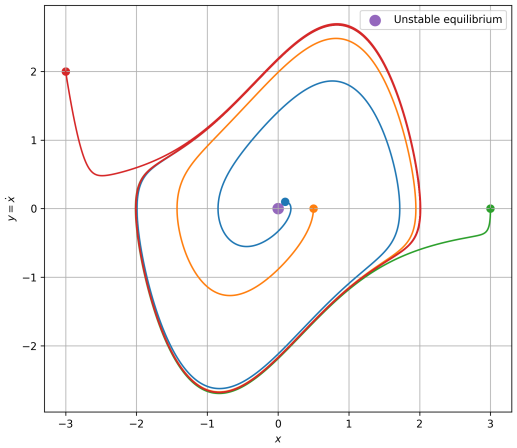
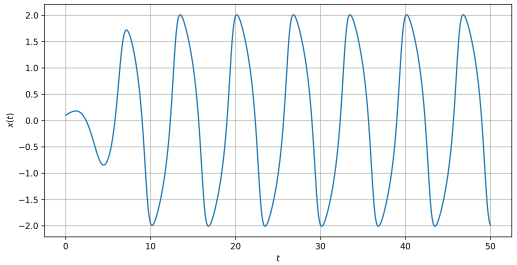
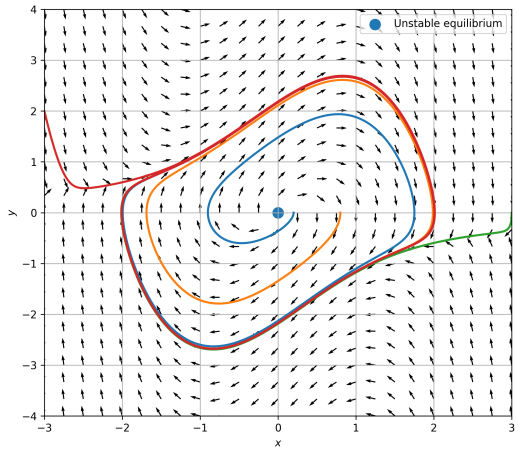
The trajectory starting inside the closed orbit moves outward because small oscillations gain energy. The trajectory starting outside the closed orbit moves inward because large oscillations lose energy. Eventually, both trajectories approach the same closed periodic orbit. Thus, an attractor need not be a point. In this example, the attracting set is a closed periodic trajectory rather than an equilibrium point.

Transient and Steady Periodic Motion Before reaching the limit cycle, the trajectory undergoes transient motion. During this stage, the oscillation amplitude changes with time. As $t \to \infty$, the transient behavior disappears, and the trajectory approaches the stable limit cycle. The resulting motion is periodic $X(t+T) = X(t)$. Thus, the long-term behavior of the system is a sustained oscillation with a well-defined period and amplitude. The stable limit cycle therefore acts as a Periodic Attractor.







2.10 Poincaré-Bendixson Theorem

Theorem 1 (Bendixson's Criterion)**.** *Consider*

$$\dot{x} = P(x, y), \qquad \dot{y} = Q(x, y),$$

where

$$P, Q \in C^1(D).$$

Define the divergence

$$s(x, y) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}.$$

If $s(x, y)$ does not change sign and is not identically zero throughout a simply connected region D, then no closed periodic orbit lies completely inside D.

(The quantity

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$$

measures the rate of expansion or contraction of phase-space area.

- $\nabla \cdot F < 0$ indicates area contraction.
- $\nabla \cdot F > 0$ indicates area expansion.
- If the divergence keeps the same sign throughout a region, closed periodic motion cannot persist inside that region.)

Proof. Assume, contrary to the statement, that a periodic orbit C exists inside the region D. Let R denote the region enclosed by C. By Green's theorem,

$$\iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA = \oint_C (P\, dy - Q\, dx).$$

Since C is a trajectory of the differential equation,

$$dx = P\, dt, \qquad dy = Q\, dt.$$

Therefore,

$$P\, dy - Q\, dx = PQ\, dt - QP\, dt = 0.$$

Hence

$$\iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA = 0.$$

However, if

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$$

has the same nonzero sign throughout R, its integral cannot be zero. This contradiction proves that no periodic orbit can exist. □

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A System with a Stable Limit Cycle Consider the nonlinear system in polar coordinates

$$\dot{r} = r(1 - r^2),$$
$$\dot{\theta} = 1.$$

This system possesses a stable limit cycle at

$$r = 1.$$

To examine Bendixson's Criterion, we first write the system in Cartesian coordinates:

$$\dot{x} = x(1 - r^2) - y,$$
$$\dot{y} = y(1 - r^2) + x,$$

where

$$r^2 = x^2 + y^2.$$

Thus,

$$P(x, y) = x(1 - r^2) - y,$$

and

$$Q(x, y) = y(1 - r^2) + x.$$

The divergence is

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 2 - 4(x^2 + y^2).$$

Equivalently,

$$\nabla \cdot F = 2 - 4r^2.$$

Observe that

$$\nabla \cdot F > 0 \quad \text{when} \quad r < \frac{1}{\sqrt{2}},$$

whereas

$$\nabla \cdot F < 0 \quad \text{when} \quad r > \frac{1}{\sqrt{2}}.$$

Hence, the divergence changes sign within the plane. Since the hypothesis of Bendixson's Criterion is not satisfied, the theorem cannot be applied. Consequently, the existence of the stable limit cycle does not contradict Bendixson's Criterion. This example illustrates that Bendixson's Criterion provides only a sufficient condition for the nonexistence of periodic orbits. If the divergence changes sign, the criterion gives no conclusion regarding the existence or nonexistence of periodic solutions.

Application of Bendixson's Criterion Consider the linear system

$$\dot{x} = -x,$$
$$\dot{y} = -y.$$

Here,

$$P(x, y) = -x, \qquad Q(x, y) = -y.$$

Therefore,

$$\frac{\partial P}{\partial x} = -1, \qquad \frac{\partial Q}{\partial y} = -1.$$

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Hence, the divergence is

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -2.$$

Since

$$\nabla \cdot F = -2 < 0$$

throughout the entire plane, the divergence has a constant negative sign. Therefore, by Bendixson's Criterion, the system possesses no closed periodic orbit. Indeed, every trajectory satisfies

$$x(t) = x_0 e^{-t}, \qquad y(t) = y_0 e^{-t},$$

which approaches the origin as $t \to \infty$. Thus, all trajectories converge to the stable equilibrium point at the origin, and no periodic motion is possible.

Damped Cubic Oscillator Consider

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= -dy - x + x^3. \end{aligned}$$

Here, $P(x, y) = y,$ and $Q(x, y) = -dy - x + x^3$. Therefore,

$$\frac{\partial P}{\partial x} = 0, \text{ and } \frac{\partial Q}{\partial y} = -d.$$

Hence,

$$\nabla \cdot F = -d.$$

Since $d > 0$, for the system, the divergence is always negative. It never changes sign. Therefore, by Bendixson's Criterion, No limit cycle exists.

Van der Pol Oscillator The state equations are

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= b(1 - x^2)y - x, \qquad b > 0. \end{aligned}$$

The divergence is

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0 + b(1 - x^2).$$

Thus,

$$\nabla \cdot F = b(1 - x^2).$$

Since

$$\nabla \cdot F > 0, \qquad |x| < 1,$$

and

$$\nabla \cdot F < 0, \qquad |x| > 1,$$

the divergence changes sign. Hence Bendixson's Criterion cannot rule out periodic motion. Indeed, the Van der Pol oscillator possesses a stable limit cycle.

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Bonhoeffer-Van der Pol Oscillator

$$\dot{x} = x - \frac{x^3}{3} - y + A_0,$$
$$\dot{y} = c(x + a - by).$$

The divergence is

$$\nabla \cdot F = 1 - x^2 - bc.$$

Since this expression depends on x, it generally changes sign. Therefore, Bendixson's criterion is inconclusive. This system is well known to possess limit cycles for suitable parameter values.

Brusselator

$$\dot{x} = a - (b+1)x + x^2 y,$$
$$\dot{y} = bx - x^2 y, \quad a, b > 0$$

The divergence is

$$\nabla \cdot F = (b-1) - 2xy.$$

Since the sign depends on (x, y), Bendixson's criterion is inconclusive. The Brusselator is known to exhibit stable limit cycles for appropriate parameters.

Duffing-Van der Pol Oscillator

$$\ddot{x} - b(1 - x^2)\dot{x} + ax + \beta x^3 = 0.$$

State equations are

$$\dot{x} = y,$$
$$\dot{y} = b(1 - x^2)y - ax - \beta x^3.$$

Hence,

$$\nabla \cdot F = b(1 - x^2).$$

The divergence changes sign. Therefore, Bendixson's criterion cannot exclude limit cycles. Stable periodic oscillations occur for suitable parameters.

Example

$$\dot{x} = y - x(x^2 + y^2 - Q),$$
$$\dot{y} = -x - y(x^2 + y^2 - Q).$$

The divergence is

$$\begin{aligned} \nabla \cdot F &= (-3x^2 - y^2 + Q) + (-x^2 - 3y^2 + Q) \\ &= 2Q - 4(x^2 + y^2). \end{aligned}$$

Since this changes sign depending on the distance from the origin, Bendixson's criterion is inconclusive. In fact, this system possesses $x^2 + y^2 = Q$ as a stable limit cycle whenever $Q > 0$.

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2.11 Higher-Dimensional Dynamical Systems

The linear stability analysis developed for two-dimensional dynamical systems can be extended naturally to systems with n state variables. The procedure remains the same: locate the equilibrium points, linearize the nonlinear system about an equilibrium, and analyze the eigenvalues of the Jacobian matrix.

General n-Dimensional System Consider the autonomous dynamical system

$$\dot{\mathbf{X}} = \mathbf{F}(\mathbf{X}),$$

where

$$\mathbf{X} = (x_1, x_2, \ldots, x_n)^T$$

is the state vector and

$$\mathbf{F} = (F_1, F_2, \ldots, F_n)^T.$$

An equilibrium point satisfies

$$\mathbf{F}(\mathbf{X}^*) = 0.$$

To study the local behavior, introduce a small perturbation

$$\mathbf{X} = \mathbf{X}^* + \delta\mathbf{X}, \qquad \|\delta\mathbf{X}\| \ll 1.$$

where

$$\delta\mathbf{X} = (\delta x_1, \delta x_2, \ldots, \delta x_n)^T.$$

Expanding the vector field about the equilibrium point using the first-order Taylor series and neglecting higher-order terms gives

$$\delta\dot{\mathbf{X}} = A\,\delta\mathbf{X},$$

where

$$A = \left[\frac{\partial F_i}{\partial x_j}\right]_{\mathbf{X}=\mathbf{X}^*}$$

is called the Jacobian (stability) matrix evaluated at the equilibrium point. The solution of the linearized system is

$$\delta\mathbf{X}(t) = e^{At}\delta\mathbf{X}(0),$$

where

$$e^{At} = I + tA + \frac{t^2A^2}{2!} + \frac{t^3A^3}{3!} + \cdots$$

is the matrix exponential. Suppose the Jacobian matrix possesses eigenvalues

$$\lambda_1, \lambda_2, \ldots, \lambda_n$$

with corresponding eigenvectors

$$\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n.$$

If these eigenvectors are linearly independent, then

$$\delta\mathbf{X}(0) = \sum_{i=1}^{n} c_i\mathbf{e}_i.$$

Consequently,

$$\delta\mathbf{X}(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t}\mathbf{e}_i.$$

Thus, the evolution of each perturbation mode is governed by its corresponding eigenvalue.

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Stability Criterion The stability of an equilibrium point depends entirely on the real parts of the eigenvalues.

- If
$$\mathrm{Re}(\lambda_i) < 0, \qquad \forall\, i,$$
all perturbations decay exponentially and the equilibrium is **asymptotically stable**.

- If at least one eigenvalue satisfies
$$\mathrm{Re}(\lambda_i) > 0,$$
the corresponding perturbation grows exponentially and the equilibrium is **unstable**.

- If one or more eigenvalues have zero real part, linearization alone is insufficient and higher-order nonlinear terms must be examined.

Hence, the stability of an equilibrium can often be determined simply from the signs of the real parts of the eigenvalues.

Routh-Hurwitz Criterion For higher-dimensional systems, finding eigenvalues analytically may be difficult because the characteristic polynomial can have a high degree. Suppose the characteristic polynomial of the Jacobian matrix is

$$\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_n = 0.$$

The Routh-Hurwitz criterion provides algebraic conditions that ensure all roots of the characteristic polynomial have negative real parts. Therefore,

- it is not always necessary to compute eigenvalues explicitly,
- stability can be determined directly from the coefficients of the characteristic polynomial.

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2.12 Lorenz Equations

The Lorenz system is one of the most celebrated nonlinear dynamical systems. Originally proposed by Edward Lorenz (1963) as a simplified model for atmospheric convection, it later became the first mathematical model shown to exhibit deterministic chaos. The governing equations are

$$\begin{aligned}\dot{x} &= \sigma(y - x),\\ \dot{y} &= rx - y - xz,\\ \dot{z} &= xy - bz,\end{aligned}$$

where $\sigma > 0$ is the Prandtl number, $r > 0$ is the Rayleigh parameter, and $b > 0$ is a geometric constant. At an equilibrium point,

$$\dot{x} = \dot{y} = \dot{z} = 0.$$

Hence,

$$\begin{aligned} \sigma(y-x) &= 0, \\ rx - y - xz &= 0, \\ xy - bz &= 0. \end{aligned}$$

From the first equation,

$$y = x.$$

Substituting this into the second equation,

$$rx - x - xz = 0,$$

or

$$x(r - 1 - z) = 0.$$

Thus two possibilities arise.

Case I: $x = 0$ Thus

$$y = 0.$$

Therefore,

$$C_1 = (0, 0, 0).$$

Case II: $x \neq 0$ Then

$$z = r - 1.$$

Substituting into the third equation,

$$x = \pm\sqrt{b(r-1)}.$$

Since

$$y = x,$$

the remaining equilibrium points are

$$C_2 = \left(\sqrt{b(r-1)}, \sqrt{b(r-1)}, r-1\right),$$

and

$$C_3 = \left(-\sqrt{b(r-1)}, -\sqrt{b(r-1)}, r-1\right).$$

Now observe that $x^2 = b(r-1)$. Therefore,

- if

  $$r < 1,$$

  then

  $$b(r-1) < 0,$$

  so no additional real equilibria exist. Only

  $$C_1 = (0, 0, 0)$$

  exists.

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- if
$$r > 1,$$
three real equilibrium points exist.

Also, the Jacobian matrix of the Lorenz system is

$$J = \begin{bmatrix} -\sigma & \sigma & 0 \\ r - z & -1 & -x \\ y & x & -b \end{bmatrix}.$$

The stability of each equilibrium point is determined from the eigenvalues of this matrix.

Case I: Stability of the $(0, 0)$ The Jacobian

$$J(C_1) = \begin{bmatrix} -\sigma & \sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{bmatrix}.$$

The characteristic equation is

$$\det(J - \lambda I) = (-b - \lambda)\left[(-\sigma - \lambda)(-1 - \lambda) - \sigma r\right] = 0.$$

Therefore,

$$\lambda_1 = -b.$$

Hence,

$$\lambda_{2,3} = \frac{-(\sigma + 1) \pm \sqrt{(\sigma + 1)^2 + 4\sigma(r - 1)}}{2}.$$

When $r < 1$ The constant term

$$\sigma(1 - r) > 0.$$

Both eigenvalues have negative real parts. Since

$$\lambda_1 = -b < 0,$$

all three eigenvalues are negative. Therefore,

$$C_1 \text{ is asymptotically stable.}$$

When $r > 1$ The constant term becomes negative. Hence one eigenvalue is positive. Therefore, C_1 is unstable. Suppose the characteristic polynomial is

$$\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3 = 0.$$

For a cubic equation, the equilibrium is asymptotically stable if

$$a_1 > 0, \qquad a_2 > 0, \qquad a_3 > 0, \qquad a_1 a_2 > a_3.$$

At the origin,

$$\begin{aligned} a_1 &= \sigma + b + 1, \\ a_2 &= \sigma(1 + b - r) + b, \\ a_3 &= \sigma b(1 - r). \end{aligned}$$

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When

$$r < 1,$$

all four inequalities are satisfied. Hence the origin is stable. When

$$r > 1, \text{ and } a_3 < 0,$$

which immediately violates the Routh-Hurwitz criterion. Therefore,

$$C_1$$

becomes unstable.

Case II: Stability of C_2 and C_3 For

$$C_2,\ C_3,$$

the characteristic polynomial is again cubic. Its coefficients are

$$\begin{aligned} a_1 &= \sigma + b + 1, \\ a_2 &= b(\sigma + r), \\ a_3 &= 2\sigma b(r-1). \end{aligned}$$

Since

$$r > 1,$$

the first three Routh-Hurwitz conditions are automatically satisfied. The remaining condition,

$$a_1 a_2 > a_3,$$

after simplification yields

$$r < r_c = \frac{\sigma(\sigma + b + 3)}{\sigma - b - 1}.$$

Thus,

$$1 < r < r_c$$

is the stability region of the two nonzero equilibrium points.

Classical Lorenz Parameters Lorenz used

$$\sigma = 10, \qquad b = \frac{8}{3}.$$

Hence

$$r_c = \frac{470}{19} \approx 24.74.$$

Therefore,

$$1 < r < 24.74 \implies C_2 \text{ and } C_3 \text{ are asymptotically stable.}$$

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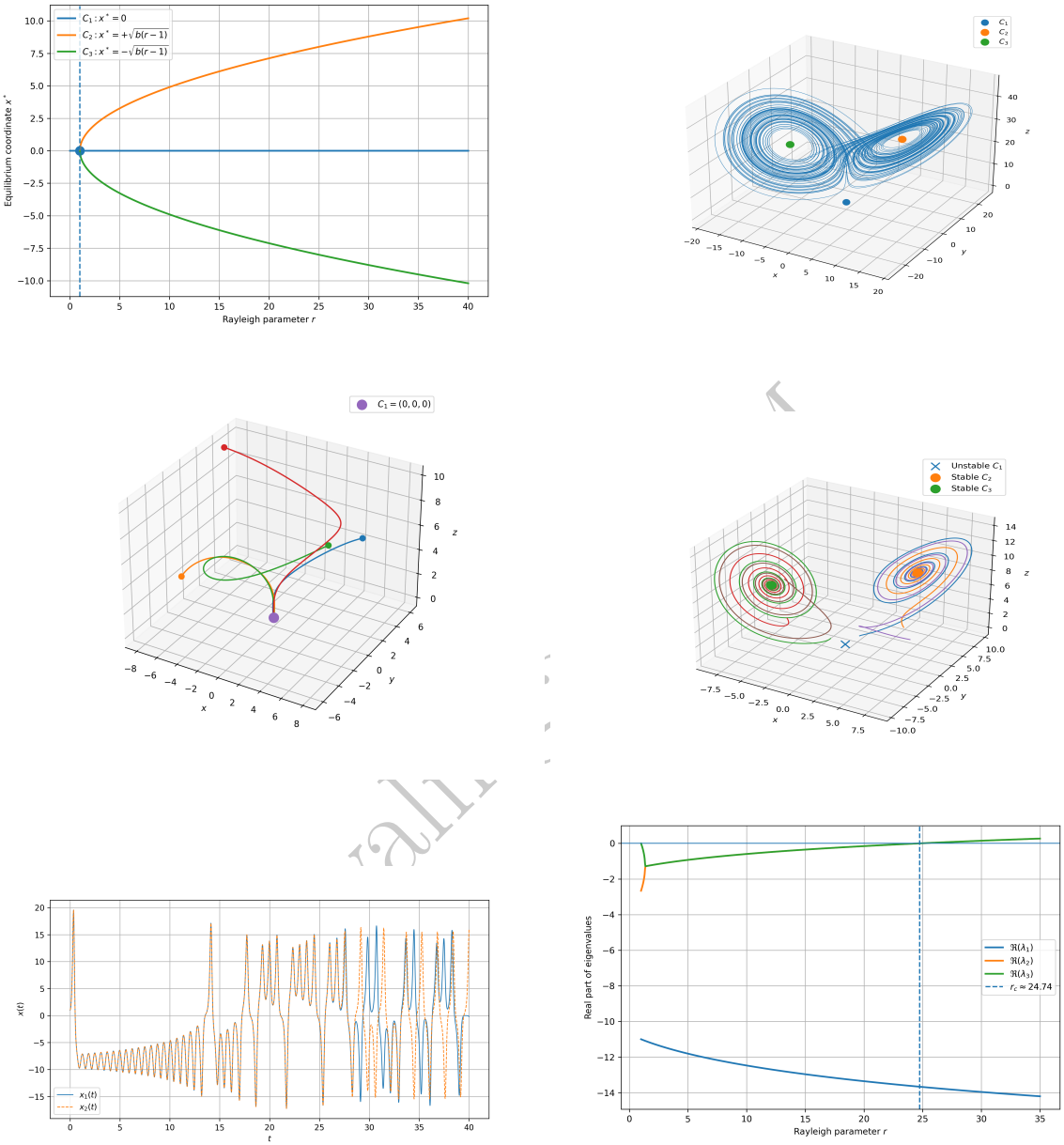
For $r > 24.74$ The Routh-Hurwitz criterion fails. One real eigenvalue remains negative, while the remaining two become complex conjugates with positive real parts. Hence C_2 and C_3 become unstable.



Graphical Interpretation of the Lorenz Dynamics For the classical parameters

$$\sigma = 10, \qquad b = \frac{8}{3}, \qquad r_c = \frac{470}{19} \approx 24.74,$$

the phase-space behavior changes significantly with the Rayleigh parameter r. For $r < 1$, the origin $C_1 = (0, 0, 0)$ is the only equilibrium and all nearby trajectories approach it. For $1 < r < r_c$, the origin becomes unstable, while the two nonzero equilibria C_2 and C_3 are asymptotically stable; trajectories approach one of these two attractors depending on the initial condition. The corresponding regions of initial conditions are called *basins of attraction*. At $r = r_c$, a complex conjugate pair of eigenvalues of C_2 and C_3 crosses the

imaginary axis, $\Re(\lambda) = 0$, and the nonzero equilibria lose stability. For $r = 28 > r_c$, all three equilibria are unstable, but trajectories may remain bounded and exhibit complicated recurrent motion around C_2 and C_3. The resulting structure is the famous **Lorenz attractor** and illustrates **deterministic chaos**. A characteristic feature of chaotic motion is **sensitive dependence on initial conditions**. For example, trajectories starting from

$$X_1(0) = (1, 1, 1), \qquad X_2(0) = (1.000001, 1, 1)$$

are initially almost indistinguishable, since

$$\|X_1(0) - X_2(0)\| = 10^{-6},$$

but eventually separate significantly. This phenomenon is commonly referred to as the *butterfly effect*. Thus, deterministic system $\not\Rightarrow$ predictable long-term motion.

2.13 Quasiperiodic Attractors

In nonlinear dynamical systems, the long-term motion is not always confined to an equilibrium point or a periodic orbit. Another important type of motion is called **quasiperiodic motion**. Such motions commonly arise in mechanics, electrical circuits, celestial mechanics, and nonlinear oscillators.

Quasiperiodic Motion A motion is said to be **quasiperiodic** if it is composed of two or more independent oscillations whose frequencies are incommensurate (i.e., their ratios are irrational). Mathematically, a quasiperiodic solution can be expressed as

$$x(t) = \Phi(\omega_1 t, \omega_2 t, \ldots, \omega_n t),$$

where

- Φ is periodic in each of its arguments,
- $\omega_1, \omega_2, \ldots, \omega_n$ are independent frequencies.

Conditions on the Frequencies For quasiperiodic motion, the frequencies must satisfy the following conditions.

1. Linear Independence There should not exist integers $k_1, k_2, \ldots, k_n$, not all zero, such that $k_1\omega_1 + k_2\omega_2 + \cdots + k_n\omega_n = 0$. Hence, the frequencies are linearly independent over the integers.

2. Irrational Frequency Ratios The ratios between the frequencies must be irrational. For two frequencies, $\frac{\omega_1}{\omega_2} \notin \mathbb{Q}$. Consequently, the motion never repeats exactly, and the trajectory never closes on itself.

Geometric Interpretation Each independent frequency contributes one angular coordinate. Therefore,

- one frequency corresponds to motion on a circle,
- two independent frequencies correspond to motion on a two-dimensional torus,
- three independent frequencies correspond to motion on a three-dimensional torus.

In general, n independent frequencies $\Longrightarrow$ n-dimensional torus (T^n).

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Quasiperiodic Attractor If nearby trajectories approach a torus carrying quasiperiodic motion, then the torus acts as an attractor. Stable Torus $\implies$ Quasiperiodic Attractor. Unlike a limit cycle, the motion never repeats exactly, infinitely many oscillation phases coexist, and trajectories densely cover the attracting torus.

Periodic or Quasiperiodic Motion Consider the motion

$$x(t) = b_1 \cos(\omega_1 t) + b_2 \cos(\omega_2 t),$$

where b_1 and b_2 are constants. Two cases arise.

Case 1: Rational Frequency Ratio If

$$\frac{\omega_1}{\omega_2} = \frac{m}{n},$$

where m and n are integers, then both oscillations repeat after a common period. Hence, motion is periodic. The corresponding phase trajectory is a closed curve.

Case 2: Irrational Frequency Ratio If

$$\frac{\omega_1}{\omega_2}$$

is irrational, the oscillations never repeat simultaneously. Therefore, the motion is quasiperiodic. The phase trajectory never closes and gradually fills the invariant torus.

Example Consider

$$x(t) = \cos t + \cos(\sqrt{2}\, t).$$

Here,

$$\omega_1 = 1, \qquad \omega_2 = \sqrt{2}.$$

Since

$$\frac{\omega_1}{\omega_2} = \frac{1}{\sqrt{2}}$$

is irrational, the motion is quasiperiodic. The corresponding phase portrait is not a closed orbit.

Example: Quasiperiodic Motion on an Attracting Torus Consider the dynamical system in toroidal coordinates

$$\dot{\rho} = 1 - \rho,$$
$$\dot{\theta} = 1,$$
$$\dot{\phi} = \sqrt{2},$$

where ρ measures the distance from an invariant torus and θ, ϕ are angular variables. The first equation has the solution

$$\rho(t) = 1 + \big(\rho_0 - 1\big)e^{-t}.$$

Hence,

$$\lim_{t\to\infty} \rho(t) = 1.$$

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Therefore, all nearby trajectories approach the surface$\rho = 1$. Thus, the torus $\rho = 1$ is an attracting invariant set. On the torus,

$$\theta(t) = \theta_0 + t, \qquad \phi(t) = \phi_0 + \sqrt{2}\, t.$$

The two angular frequencies are

$$\omega_1 = 1, \qquad \omega_2 = \sqrt{2}.$$

Since

$$\frac{\omega_1}{\omega_2} = \frac{1}{\sqrt{2}} \notin \mathbb{Q},$$

the two angular motions never repeat simultaneously. Consequently, the trajectory never closes on itself. Instead, as $t \to \infty$, the trajectory moves quasiperiodically and densely covers the invariant torus. Hence, Attraction to a torus + incommensurate frequencies $\Longrightarrow$ quasiperiodic attractor.

To visualize the motion, a torus may be represented in Cartesian coordinates by

$$\begin{aligned} x &= (R + \rho\cos\phi)\cos\theta, \\ y &= (R + \rho\cos\phi)\sin\theta, \\ z &= \rho\sin\phi, \end{aligned}$$

where $R > 1$ is the major radius of the torus. A trajectory starting with $\rho_0 \neq 1$ first approaches the torus because $\rho(t) \to 1$. After the transient motion disappears, the trajectory continues to move with the two independent frequencies 1 and $\sqrt{2}$. Unlike a stable equilibrium, the trajectory does not approach a point. Unlike a stable limit cycle, the motion does not approach a single closed orbit. Instead, the attracting set is an entire torus carrying quasiperiodic motion. Thus,

Stable equilibrium $\to$ point attractor,
Stable limit cycle $\to$ periodic attractor,
Stable invariant torus $\to$ quasiperiodic attractor.

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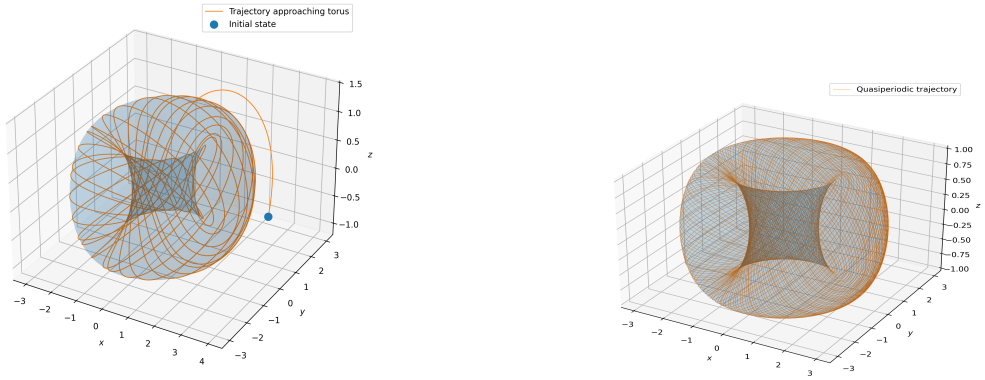

2.14 Chaotic Attractors

The attractors studied so far, namely equilibrium points, limit cycles and quasiperiodic attractors, describe motions that are stationary, periodic or almost periodic. In contrast, many nonlinear dynamical systems exhibit a more complicated type of long-term behavior known as **chaotic motion**.

Definition A **chaotic attractor** is a bounded attracting set on which the system evolves in an apparently irregular and unpredictable manner. Although the motion is deterministic (governed entirely by differential equations), it appears random because of its extreme sensitivity to initial conditions.

Unlike equilibrium points or periodic orbits, a chaotic attractor is neither a single point, a closed orbit, nor an invariant torus.

Characteristics of Chaotic Attractors A chaotic attractor possesses the following important properties.

1. **Bounded Motion** The trajectory always remains inside a finite region of the phase space.
2. **Aperiodic Motion** The trajectory never repeats itself exactly and does not possess a finite period.
3. **Sensitive Dependence on Initial Conditions** Two trajectories starting from very close initial conditions separate exponentially with time.
4. **Deterministic Dynamics** Although the motion appears random, it is completely governed by the underlying differential equations.

Sensitive Dependence on Initial Conditions The defining feature of chaos is the sensitive dependence on initial conditions. Suppose two nearby trajectories have initial conditions

$$\mathbf{x}_1(0), \qquad \mathbf{x}_2(0),$$

with initial separation

$$\delta(0) = \|\mathbf{x}_1(0) - \mathbf{x}_2(0)\|.$$

For a chaotic system,

$$\delta(t) \approx \delta(0)e^{\lambda t},$$

where $\lambda > 0$ is called the **largest Lyapunov exponent**. Since $e^{\lambda t}$ grows rapidly, even extremely small differences in the initial conditions eventually produce completely different trajectories. Although the governing equations are deterministic, measurements of the initial conditions always contain small errors. If $\delta(0)$ denotes the measurement error, then after time t,

$$\delta(t) = \delta(0)e^{\lambda t}.$$

Hence the prediction error grows exponentially. Consequently, long-term prediction becomes practically impossible even though the system itself is deterministic.

Lorentz Oscillator See the explanation for the Lorentz oscillator previously after the section on higher dimensional dynamical systems.

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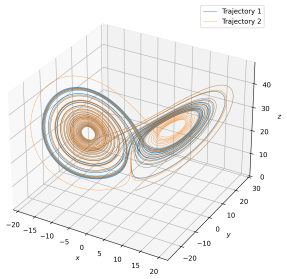

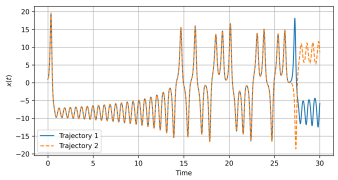

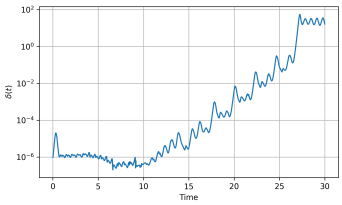

2.15 Poincaré Map

The continuous evolution of a nonlinear dynamical system is often difficult to analyze, especially when the trajectories become complicated. A useful technique for simplifying the study of such systems is the **Poincaré map**, which converts a continuous-time dynamical system into a discrete-time system.

Definition A **Poincaré map** is obtained by observing the state of a dynamical system at fixed intervals of time or whenever the trajectory intersects a chosen surface (called the Poincaré section). Instead of studying a continuous trajectory, we analyze the sequence of points

$$X_0,\ X_1,\ X_2,\ \ldots$$

generated by successive observations. Thus, a continuous flow is transformed into a discrete map.

Why Use a Poincaré Map? The Poincaré map reduces the dimensionality of the problem and makes the long-term behavior of the system easier to visualize. It is particularly useful for identifying periodic motion, detecting quasiperiodic motion, distinguishing chaotic attractors, and studying bifurcations.

Poincaré Map of a Quasiperiodic Motion Consider the quasiperiodic solution

$$x(t) = b_1 \cos(\omega_1 t) + b_2 \cos(\omega_2 t),$$

where $\frac{\omega_1}{\omega_2}$ is irrational. To construct the Poincaré map, sample the solution at the times

$$t_n = \frac{2\pi n}{\omega_1}, \qquad n = 0, 1, 2, \ldots$$

Since $\cos(\omega_1 t_n) = 1$, the sampled coordinates become

$$x_n = b_1 + b_2 \cos\left(\frac{2\pi n \omega_2}{\omega_1}\right),$$

and

$$y_n = -b_2 \omega_2 \sin\left(\frac{2\pi n \omega_2}{\omega_1}\right).$$

The collection of points (x_n, y_n) forms the Poincaré map. Now to analyze ther shape of the Poincaré map, We square the above equations,

$$(x_n - b_1)^2 = b_2^2 \cos^2\theta,$$

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where $\theta = \frac{2\pi n\omega_2}{\omega_1}$, and

$$\left(\frac{y_n}{\omega_2}\right)^2 = b_2^2 \sin^2\theta.$$

Adding the two equations,

$$(x_n - b_1)^2 + \left(\frac{y_n}{\omega_2}\right)^2 = b_2^2.$$

This is the equation of an ellipse. Hence, the Poincaré map of a quasiperiodic motion is a closed ellipse (or, in general, a closed invariant curve).

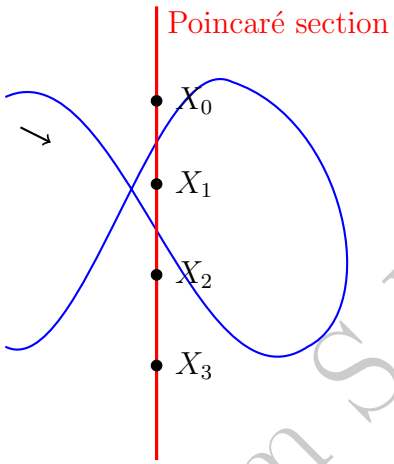

Figure 7: A continuous trajectory intersects the Poincaré section repeatedly. Only the intersection points are retained to construct the Poincaré map.

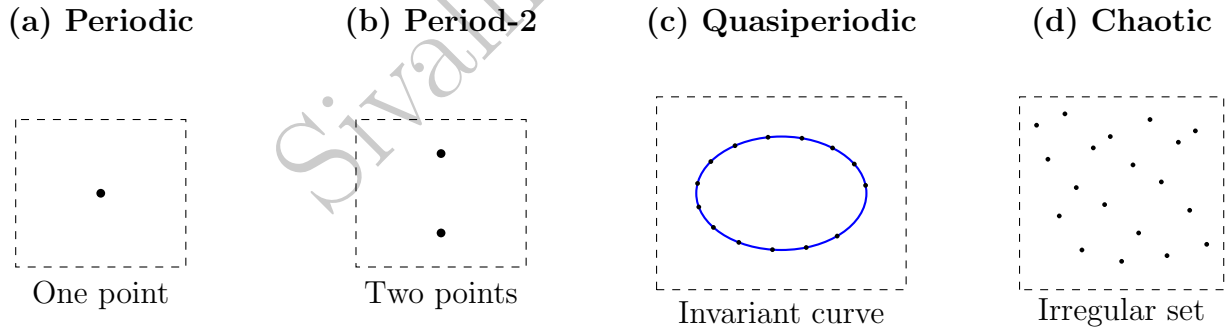

Figure 8: Typical Poincaré maps. (a) A periodic orbit produces a single point. (b) A period-two orbit produces two alternating points. (c) A quasiperiodic orbit generates an invariant closed curve. (d) A chaotic orbit produces an irregular (fractal) set of points.

2.16 Dissipative and Conservative Systems

In dynamical systems, an important classification is based on whether the system conserves energy or dissipates it during motion. This leads to the concepts of **conservative** and **dissipative** systems.

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Conservative Systems A dynamical system is called **conservative** if the total mechanical energy remains constant during the motion. Suppose a particle is subjected to a force

$$\mathbf{F} = -\nabla V,$$

where $V(x, y, z)$ is the potential energy, V depends only on position, and ∇ denotes the gradient operator. Such a force is called a **conservative force**.

Work Done by a Conservative Force Let the particle undergo an infinitesimal displacement $d\mathbf{r}$. The elementary work done is $dW = \mathbf{F} \cdot d\mathbf{r}$. Using

$$\mathbf{F} = -\nabla V,$$

we obtain

$$dW = -\nabla V \cdot d\mathbf{r}.$$

Since

$$dV = \nabla V \cdot d\mathbf{r},$$

it follows that

$$dW = -dV.$$

Integrating between two points A and B,

$$W = \int_A^B \mathbf{F} \cdot d\mathbf{r} = -\int_A^B dV.$$

Therefore,

$$W = V(A) - V(B).$$

Hence, the work done depends only on the initial and final points, and it is independent of the path followed. If the particle returns to its starting point, $A = B$, then $W = V(A) - V(A) = 0$. Therefore,

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0.$$

This proves that the work done around any closed path is zero. Hence the total mechanical energy $E = T + V$ remains constant. Such systems are called conservative systems.

Phase Space Volume Consider an n-dimensional dynamical system

$$\dot{\mathbf{X}} = \mathbf{F}(\mathbf{X}).$$

Suppose a small rectangular volume around a point in phase space is

$$\Delta T = \prod_{i=1}^{n} \Delta x_i.$$

As time evolves, this volume may remain constant, or shrink, or expand. Studying this behavior helps classify dynamical systems.

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Rate of Change of Phase Volume Taking logarithms,

$$\ln(\Delta T) = \sum_{i=1}^{n} \ln(\Delta x_i).$$

Differentiating with respect to time,

$$\frac{1}{\Delta T}\frac{d(\Delta T)}{dt} = \sum_{i=1}^{n} \frac{1}{\Delta x_i}\frac{d(\Delta x_i)}{dt}.$$

For small perturbations,

$$\frac{d(\Delta x_i)}{dt} = \sum_{j=1}^{n} \frac{\partial F_i}{\partial x_j}\Delta x_j.$$

Evaluating the instantaneous change at the chosen point,

$$\frac{1}{\Delta T}\frac{d(\Delta T)}{dt} = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i}.$$

The quantity

$$\nabla \cdot \mathbf{F} = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i}$$

is called the **divergence** of the vector field.

Classification Using Divergence The divergence completely determines the local behavior of phase-space volume.

Case 1 If

$$\nabla \cdot \mathbf{F} = 0,$$

then

$$\frac{d(\Delta T)}{dt} = 0.$$

Hence,

$$\Delta T = \text{constant}.$$

The phase-space volume is preserved. The system is called Conservative.

Case 2 If

$$\nabla \cdot \mathbf{F} < 0,$$

then

$$\Delta T$$

decreases continuously. Nearby trajectories move closer together. The system loses energy. Hence, a dissipative system

Case 3 If

$$\nabla \cdot \mathbf{F} > 0,$$

phase-space volume increases. Nearby trajectories move apart. Such systems are called Volume Expanding Systems

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Example Consider

$$\ddot{x} + f(x) = 0.$$

Introduce

$$x_1 = x, \qquad x_2 = \dot{x}.$$

Then

$$\dot{x}_1 = x_2, \text{ and } \dot{x}_2 = -f(x_1).$$

Hence

$$F_1 = x_2, \qquad F_2 = -f(x_1).$$

Now compute the divergence.

$$\frac{\partial F_1}{\partial x_1} = 0,$$

because x_2 is independent of x_1. Also,

$$\frac{\partial F_2}{\partial x_2} = 0,$$

because F_2 depends only on x_1. Therefore,

$$\nabla \cdot F = 0 + 0 = 0.$$

Hence, the system is conservative.

Example Consider

$$\ddot{x} + d\dot{x} + f(x) = 0, \qquad d > 0.$$

Let

$$x_1 = x, \qquad x_2 = \dot{x}.$$

Then

$$\dot{x}_1 = x_2, \text{ and} \dot{x}_2 = -dx_2 - f(x_1).$$

Hence,

$$F_1 = x_2, \text{ and } F_2 = -dx_2 - f(x_1).$$

Now,

$$\frac{\partial F_1}{\partial x_1} = 0,$$

and

$$\frac{\partial F_2}{\partial x_2} = -d.$$

Therefore,

$$\nabla \cdot F = -d < 0.$$

Hence,the system is dissipative. Since $d > 0$, the phase-space volume decreases exponentially.

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Example Show that the Lorenz system is dissipative. The Lorenz equations are

$$\begin{aligned}\dot{x} &= \sigma(y - x),\\ \dot{y} &= rx - y - xz,\\ \dot{z} &= xy - bz.\end{aligned}$$

The vector field is

$$\begin{aligned}F_1 &= \sigma(y - x),\\ F_2 &= rx - y - xz,\\ F_3 &= xy - bz.\end{aligned}$$

Compute each partial derivative.

$$\frac{\partial F_1}{\partial x} = -\sigma, \quad \frac{\partial F_2}{\partial y} = -1, \text{ and } \frac{\partial F_3}{\partial z} = -b.$$

Therefore,

$$\nabla \cdot F = -\sigma - 1 - b.$$

Since

$$\sigma > 0, \qquad b > 0,$$

we obtain

$$\nabla \cdot F < 0.$$

Hence, the Lorenz system is dissipative. The phase-space volume contracts exponentially with time.

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Remark Many nonlinear systems possessing stable equilibrium points, stable limit cycles, or chaotic attractors are dissipative. The contraction of phase-space volume allows trajectories starting from different initial conditions to approach the same attracting set.

In contrast, conservative systems preserve phase-space volume. Consequently, they cannot possess attracting or repelling limit cycles. Instead, they usually contain continuous families of periodic orbits, each corresponding to a different initial condition.

That is, conservative systems generally exhibit regular periodic motion, whereas dissipative systems may possess stable equilibrium points, limit cycles, quasiperiodic attractors, and even chaotic attractors.

Comparison of Conservative and Dissipative Systems **Conservative System.** Consider the simple harmonic oscillator

$$\ddot{x} + x = 0. \quad \text{the system becomes} \quad \dot{x}_1 = x_2, \qquad \dot{x}_2 = -x_1.$$

Thus, the divergence is

$$\nabla \cdot F = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} = 0 + 0 = 0.$$

Hence, the phase-space volume is preserved, and the system is conservative. The phase portrait consists of a family of closed curves (circles or ellipses), each corresponding to a different initial condition.

Dissipative System. Consider the damped harmonic oscillator

$$\ddot{x} + c\dot{x} + x = 0, \qquad c > 0. \quad \text{Thus} \quad \dot{x}_1 = x_2, \qquad \dot{x}_2 = -x_1 - cx_2.$$

Hence, the divergence is

$$\nabla \cdot F = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} = 0 - c = -c < 0.$$

Therefore, the phase-space volume contracts with time, and the system is dissipative. All trajectories spiral toward the stable equilibrium at the origin because energy is continuously lost due to damping.

Property	Conservative System	Dissipative System
Mechanical Energy	Constant	Decreases with time
Divergence	$\nabla \cdot F = 0$	$\nabla \cdot F < 0$
Phase-Space Volume	Preserved	Contracts
Typical Trajectories	Families of closed orbits	Spirals toward attractors
Examples	Simple harmonic oscillator	Damped oscillator, Lorenz system
Stable Limit Cycle	Not possible	Possible
Chaotic Attractor	Not possible	Possible

2.17 Hamiltonian Systems

Hamiltonian systems form one of the most important classes of conservative dynamical systems. They arise naturally in classical mechanics and describe the evolution of systems in which the total energy is conserved. Unlike dissipative systems, Hamiltonian systems preserve phase-space volume during their evolution.

Generalized Coordinates and Momenta Consider a mechanical system with n degrees of freedom. Instead of using only the generalized coordinates

$$q_1, q_2, \ldots, q_n,$$

Hamiltonian mechanics introduces the corresponding generalized momenta

$$p_1, p_2, \ldots, p_n.$$

Thus, the complete state vector becomes

$$X = (q_1, q_2, \ldots, q_n, p_1, p_2, \ldots, p_n)^T,$$

which belongs to a $2n$-dimensional phase space.

Hamilton's Equations The dynamics of a Hamiltonian system are governed by Hamilton's equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i},$$

and

$$\dot{p}_i = -\frac{\partial H}{\partial q_i},$$

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where

$$i = 1, 2, \ldots, n,$$

and

$$H(q, p)$$

is called the Hamiltonian function. In many mechanical systems,

$$H = T + V,$$

where

- T is the kinetic energy,
- V is the potential energy.

Phase-Space Volume Conservation One of the most remarkable properties of Hamiltonian systems is that the phase-space volume remains constant during the motion. To prove this, consider the vector field

$$F = (\dot{q}_1, \ldots, \dot{q}_n, \dot{p}_1, \ldots, \dot{p}_n).$$

The divergence of the vector field is

$$\nabla \cdot F = \sum_{i=1}^{n} \frac{\partial \dot{q}_i}{\partial q_i} + \sum_{i=1}^{n} \frac{\partial \dot{p}_i}{\partial p_i}.$$

Substituting Hamilton's equations,

$$\dot{q}_i = \frac{\partial H}{\partial p_i},$$

and

$$\dot{p}_i = -\frac{\partial H}{\partial q_i},$$

gives

$$\nabla \cdot F = \sum_{i=1}^{n} \frac{\partial}{\partial q_i}\left(\frac{\partial H}{\partial p_i}\right) + \sum_{i=1}^{n} \frac{\partial}{\partial p_i}\left(-\frac{\partial H}{\partial q_i}\right).$$

Hence,

$$\nabla \cdot F = \sum_{i=1}^{n} \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i}\right).$$

If the Hamiltonian possesses continuous second-order partial derivatives, Clairaut's theorem gives

$$\frac{\partial^2 H}{\partial q_i \partial p_i} = \frac{\partial^2 H}{\partial p_i \partial q_i}.$$

Therefore,

$$\nabla \cdot F = 0.$$

Hence,

$$\frac{d(\Delta V)}{dt} = 0,$$

which means that every infinitesimal phase-space volume remains unchanged. This result is known as Liouville's Theorem.

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Physical Interpretation Suppose a closed surface encloses a small volume in phase space. As every point on the surface evolves according to Hamilton's equations, the shape of the volume may deform, the volume may stretch or bend, but its total volume always remains constant. Hence Hamiltonian systems preserve phase-space volume. The types of equilibrium points also differ. Hamiltonian systems can possess Centers, or Saddles. They cannot possess Stable nodes, Stable stars, Stable spirals (foci), Unstable nodes, Unstable stars,or Unstable spirals. Since

$$\nabla \cdot F = 0,$$

the sum of the eigenvalues of the Jacobian matrix is zero. Therefore,

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = 0.$$

Consequently, all eigenvalues cannot simultaneously have negative real parts. Hence asymptotically stable equilibria cannot exist. Similarly, all eigenvalues cannot simultaneously have positive real parts. Thus, only centers and saddles are possible equilibrium points.

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3. Bifurcations and Onset of Chaos in Dissipative Systems

Many engineering and physical systems possess parameters such as damping coefficient, forcing amplitude, forcing frequency, nonlinear stiffness, control gain, growth rate, and reaction rate. These parameters are called **control parameters** because changing them alters the system's qualitative behaviour.

What is a Bifurcation? A **bifurcation** is a sudden qualitative change in the behaviour of a dynamical system caused by a smooth variation of a control parameter. Unlike ordinary parameter changes, bifurcations alter the structure of the solutions themselves. Examples creation or disappearance of equilibrium points, change in stability of equilibrium points, birth of periodic oscillations, or transition from periodic to chaotic motion.

Control Parameter A control parameter is a quantity whose variation influences the dynamics of the system without being a state variable. Typical examples are $\mu,\ a,\ b,\ f,\ \omega,\ r,\ \sigma$. As the control parameter changes continuously, the system may remain qualitatively unchanged over an interval of values. However, at certain critical parameter values, a sudden change in the dynamics occurs.

Critical (Bifurcation) Value A **critical value** or **bifurcation value** is the value of the control parameter at which the qualitative nature of the solution changes. Mathematically,

$$\mu = \mu_c$$

is called a bifurcation point if the phase portrait for

$$\mu < \mu_c$$

is topologically different from that for

$$\mu > \mu_c.$$

A bifurcation occurs because an existing solution loses stability. As the parameter changes, a stable solution $\rightarrow$ loses stability $\rightarrow$ a new stable solution appears. Thus, bifurcations represent transitions between different types of stable motions.

Definition of Bifurcation A **bifurcation** is a sudden qualitative change in the dynamics of a nonlinear system caused by a smooth variation of a control parameter. The parameter value at which this change occurs is called the **bifurcation value** or **critical parameter**. Mathematically, say let $\mu = \mu_c$ be called a bifurcation point if the phase portrait or stability of the system changes as the parameter μ passes through μ_c.

Example Consider the general damped Duffing oscillator:

$$\ddot{x} + d\dot{x} - ax + bx^3 = 0, \qquad a > 0,\ b > 0, \quad \text{equivalently} \quad \begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= ax_1 - bx_1^3 - dx_2. \end{aligned}$$

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Its equilibrium points are

$$(0,0), \left(\pm\sqrt{\frac{a}{b}}, 0\right)$$

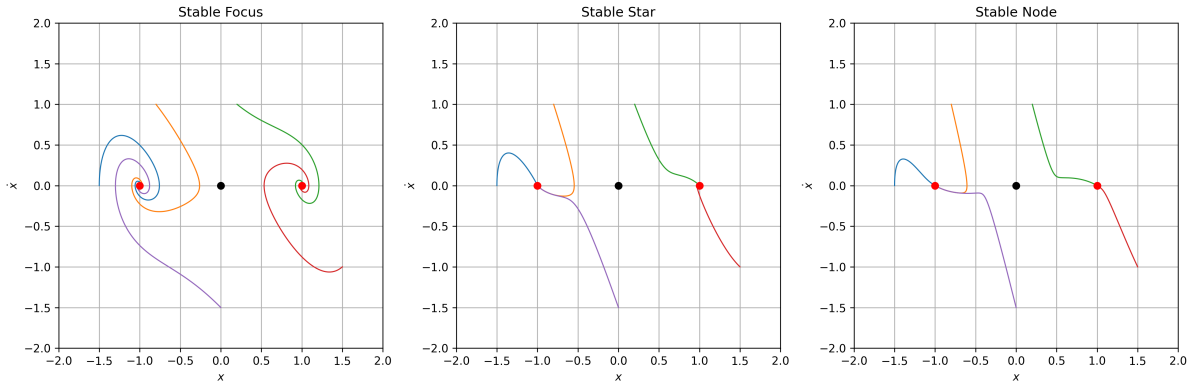
The origin is always a saddle point. But $\left(\pm\sqrt{\frac{a}{b}}, 0\right)$, change their nature depending on the damping parameter d:

$$d^2 \begin{cases} < 8a, & \text{Stable Focus,} \\ = 8a, & \text{Stable Star,} \\ > 8a, & \text{Stable Node.} \end{cases}$$

Thus, the critical value

$$d = \sqrt{8a}$$

marks a qualitative change in the system dynamics and is therefore a **bifurcation value**. The figure below shows this phenomenon for $a = b = 1$.



A very small perturbation near the bifurcation value can lead to a completely different evolution of the system. The four fundamental local bifurcations are

1. Saddle-node bifurcation,
2. Pitchfork bifurcation,
3. Transcritical bifurcation,
4. Hopf bifurcation.

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3.1 Saddle-Node Bifurcation

A saddle-node bifurcation describes the creation or annihilation of two equilibrium points as a control parameter crosses a critical value. Typically, for one parameter region, no equilibrium points exist, at the critical parameter value, a single degenerate tranquility appears, and beyond the critical value, two equilibrium points emerge: one stable and one unstable. Hence a saddle-node bifurcation is also called a **fold bifurcation**.

Example Consider the nonlinear system

$$\dot{x} = \mu - x^2,$$
$$\dot{y} = -y,$$

where μ is the bifurcation (control) parameter, and x and y are the state variables. Observe that the y-equation is independent of x, whereas the x-equation contains the nonlinear term x^2. Therefore the equilibrium are,

$$E_1 = (\sqrt{\mu}, 0)\,, \qquad E_2 = (-\sqrt{\mu}, 0)\,.$$

Case 1: $\mu < 0$ There is no real solution. Hence, no real equilibrium point exists.

Case 2: $\mu = 0$ Thus only one equilibrium point exists, $(0, 0)$.

Case 3: $\mu > 0$ There are two real equilibrium points,

$$(\sqrt{\mu}, 0), \qquad (-\sqrt{\mu}, 0).$$

Now, consider

$$F(x, y) = \begin{bmatrix} \mu - x^2 \\ -y \end{bmatrix}.$$

The Jacobian matrix is

$$J = \begin{bmatrix} -2x & 0 \\ 0 & -1 \end{bmatrix}.$$

Substituting $x = \sqrt{\mu}$, gives

$$\lambda_1 = -2\sqrt{\mu}, \qquad \lambda_2 = -1.$$

Both eigenvalues are negative. Therefore,E_1 is a stable node. now, stbstituting $x = -\sqrt{\mu}$, gives

$$\lambda_1 = 2\sqrt{\mu}, \qquad \lambda_2 = -1.$$

One eigenvalue is positive while the other is negative. Hence E_2 is a saddle point. The system at $\mu = 0$ becomes

$$\dot{x} = -x^2, \text{ and } \dot{y} = -y.$$

Hence

$$y(t) = y_0 e^{-t} \text{ and } x(t) = \frac{x_0}{1 + x_0 t}.$$

Therefore,

$$y(t) \to 0 \qquad (t \to \infty).$$

If $x_0 > 0$ The denominator increases with time. Hence $x(t) \to 0$. Trajectories approach the origin.

If $x_0 < 0$ The denominator $1 + x_0 t$ becomes zero at $t = -\frac{1}{x_0} > 0$. Thus $x(t) \to -\infty$. Trajectories move away from the origin.

Thus at $\mu = 0$, the equilibrium point is neither a node nor a saddle. Instead, trajectories approach from one side, and trajectories diverge from the other side. This degenerate equilibrium is called a Saddle-Node.
If

$$\mu < 0$$

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No equilibrium points exist and if

$$\mu = 0$$

one saddle-node equilibrium exists.

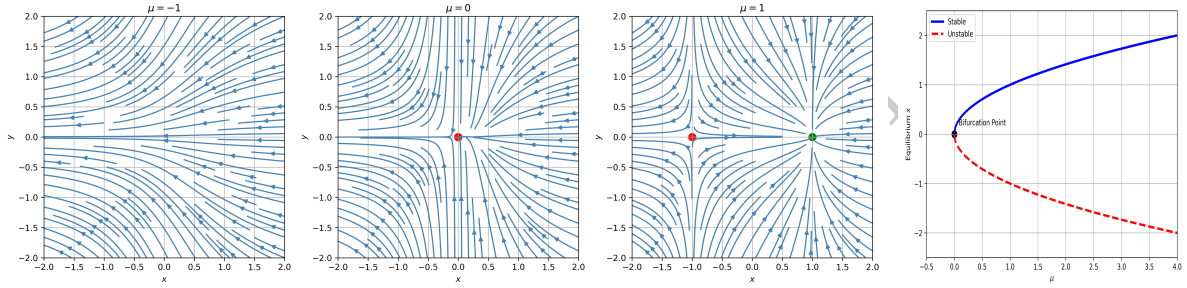
$$\mu > 0$$

Two equilibrium points appear: one stable node, and one unstable saddle.

Bifurcation Diagram The equilibrium solutions satisfy

$$x^2 = \mu.$$

The upper branch $x = \sqrt{\mu}$ corresponds to the stable equilibrium, whereas $x = -\sqrt{\mu}$ corresponds to the unstable saddle. Both branches meet at $(\mu, x) = (0, 0)$, which is the bifurcation point.



Example Consider the nonlinear system

$$\dot{x} = \mu - x^2, \qquad \dot{y} = -y.$$

Hence

$$y(t) = y_0 e^{-t}.$$

Thus,

$$\lim_{t\to\infty} y(t) = 0.$$

Case I: $\mu > 0$ The first equation becomes

$$\frac{dx}{dt} = \mu - x^2.$$

Let

$$a = \sqrt{\mu} > 0.$$

Then

$$\frac{dx}{dt} = a^2 - x^2.$$

Separating variables,

$$\frac{dx}{a^2 - x^2} = dt.$$

Using the standard integral

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a + x}{a - x} \right|,$$

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we obtain

$$\frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| = t + C.$$

Multiplying by $2a$,

$$\ln\left|\frac{a+x}{a-x}\right| = 2at + C_1.$$

Exponentiating,

$$\frac{a+x}{a-x} = Ce^{2at}.$$

Using the initial condition

$$x(0) = x_0,$$

we obtain

$$C = \frac{a+x_0}{a-x_0}.$$

After algebraic simplification,

$$x(t) = a\,\frac{x_0 + a\tanh(at)}{a + x_0\tanh(at)}.$$

Replacing $a = \sqrt{\mu}$,

$$x(t) = \sqrt{\mu}\,\frac{x_0 + \sqrt{\mu}\tanh(\sqrt{\mu}\,t)}{\sqrt{\mu} + x_0\tanh(\sqrt{\mu}\,t)}.$$

Since

$$\tanh(\sqrt{\mu}t) \to 1,$$

we obtain

$$\lim_{t\to\infty} x(t) = \sqrt{\mu}.$$

Hence every trajectory approaches

$$(\sqrt{\mu}, 0),$$

which confirms that it is the stable equilibrium.

Case II: $\mu < 0$ Let

$$\mu = -a^2, \qquad a = \sqrt{-\mu}.$$

Then

$$\dot{x} = -(a^2 + x^2).$$

Separating variables,

$$\frac{dx}{a^2+x^2} = -dt.$$

Since

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right),$$

we obtain

$$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) = -t + C.$$

Therefore,

$$\tan^{-1}\left(\frac{x}{a}\right) = -at + C_1.$$

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Taking tangent,

$$\frac{x}{a} = \tan(C_1 - at).$$

Using

$$x(0) = x_0,$$

gives

$$C_1 = \tan^{-1}\left(\frac{x_0}{a}\right).$$

Applying the tangent subtraction formula,

$$x(t) = a\,\frac{x_0 - a\tan(at)}{a + x_0\tan(at)}.$$

Finally,

$$x(t) = \sqrt{-\mu}\,\frac{x_0 - \sqrt{-\mu}\tan(\sqrt{-\mu}\,t)}{\sqrt{-\mu} + x_0\tan(\sqrt{-\mu}\,t)}.$$

This agrees with the required expression. For

$$\mu > 0,$$

the solution approaches

$$x = \sqrt{\mu},$$

confirming the linear stability analysis. For

$$\mu < 0,$$

there is no equilibrium point and the solution does not settle to a steady state, again agreeing with the theoretical prediction.

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Example Consider

$$\dot{x} = \mu + x^2, \qquad \dot{y} = -y.$$

Setting

$$\dot{x} = 0, \qquad \dot{y} = 0,$$

gives

$$E_1 = (\sqrt{-\mu}, 0), \qquad E_2 = (-\sqrt{-\mu}, 0).$$

These exist only when

$$\mu < 0.$$

The Jacobian is

$$J = \begin{bmatrix} 2x & 0 \\ 0 & -1 \end{bmatrix}.$$

At

$$(\sqrt{-\mu}, 0),$$

the eigenvalues are

$$2\sqrt{-\mu}, \qquad -1.$$

Hence Saddle. At

$$(-\sqrt{-\mu}, 0),$$

the eigenvalues are

$$-2\sqrt{-\mu}, \qquad -1.$$

Hence Stable node. The equation

$$\dot{x} = \mu + x^2$$

is solved exactly in the same manner as the previous exercise. For

$$\mu < 0,$$

the solution involves hyperbolic tangent, whereas for

$$\mu > 0,$$

it involves the tangent function. The long-time behaviour agrees with the linear stability analysis. The system

$$\dot{x} = \mu - x^2$$

creates equilibria for

$$\mu > 0,$$

whereas

$$\dot{x} = \mu + x^2$$

creates equilibria for

$$\mu < 0.$$

Thus, the bifurcation diagram is the mirror image of the previous one about the vertical axis.

Example Consider

$$\dot{x} = y + \nu_2 x + x^2, \text{ and } \dot{y} = \nu_1 + x^2.$$

Equilibrium points satisfy

$$\nu_1 + x^2 = 0,$$

giving

$$x = \pm\sqrt{-\nu_1}.$$

Also,

$$y = -(\nu_2 x + x^2).$$

A saddle-node bifurcation occurs when the two equilibrium points merge. This happens when

$$x = 0.$$

Substituting into

$$\nu_1 + x^2 = 0$$

gives

$$\nu_1 = 0.$$

Thus the saddle-node bifurcation occurs along the line

$$\nu_1 = 0$$

in the parameter plane.

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Exercise 5 Consider

$$\dot{x} = \mu x - x^2 + a, \qquad \dot{y} = -y.$$

The equilibrium points satisfy

$$\mu x - x^2 + a = 0.$$

Rearranging,

$$x^2 - \mu x - a = 0.$$

Using the quadratic formula,

$$x = \frac{\mu \pm \sqrt{\mu^2 + 4a}}{2}.$$

Real equilibria exist only if

$$\mu^2 + 4a \geq 0.$$

Therefore,

$$\mu^2 \geq -4a.$$

Since

$$a < 0,$$

we obtain

$$|\mu| \geq 2\sqrt{-a}.$$

At

$$|\mu| = 2\sqrt{-a},$$

the discriminant becomes zero, both equilibrium points coincide, and a saddle-node bifurcation occurs. Hence

$$\mu = \pm 2\sqrt{-a}$$

are the bifurcation values. For

$$|\mu| < 2\sqrt{-a},$$

the discriminant is negative. Therefore, no real equilibrium points exist.

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3.2 Pitchfork Bifurcation

A **pitchfork bifurcation** is a local bifurcation in which the stability of an equilibrium point changes at a critical parameter value, leading to the creation of two additional equilibrium points. The name *pitchfork* comes from the characteristic shape of the bifurcation diagram, which resembles the prongs of a pitchfork. Pitchfork bifurcations commonly occur in systems possessing **reflection symmetry**, i.e., systems that remain unchanged under the transformation

$$f(-x) \to -f(x).$$

There are two types of pitchfork bifurcations: Supercritical Pitchfork Bifurcation and Subcritical Pitchfork Bifurcation.

Supercritical Pitchfork Bifurcation Consider the nonlinear dynamical system

$$\dot{x} = \mu x - x^3,$$
$$\dot{y} = -y,$$

where μ is the bifurcation parameter. The equilibrium points are

$$E_0 = (0,0), \qquad E_1 = (-\sqrt{\mu}, 0), \qquad E_2 = (\sqrt{\mu}, 0).$$

The number of equilibrium points depends on the value of μ.

- $\mu < 0$ Only the origin exists because $\pm\sqrt{\mu}$ is imaginary.
- $\mu = 0$ Only one equilibrium point exists, $(0,0)$.
- $\mu > 0$ Three real equilibrium points exist, $(0,0), \qquad (\pm\sqrt{\mu}, 0)$.

The Jacobian matrix is

$$J = \begin{bmatrix} \mu - 3x^2 & 0 \\ 0 & -1 \end{bmatrix}.$$

Hence, the eigenvalues are

$$\lambda_1 = \mu - 3x^2, \qquad \lambda_2 = -1.$$

Case 1: $\mu < 0$ The only equilibrium is $(0,0)$. Its eigenvalues are

$$\lambda_1 = \mu < 0, \qquad \lambda_2 = -1.$$

Hence, $(0,0)$ is a stable node.

Case 2: $\mu = 0$ The system reduces to

$$\dot{x} = -x^3, \qquad \dot{y} = -y.$$

The exact solution of the x-equation is

$$x(t) = \frac{x_0}{\sqrt{1 + 2x_0^2 t}}.$$

Since

$$x(t) \to 0,$$

the origin remains stable.

Case 3: $\mu > 0$ The origin has eigenvalues

$$\lambda_1 = \mu > 0, \qquad \lambda_2 = -1.$$

Therefore, $(0,0)$ becomes unstable (saddle). At

$$x = \pm\sqrt{\mu},$$

the eigenvalues become

$$\lambda_1 = \mu - 3\mu = -2\mu,$$

and

$$\lambda_2 = -1.$$

Hence, $(\pm\sqrt{\mu}, 0)$ are stable nodes.

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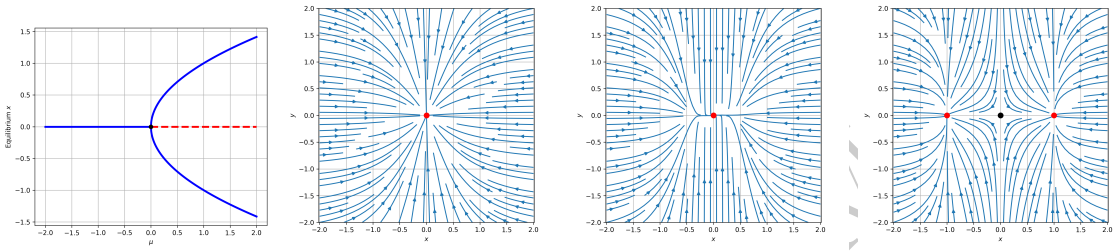
Supercritical Pitchfork Bifurcation Diagram The qualitative behaviour is

$$\begin{aligned} \mu < 0 &\implies \text{One stable equilibrium} \\ \mu = 0 &\implies \text{Critical point or bifurcation point} \\ \mu > 0 &\implies \text{One unstable} + \text{Two stable equilibria.} \end{aligned}$$

Thus, the original stable equilibrium loses its stability at

$$\mu = 0$$

and two new stable equilibrium points emerge. This is called a Supercritical Pitchfork Bifurcation.



Exact Solution For

$$\mu \neq 0,$$

the exact solution is

$$x(t) = \left[\frac{1}{\mu} + \left(\frac{1}{x_0^2} - \frac{1}{\mu}\right) e^{-2\mu t}\right]^{-1/2},$$

while

$$y(t) = y_0 e^{-t}.$$

When

$$\mu = 0,$$

the solution reduces to

$$x(t) = \frac{x_0}{\sqrt{1 + 2x_0^2 t}}, \qquad y(t) = y_0 e^{-t}.$$

The exact solution confirms the predictions obtained from linear stability analysis.

Subcritical Pitchfork Bifurcation Consider the modified system

$$\begin{aligned} \dot{x} &= \mu x + x^3, \\ \dot{y} &= -y. \end{aligned}$$

The equilibrium condition is

$$\mu x + x^3 = 0.$$

Hence,

$$x(\mu + x^2) = 0.$$

Therefore,

$$x = 0, \qquad x = \pm\sqrt{-\mu}.$$

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For

$$\mu > 0,$$

only the origin exists and it is unstable. For

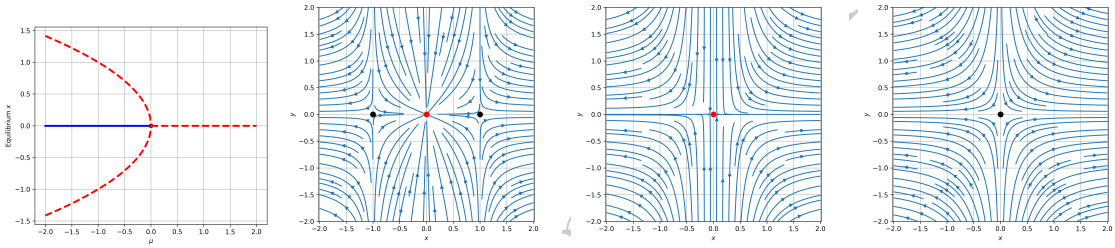
$$\mu < 0,$$

three equilibrium points exist. The origin is stable, while the two nonzero equilibrium points are unstable.

Subcritical Pitchfork Bifurcation Diagram The qualitative behaviour is

$$\begin{aligned} \mu > 0 &\implies \text{One unstable equilibrium.} \\ \mu = 0 &\implies \text{Critical point or bifurcation point.} \\ \mu < 0 &\implies \text{One stable} + \text{Two unstable equilibria.} \end{aligned}$$

Thus, at the critical parameter value, the stable equilibrium coexists with two unstable branches. This is called a Subcritical Pitchfork Bifurcation.



Supercritical	Subcritical
Occurs in $\dot{x} = \mu x - x^3$	Occurs in $\dot{x} = \mu x + x^3$
Stable equilibrium loses stability	Unstable equilibrium becomes stable
Two stable branches appear	Two unstable branches appear
Stable after bifurcation	Unstable after bifurcation

Pitchfork bifurcations occur in systems possessing reflection symmetry. Both systems satisfy

$$f(-x) = -f(x),$$

which means that the governing equations remain unchanged under the transformation

$$x \to -x.$$

This symmetry is responsible for the simultaneous appearance of the two equilibrium points

$$x = \pm\sqrt{\mu} \quad \text{or} \quad x = \pm\sqrt{-\mu}.$$

3.3 Transcritical Bifurcation

A **transcritical bifurcation** is a local bifurcation in which two equilibrium points exist for all values of the control parameter and exchange their stability as the parameter passes through a critical value. Unlike the saddle-node bifurcation, no equilibrium points are created or destroyed. Instead, the two equilibrium points intersect and interchange their stable and unstable nature. For this reason, the transcritical bifurcation is also known as the **exchange of stability bifurcation**. Consider the nonlinear dynamical system

$$\begin{aligned}\dot{x} &= -\mu x + x^2,\\ \dot{y} &= -y,\end{aligned}$$

where x, y are the state variables, and μ is the bifurcation parameter. Therefore, the equilibrium points are

$$E_0 = (0,0), \text{ and } E_1 = (\mu, 0).$$

Unlike the saddle-node bifurcation, both equilibrium points exist for every value of μ. The Jacobian matrix is

$$J = \begin{bmatrix} -\mu + 2x & 0 \\ 0 & -1 \end{bmatrix}.$$

Therefore, the eigenvalues are

$$\lambda_1 = -\mu + 2x, \qquad \lambda_2 = -1.$$

Substituting $x = 0$, gives $\lambda_1 = -\mu, \qquad \lambda_2 = -1$. Hence,

- If $\mu < 0$, $\lambda_1 > 0$, so the equilibrium point is Unstable (Saddle).
- If $\mu > 0$, both eigenvalues are negative. Hence, Stable Node.

For $E_1 = (\mu, 0)$ Substituting $x = \mu$, in Jacobian gives

$$\lambda_1 = -\mu + 2\mu = \mu, \text{ and } \lambda_2 = -1.$$

Therefore,

- If $\mu < 0$, both eigenvalues are negative. Hence, Stable Node.
- If $\mu > 0$, one eigenvalue becomes positive. Hence, Unstable (Saddle).

Case 1: $\mu < 0$ Two equilibrium points exist. $E_0 = (0,0)$ is unstable, while $E_1 = (\mu, 0)$ is stable.

Case 2: $\mu = 0$ Both equilibrium points merge at $(0,0)$. This is the bifurcation point where the exchange of stability occurs.

Case 3: $\mu > 0$ Again two equilibrium points exist. However, their stability has been interchanged. Now $E_0 = (0,0)$ becomes stable, while $E_1 = (\mu, 0)$ becomes unstable.

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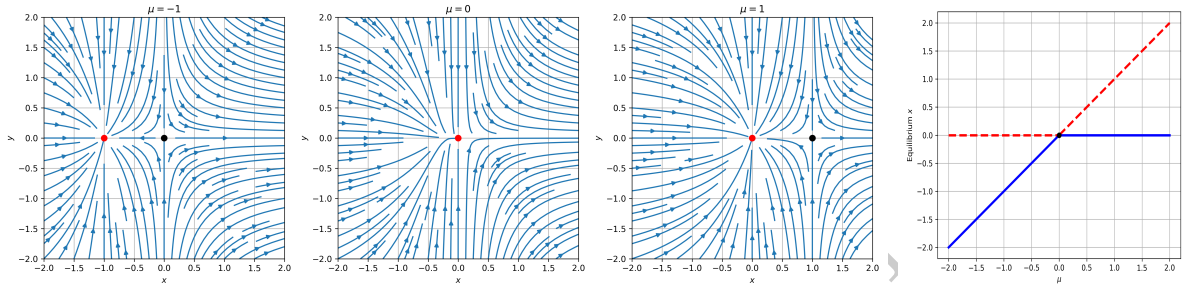
Bifurcation Diagram The equilibrium solutions are

$$x = 0, \quad \text{and} \quad x = \mu.$$

These two straight-line branches intersect at

$$(\mu, x) = (0, 0).$$

At this point, the stable branch becomes unstable, while the unstable branch becomes stable. This exchange of stability is the characteristic feature of a transcritical bifurcation.



3.4 Hopf Bifurcation

A **Hopf bifurcation** is a local bifurcation in which a stable equilibrium point loses its stability and a **periodic solution (limit cycle)** is created as a control parameter crosses a critical value.

Unlike the saddle-node, pitchfork and transcritical bifurcations, which involve only equilibrium points, the Hopf bifurcation gives rise to a new oscillatory motion. Hence, it is one of the most important bifurcations in nonlinear dynamical systems. Consider a nonlinear dynamical system

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, \mu),$$

where $\mathbf{x}$ is the state vector, and μ is the control (bifurcation) parameter. Suppose the system has an equilibrium point

$$\mathbf{x} = \mathbf{x}^*.$$

As the parameter μ varies, the stability of this equilibrium point changes. In a Hopf bifurcation,

- the equilibrium point is initially stable,
- a pair of complex conjugate eigenvalues approaches the imaginary axis,
- at a critical value $\mu = \mu_0$, the real parts of the eigenvalues become zero,
- beyond this point the equilibrium becomes unstable,
- simultaneously a periodic orbit (limit cycle) is created.

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Example: A classical example of a Hopf bifurcation is provided by the Van der Pol oscillator

$$\dot{x} = y,$$
$$\dot{y} = b(1 - x^2)y - x,$$

where b is the bifurcation parameter. The equilibrium points

$$(0, 0).$$

The Jacobian matrix is

$$J(x, y) = \begin{bmatrix} 0 & 1 \\ -2bxy - 1 & b(1 - x^2) \end{bmatrix}.$$

Evaluating at the equilibrium point,

$$J(0, 0) = \begin{bmatrix} 0 & 1 \\ -1 & b \end{bmatrix}.$$

The characteristic equation is

$$\lambda^2 - b\lambda + 1 = 0.$$

Hence,

$$\lambda_{1,2} = \frac{b \pm \sqrt{b^2 - 4}}{2}.$$

The nature of the equilibrium depends upon the value of the parameter b.

Case 1: $-2 < b < 0$ Thus for

$$-2 < b < 0, \implies b^2 < 4,$$

so the eigenvalues are

$$\lambda_{1,2} = \frac{b}{2} \pm i\frac{\sqrt{4 - b^2}}{2}.$$

The real part is

$$\mathrm{Re}(\lambda) = \frac{b}{2} < 0.$$

Therefore, all nearby trajectories spiral toward the origin. Hence, $(0, 0)$ is a stable focus. The oscillations gradually decay because the effective damping is positive near the equilibrium. Thus the system becomes

$$\dot{x} = y, \qquad \dot{y} = -x.$$

Eliminating y,

$$\ddot{x} + x = 0,$$

which is the simple harmonic oscillator. The eigenvalues become

$$\lambda_{1,2} = \pm i.$$

Their real parts are zero. Hence, trajectories neither approach nor move away from the origin, every nearby trajectory is a closed periodic orbit, and the origin is a center. The parameter value

$$b = 0$$

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marks the point where the real part of the eigenvalues changes sign. Therefore,

$$b = 0$$

is the Hopf bifurcation point. Again $b^2 < 4$, so

$$\lambda_{1,2} = \frac{b}{2} \pm i\frac{\sqrt{4-b^2}}{2}.$$

Now

$$\mathrm{Re}(\lambda) = \frac{b}{2} > 0.$$

Hence, trajectories initially spiral away from the origin. Linear analysis therefore predicts that the equilibrium is unstable. However, the nonlinear term $-bx^2y$ becomes increasingly important as the amplitude grows. It provides stronger damping for large oscillations and prevents the trajectories from diverging to infinity.Consequently, trajectories settle onto a closed periodic orbit known as a **stable limit cycle**. Thus, $(0, 0)$ becomes an unstable focus, and a stable periodic oscillation is created around it.

As the parameter b increases through the critical value $b = 0$, the equilibrium point changes from a stable focus to an unstable focus while a stable limit cycle emerges. This qualitative change in the dynamics is called a Supercritical Hopf Bifurcation.

Definition A Hopf bifurcation occurs when

1. an equilibrium point possesses a pair of complex conjugate eigenvalues,

   $$\lambda_{1,2} = \alpha(\mu) \pm i\beta(\mu),$$

   where $\beta(\mu) > 0$,

2. the real part satisfies

   $$\alpha(\mu_0) = 0,$$

   at the critical parameter value $\mu = \mu_0$,

3. the real part changes sign as the parameter crosses μ_0,

4. a periodic solution (limit cycle) is created or destroyed.

For a Hopf bifurcation,

$$\lambda_{1,2} = \alpha \pm i\beta, \qquad \beta \neq 0.$$

The stability depends on the sign of α.

Condition	Behaviour
$\alpha < 0$	Stable focus
$\alpha = 0$	Center (critical point)
$\alpha > 0$	Unstable focus

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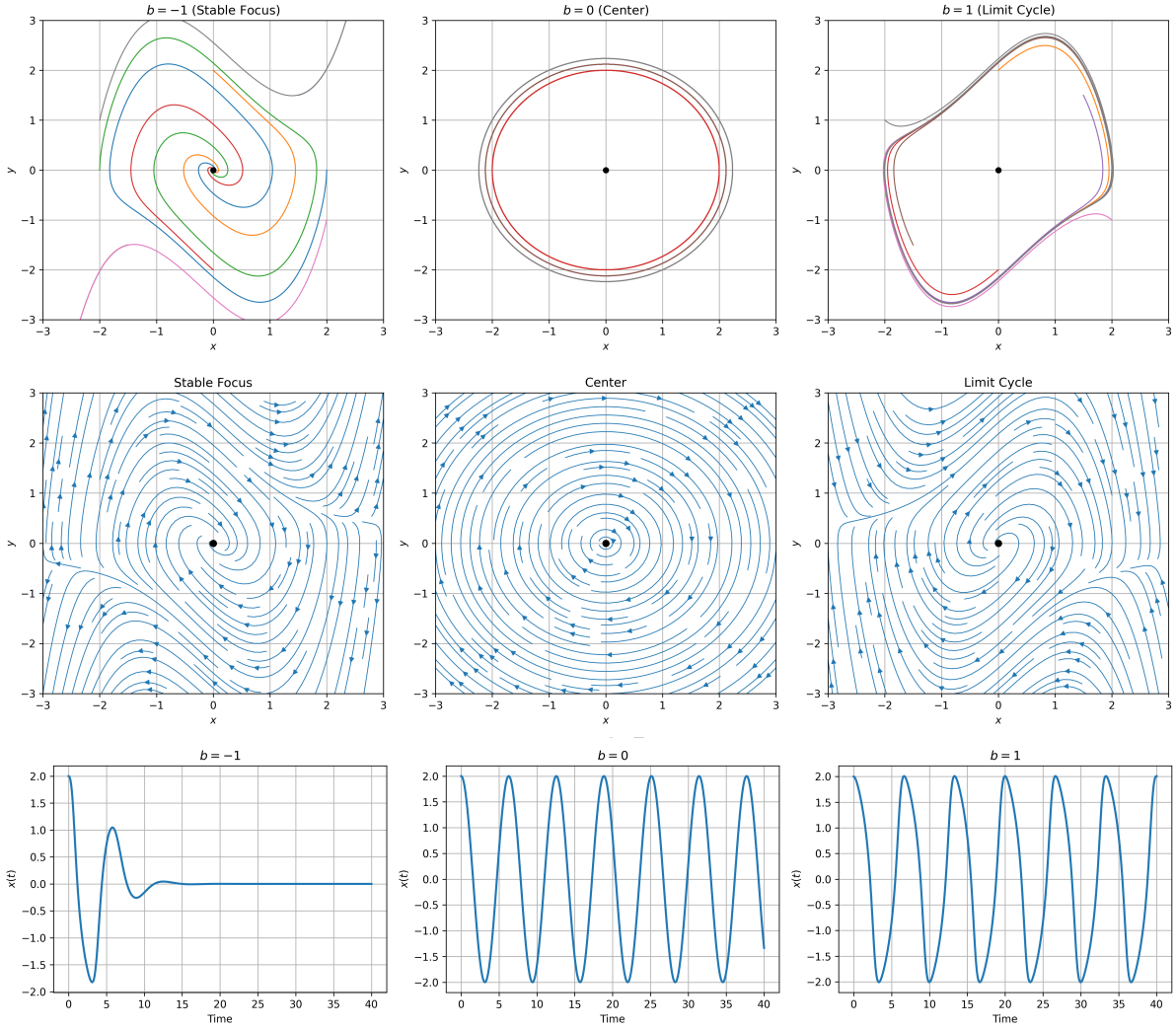
The crossing of the imaginary axis by the complex eigenvalues produces the Hopf bifurcation.



3.5 Discrete Dynamical Systems: Logistic Map - Equilibrium Points and Stability

Unlike continuous dynamical systems, where the state evolves continuously with time according to differential equations, a **discrete dynamical system** evolves only at discrete instants of time. Such systems are described by **difference equations**, **recurrence relations**, or **iterated maps**. A general one-dimensional discrete dynamical system is written as

$$x_{n+1} = f(x_n),$$

where x_n denotes the state at the discrete instant n, $f(x)$ is a nonlinear function, and $n = 0, 1, 2, \ldots$. Starting from an initial value x_0, successive iterations generate the orbit

$$x_0,\ x_1 = f(x_0),\ x_2 = f(x_1),\ x_3 = f(x_2), \ldots$$

The long-term behaviour of this sequence determines the dynamics of the system.

Logistic Map One of the most important nonlinear maps is the **logistic map**,

$$x_{n+1} = a x_n (1 - x_n),$$

where

$$0 \leq x_n \leq 1, \qquad 0 \leq a \leq 4.$$

Here x_n represents the normalized population at generation n, and a is the growth parameter. The function

$$f(x) = ax(1-x)$$

is a parabola having maximum value

$$f\left(\frac{1}{2}\right) = \frac{a}{4}.$$

The parameter interval

$$0 \leq a \leq 4$$

ensures that every iterate remains inside the interval $[0, 1]$.

Equilibrium (Fixed) Points A fixed point (equilibrium point) satisfies

$$x_{n+1} = x_n = x^* \quad \text{means the dynamics becomes zero}$$

Substituting into the logistic map,

$$x^* = ax^*(1-x^*).$$

Rearranging,

$$x^*\left(a - 1 - ax^*\right) = 0.$$

Hence,

$$x_1^* = 0, \text{ and } x_2^* = \frac{a-1}{a}.$$

Therefore the logistic map possesses two equilibrium points. $x_1^* = 0$ exists for every value of a. The second equilibrium point $x_2^* = \frac{a-1}{a}$ lies inside the interval $[0, 1]$ only when $a \geq 1$.

Linear Stability Analysis Suppose the equilibrium point is slightly perturbed,

$$x_n = x^* + \delta_n,$$

where

$$|\delta_n| \ll 1.$$

Substituting,

$$x_{n+1} = f(x^* + \delta_n).$$

Expanding by Taylor series,

$$f(x^* + \delta_n) = f(x^*) + f'(x^*)\delta_n + O(\delta_n^2).$$

Since

$$f(x^*) = x^*,$$

we obtain

$$x_{n+1} = x^* + f'(x^*)\delta_n.$$

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Hence

$$\delta_{n+1} = f'(x^*)\delta_n.$$

Repeating,

$$\delta_n = [f'(x^*)]^n \delta_0.$$

For stability,

$$\delta_n \to 0 \qquad (n \to \infty).$$

Therefore,

$$|f'(x^*)| < 1.$$

This is the stability criterion for one-dimensional discrete maps.

For the Logistic Map For $f(x) = ax(1-x)$, the derivative is

$$f'(x) = a(1-2x).$$

For $x_1^* = 0$,

$$f'(0) = a.$$

The stability condition becomes $|a| < 1$. Since $a \geq 0$, we obtain $0 \leq a < 1$. Thus, $x^* = 0$ is stable for $0 \leq a < 1$. For $a > 1$, $|f'(0)| > 1$, and therefore $x^* = 0$ becomes unstable. For $x_2^* = \frac{a-1}{a}$, the derivative becomes

$$f'(x_2^*) = a\left(1 - 2\frac{a-1}{a}\right) = 2 - a.$$

Hence

$$|2-a| < 1.$$

This gives

$$1 < a < 3.$$

Therefore, $x^* = \frac{a-1}{a}$ is stable for $1 < a < 3$. For $a > 3$, the equilibrium becomes unstable. At $a = 1$, both equilibrium points coincide,

$$0 = \frac{a-1}{a}.$$

For $a < 1$, the origin is stable. For $a > 1$, the origin becomes unstable, while $\frac{a-1}{a}$ becomes stable. Thus the two equilibrium points exchange stability. Hence, $a = 1$ is a **transcritical bifurcation point**.

Critical Cases For the one-dimensional discrete dynamical system

$$x_{n+1} = f(x_n),$$

let x^* be an equilibrium point satisfying

$$f(x^*) = x^*.$$

The usual linear stability criterion states that x^* is locally asymptotically stable if

$$|f'(x^*)| < 1$$

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and unstable if

$$|f'(x^*)| > 1.$$

However, when

$$|f'(x^*)| = 1,$$

the linearization is inconclusive. In such critical cases, higher-order terms in the Taylor expansion must be considered. Let

$$x_n = x^* + \delta_n, \qquad |\delta_n| \ll 1,$$

where δ_n represents a small perturbation from the equilibrium. Expanding $f(x_n)$ about x^*, we obtain

$$f(x^* + \delta_n) = f(x^*) + f'(x^*)\delta_n + \frac{f''(x^*)}{2}\delta_n^2 + \frac{f'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

Since $f(x^*) = x^*$,

$$x_{n+1} = x^* + f'(x^*)\delta_n + \frac{f''(x^*)}{2}\delta_n^2 + \frac{f'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

Therefore,

$$\delta_{n+1} = f'(x^*)\delta_n + \frac{f''(x^*)}{2}\delta_n^2 + \frac{f'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

Case 1: $f'(x^*) = 1$

When

$$f'(x^*) = 1,$$

the perturbation equation becomes

$$\delta_{n+1} = \delta_n + \frac{f''(x^*)}{2}\delta_n^2 + \frac{f'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

The linearized equation is simply

$$\delta_{n+1} = \delta_n,$$

which gives no information about whether the perturbation approaches or moves away from zero. We therefore examine the first nonzero nonlinear term.

Case 1(a): $f''(x^*) < 0$

Suppose

$$f''(x^*) < 0.$$

Writing

$$\frac{f''(x^*)}{2} = -C, \qquad C > 0,$$

we obtain

$$\delta_{n+1} = \delta_n - C\delta_n^2 + O(\delta_n^3).$$

For sufficiently small δ_n, the quadratic term dominates the higher-order terms. If

$$\delta_n > 0,$$

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then

$$\delta_{n+1} < \delta_n.$$

Thus, positive perturbations decrease toward zero. Hence, trajectories approach x^* from the right. On the other hand, if

$$\delta_n < 0,$$

then $\delta_n^2 > 0$, and therefore

$$\delta_{n+1} = \delta_n - C\delta_n^2 + O(\delta_n^3) < \delta_n.$$

Since $\delta_n < 0$, this means that δ_{n+1} becomes more negative. Thus, the perturbation moves away from zero, and trajectories move away from x^* on the left. Therefore,

$$f'(x^*) = 1, \qquad f''(x^*) < 0$$

implies that x^* is attracting from the right and repelling from the left. Such an equilibrium is called *semistable*.

Case 1(b): $f''(x^*) > 0$

Now suppose

$$f''(x^*) > 0.$$

Writing

$$\frac{f''(x^*)}{2} = C, \qquad C > 0,$$

the perturbation equation becomes

$$\delta_{n+1} = \delta_n + C\delta_n^2 + O(\delta_n^3).$$

If

$$\delta_n > 0,$$

then

$$\delta_{n+1} > \delta_n,$$

so the perturbation increases and moves away from zero. Hence, the equilibrium repels trajectories from the right. If

$$\delta_n < 0,$$

then

$$\delta_{n+1} > \delta_n.$$

Since $\delta_n < 0$, increasing δ_n toward zero means that the perturbation moves toward the equilibrium. Hence, the equilibrium attracts trajectories from the left. Therefore,

$$f'(x^*) = 1, \qquad f''(x^*) > 0$$

implies that x^* is attracting from the left and repelling from the right. Thus, the equilibrium is again semistable.

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Case 1(c): $f''(x^*) = 0$

If

$$f''(x^*) = 0,$$

the quadratic term vanishes and the perturbation equation becomes

$$\delta_{n+1} = \delta_n + \frac{f'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

Now the cubic term determines the local behavior. If

$$f'''(x^*) < 0,$$

then

$$\delta_{n+1} = \delta_n - C\delta_n^3 + O(\delta_n^4), \qquad C > 0.$$

For $\delta_n > 0$,

$$\delta_{n+1} < \delta_n,$$

while for $\delta_n < 0$,

$$\delta_{n+1} > \delta_n.$$

Thus, perturbations from both sides move toward zero, and x^* is locally asymptotically stable. If

$$f'''(x^*) > 0,$$

then

$$\delta_{n+1} = \delta_n + C\delta_n^3 + O(\delta_n^4), \qquad C > 0.$$

For $\delta_n > 0$,

$$\delta_{n+1} > \delta_n,$$

and for $\delta_n < 0$,

$$\delta_{n+1} < \delta_n.$$

Thus, perturbations move away from zero on both sides, and x^* is unstable. If

$$f'''(x^*) = 0,$$

then the cubic term also vanishes, and the first nonzero higher-order term must be examined.

Case 2: $f'(x^*) = -1$

When

$$f'(x^*) = -1,$$

the perturbation equation is

$$\delta_{n+1} = -\delta_n + \frac{f''(x^*)}{2}\delta_n^2 + \frac{f'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

The linear term gives

$$\delta_{n+1} = -\delta_n.$$

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Thus, the perturbation changes sign at every iteration. Consequently, it is more convenient to study the dynamics over two iterations. Define the second iterate

$$g(x) = f(f(x)).$$

Since

$$f(x^*) = x^*,$$

we have

$$g(x^*) = f(f(x^*)) = f(x^*) = x^*.$$

Therefore, x^* is also a fixed point of g. Using the chain rule,

$$g'(x) = f'(f(x))f'(x).$$

Evaluating at $x = x^*$,

$$g'(x^*) = f'(f(x^*))f'(x^*) = [f'(x^*)]^2 = (-1)^2 = 1.$$

Thus,

$$g'(x^*) = 1.$$

We now calculate the second derivative. Differentiating $g'(x)$,

$$g''(x) = f''(f(x))[f'(x)]^2 + f'(f(x))f''(x).$$

At $x = x^*$,

$$g''(x^*) = f''(x^*)[f'(x^*)]^2 + f'(x^*)f''(x^*).$$

Since

$$f'(x^*) = -1,$$

we obtain

$$g''(x^*) = f''(x^*) - f''(x^*) = 0.$$

Hence,

$$g''(x^*) = 0.$$

Therefore, both the linear and quadratic terms of the second iterate are inconclusive, and we must calculate the third derivative. Differentiating $g''(x)$, we obtain

$$g'''(x) = f'''(f(x))[f'(x)]^3 + 3f''(f(x))f'(x)f''(x) + f'(f(x))f'''(x).$$

Evaluating at $x = x^*$, using $f(x^*) = x^*$ and $f'(x^*) = -1$, gives

$$\begin{aligned} g'''(x^*) &= f'''(x^*)[-1]^3 + 3f''(x^*)[-1]f''(x^*) + [-1]f'''(x^*) \\ &= -f'''(x^*) - 3[f''(x^*)]^2 - f'''(x^*) \\ &= -2f'''(x^*) - 3[f''(x^*)]^2. \end{aligned}$$

Thus,

$$g'''(x^*) = -2f'''(x^*) - 3[f''(x^*)]^2.$$

Since

$$g'(x^*) = 1, \qquad g''(x^*) = 0,$$

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the Taylor expansion of g about x^* is

$$g(x^* + \delta_n) = x^* + \delta_n + \frac{g'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

Because

$$x_{n+2} = g(x_n),$$

we obtain

$$\delta_{n+2} = \delta_n + \frac{g'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

Case 2(a): $g'''(x^*) < 0$

Suppose

$$g'''(x^*) < 0.$$

Define

$$C = -\frac{g'''(x^*)}{6} > 0.$$

Then

$$\delta_{n+2} = \delta_n - C\delta_n^3 + O(\delta_n^4).$$

If

$$\delta_n > 0,$$

then $\delta_n^3 > 0$, so

$$\delta_{n+2} < \delta_n.$$

Thus, the perturbation decreases toward zero. If

$$\delta_n < 0,$$

then $\delta_n^3 < 0$, and hence

$$\delta_{n+2} > \delta_n.$$

Since $\delta_n < 0$, this means that the perturbation moves toward zero. Therefore, the perturbation approaches zero from both sides when observed every two iterations. Hence, x^* is locally asymptotically stable for the original map. Consequently,

$$g'''(x^*) < 0$$

implies local asymptotic stability. Since

$$g'''(x^*) = -2f'''(x^*) - 3[f''(x^*)]^2,$$

the stability condition can also be written as

$$-2f'''(x^*) - 3[f''(x^*)]^2 < 0.$$

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Case 2(b): $g'''(x^*) > 0$

Suppose

$$g'''(x^*) > 0.$$

Define

$$C = \frac{g'''(x^*)}{6} > 0.$$

Then

$$\delta_{n+2} = \delta_n + C\delta_n^3 + O(\delta_n^4).$$

If

$$\delta_n > 0,$$

then

$$\delta_{n+2} > \delta_n,$$

so the perturbation moves away from zero. If

$$\delta_n < 0,$$

then

$$\delta_{n+2} < \delta_n,$$

so the perturbation again moves away from zero. Hence,

$$g'''(x^*) > 0$$

implies that x^* is unstable. Equivalently,

$$-2f'''(x^*) - 3[f''(x^*)]^2 > 0$$

implies instability.

Case 2(c): $g'''(x^*) = 0$

If

$$g'''(x^*) = 0,$$

then the cubic term also vanishes. In this situation, the fourth-order or higher-order terms in the Taylor expansion of the second iterate must be examined. Therefore, no conclusion can be obtained from the first three derivatives alone.

All the Cases For

$$x_{n+1} = f(x_n),$$

the linear stability criterion is inconclusive when

$$|f'(x^*)| = 1.$$

If

$$f'(x^*) = 1,$$

then the quadratic term determines the behavior whenever $f''(x^*) \neq 0$:

$$f''(x^*) < 0 \quad \Longrightarrow \quad \text{attracting from the right and repelling from the left,}$$

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and

$$f''(x^*) > 0 \quad \Longrightarrow \quad \text{attracting from the left and repelling from the right.}$$

Thus, the equilibrium is semistable in either case. If

$$f''(x^*) = 0,$$

then the cubic term must be considered. In particular,

$$f'''(x^*) < 0 \quad \Longrightarrow \quad \text{locally asymptotically stable,}$$

while

$$f'''(x^*) > 0 \quad \Longrightarrow \quad \text{unstable.}$$

If

$$f'(x^*) = -1,$$

we consider the second iterate

$$g = f \circ f.$$

At the equilibrium,

$$g'(x^*) = 1, \qquad g''(x^*) = 0,$$

and

$$g'''(x^*) = -2f'''(x^*) - 3[f''(x^*)]^2.$$

Therefore,

$$-2f'''(x^*) - 3[f''(x^*)]^2 < 0 \quad \Longrightarrow \quad \text{locally asymptotically stable,}$$

whereas

$$-2f'''(x^*) - 3[f''(x^*)]^2 > 0 \quad \Longrightarrow \quad \text{unstable.}$$

If

$$-2f'''(x^*) - 3[f''(x^*)]^2 = 0,$$

higher-order terms must be considered.

Application to the Logistic Map Consider the logistic map

$$x_{n+1} = f(x_n) = ax_n(1 - x_n).$$

Expanding $f(x)$,

$$f(x) = ax - ax^2.$$

Therefore,

$$f'(x) = a(1 - 2x),$$
$$f''(x) = -2a,$$

and

$$f'''(x) = 0.$$

The equilibrium points satisfy

$$f(x^*) = x^*.$$

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Hence,

$$ax^*(1-x^*) = x^*.$$

Rearranging,

$$ax^*(1-x^*) - x^* = 0,$$

and factoring x^*,

$$x^*[a(1-x^*) - 1] = 0.$$

Therefore, the two equilibrium points are

$$x_1^* = 0$$

and

$$a(1-x_2^*) - 1 = 0.$$

Thus,

$$1 - x_2^* = \frac{1}{a},$$

which gives

$$x_2^* = 1 - \frac{1}{a}.$$

Hence,

$$x_1^* = 0, \qquad x_2^* = 1 - \frac{1}{a}.$$

Critical Case $a = 1$ At $a = 1$, the two equilibrium branches coincide:

$$x_1^* = 0, \qquad x_2^* = 1 - \frac{1}{1} = 0.$$

Thus,

$$x^* = 0.$$

For $a = 1$,

$$f'(x) = 1 - 2x.$$

Therefore,

$$f'(0) = 1.$$

The linear stability criterion is consequently inconclusive. We now calculate the second derivative:

$$f''(x) = -2,$$

so

$$f''(0) = -2 < 0.$$

Thus, this point satisfies

$$f'(x^*) = 1, \qquad f''(x^*) < 0.$$

From the critical-case analysis, the equilibrium is attracting from the right and repelling from the left. This can also be seen directly from the perturbation equation. Since

$$x_n = x^* + \delta_n = \delta_n,$$

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we have

$$\delta_{n+1} = \delta_n + \frac{f''(0)}{2}\delta_n^2 + O(\delta_n^3).$$

Since

$$\frac{f''(0)}{2} = -1,$$

we obtain

$$\delta_{n+1} = \delta_n - \delta_n^2 + O(\delta_n^3).$$

For

$$\delta_n > 0,$$

the quadratic term is negative, and hence

$$\delta_{n+1} < \delta_n.$$

Thus, positive perturbations move toward zero. For

$$\delta_n < 0,$$

we have

$$\delta_{n+1} < \delta_n < 0,$$

so the perturbation becomes more negative and moves away from zero. Therefore,

$$x^* = 0$$

is semistable at $a = 1$. The two equilibrium branches

$$x_1^* = 0$$

and

$$x_2^* = 1 - \frac{1}{a}$$

intersect at

$$a = 1.$$

Moreover, the stability of the two branches is exchanged as a passes through 1. Therefore,

$$a = 1$$

is the transcritical bifurcation parameter of the logistic map.

Critical Case $a = 3$ For $a > 1$, the nonzero equilibrium is

$$x^* = 1 - \frac{1}{a}.$$

At

$$a = 3,$$

we obtain

$$x^* = 1 - \frac{1}{3} = \frac{2}{3}.$$

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The derivative of the logistic map is

$$f'(x) = a(1 - 2x).$$

Therefore,

$$f'\left(\frac{2}{3}\right) = 3\left(1 - \frac{4}{3}\right) = 3\left(-\frac{1}{3}\right) = -1.$$

Thus,

$$f'(x^*) = -1,$$

and the linear stability criterion is inconclusive. We therefore use the second-iterate analysis. For the logistic map,

$$f''(x) = -2a,$$

so at $a = 3$,

$$f''\left(\frac{2}{3}\right) = -6.$$

Also,

$$f'''(x) = 0.$$

For the second iterate $g = f \circ f$,

$$g'''(x^*) = -2f'''(x^*) - 3[f''(x^*)]^2.$$

Substituting the values at $a = 3$,

$$g'''(x^*) = -2(0) - 3(-6)^2.$$

Hence,

$$g'''(x^*) = -3(36) = -108 < 0.$$

Therefore, the second iterate is locally asymptotically stable, and consequently the fixed point of the original logistic map is locally asymptotically stable at $a = 3$. More explicitly, the perturbation after two iterations satisfies

$$\delta_{n+2} = \delta_n + \frac{g'''(x^*)}{6}\delta_n^3 + O(\delta_n^4).$$

Since

$$\frac{g'''(x^*)}{6} = \frac{-108}{6} = -18,$$

we obtain

$$\delta_{n+2} = \delta_n - 18\delta_n^3 + O(\delta_n^4).$$

For $\delta_n > 0$,

$$\delta_{n+2} < \delta_n,$$

while for $\delta_n < 0$,

$$\delta_{n+2} > \delta_n.$$

Thus, the perturbation approaches zero from both sides when observed every two iterations. Hence,

$$x^* = \frac{2}{3}$$

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is locally asymptotically stable at $a = 3$. The multiplier at the fixed point is

$$f'(x^*) = -1.$$

As a increases through 3, the fixed point loses stability and a stable period-two orbit is created. Therefore,

$$a = 3$$

is the first period-doubling bifurcation parameter of the logistic map. For $a > 3$, sufficiently close to 3, the fixed point

$$x^* = 1 - \frac{1}{a}$$

is unstable, while the newly created period-two orbit is stable. Sof, for the one-dimensional discrete dynamical system

$$x_{n+1} = f(x_n),$$

the usual linear criterion

$$|f'(x^*)| < 1$$

does not determine stability when

$$|f'(x^*)| = 1.$$

When

$$f'(x^*) = 1,$$

if

$$f''(x^*) < 0,$$

the equilibrium is attracting from the right and repelling from the left, whereas if

$$f''(x^*) > 0,$$

the equilibrium is attracting from the left and repelling from the right. Hence, the equilibrium is semistable. If

$$f'(x^*) = 1, \qquad f''(x^*) = 0,$$

the cubic and, if necessary, higher-order terms must be examined. When

$$f'(x^*) = -1,$$

we consider the second iterate

$$g = f \circ f.$$

Then

$$g'(x^*) = 1, \qquad g''(x^*) = 0,$$

and

$$g'''(x^*) = -2f'''(x^*) - 3[f''(x^*)]^2.$$

Consequently,

$$g'''(x^*) < 0 \quad \Longrightarrow \quad x^* \text{ is locally asymptotically stable,}$$

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and

$$g'''(x^*) > 0 \quad \Longrightarrow \quad x^* \text{ is unstable.}$$

For the logistic map

$$x_{n+1} = ax_n(1 - x_n),$$

the critical parameter

$$a = 1$$

corresponds to a transcritical bifurcation, while

$$a = 3$$

corresponds to the first period-doubling bifurcation.

3.6 Periodic Solutions (Cycles) and Period-Doubling Phenomenon

Periodic Solutions (Cycles) Besides equilibrium (fixed-point) solutions, discrete dynamical systems admit more general solutions called **periodic solutions** or **cycles**. A fixed point satisfies

$$x_{n+1} = x_n = x^*,$$

and is therefore called a **period-1 cycle**. More generally,

$$x_{n+N} = x_n$$

defines a **period-N cycle**, where the trajectory repeats after every N iterations.

Examples:

$$\begin{aligned} \text{Period-2:} \quad & x_{n+2} = x_n, \\ \text{Period-3:} \quad & x_{n+3} = x_n, \\ \text{Period-}N: \quad & x_{n+N} = x_n. \end{aligned}$$

Thus, the orbit successively visits N distinct points before repeating.

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Period-2 Cycle of the Logistic Map For the logistic map

$$x_{n+1} = ax_n(1 - x_n),$$

let the two points of the period-2 cycle be $x_1^*, \qquad x_2^*$. They satisfy

$$f(x_1^*) = x_2^*, \qquad f(x_2^*) = x_1^*.$$

Equivalently,

$$f(f(x)) = x,$$

excluding the fixed-point solutions. Hence,

$$x_2^* = ax_1^*(1 - x_1^*), \quad \text{and} \quad x_1^* = ax_2^*(1 - x_2^*).$$

Subtracting the two equations,

$$x_2^* - x_1^* = a(x_2^* - x_1^*) - a(x_2^{*2} - x_1^{*2}),$$

which gives

$$x_1^* + x_2^* = \frac{a+1}{a}.$$

Multiplying the first equation by x_2^* and the second by x_1^*,

$$x_1^* x_2^* = \frac{x_1^* + x_2^*}{a},$$

therefore,

$$x_1^* x_2^* = \frac{a+1}{a^2}.$$

Using the sum and product of roots,

$$a^2x^2 - a(a+1)x + (a+1) = 0.$$

Hence,

$$x_{1,2}^* = \frac{(a+1) \pm \sqrt{(a+1)(a-3)}}{2a}.$$

Therefore, a real period-2 solution exists only for $a \geq 3$.

Stability of a Period-2 Cycle Suppose

$$x_n = x_1^* + \delta_n,$$

where

$$|\delta_n| \ll 1.$$

Using Taylor expansion,

$$\delta_{n+1} = f'(x_1^*)\delta_n.$$

Similarly,

$$\delta_{n+2} = f'(x_2^*)\delta_{n+1}.$$

Hence,

$$\delta_{n+2} = f'(x_1^*)f'(x_2^*)\delta_n.$$

For asymptotic stability,

$$|\delta_{n+2}| < |\delta_n|,$$

which gives

$$|f'(x_1^*)f'(x_2^*)| < 1.$$

This is the stability criterion for every period-2 orbit.

Application to the Logistic Map For

$$f'(x) = a(1-2x),$$

substituting the period-2 points gives

$$|4 + 2a - a^2| < 1.$$

Solving,

$$3 < a < 1 + \sqrt{6}.$$

Since

$$1 + \sqrt{6} \approx 3.44949,$$

the stable period-2 cycle exists in

$$3 < a < 3.44949.$$

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3.7 Period-Doubling Phenomenon

As the parameter a increases, the stability of successive periodic solutions is lost, producing new solutions with twice the period. The sequence is $(1, 2, 4, 8, 16, 32, \cdots)$. This process is known as the Period-Doubling Bifurcation. Since the period doubles after each bifurcation, it is also called the Flip Bifurcation.

Stable Orbit	Parameter Range	Bifurcation Value
Period-1	$0 < a < 3$	$a_1 = 3$
Period-2	$3 < a < 3.44949$	$a_2 = 3.44949$
Period-4	$3.44949 < a < 3.54411$	$a_3 = 3.54411$
Period-8	$3.54411 < a < 3.56445$	$a_4 = 3.56445$
Period-16	$3.56445 < a < 3.56881$	$a_5 = 3.56881$

Each stable periodic orbit loses stability and gives birth to a new orbit with twice the period.

General Stability Criterion For a period-N orbit

$$x_1^*, x_2^*, \ldots, x_N^*,$$

linearization yields

$$\delta_{n+N} = \left(\prod_{i=1}^{N} f'(x_i^*) \right) \delta_n.$$

Hence,

$$\left| \prod_{i=1}^{N} f'(x_i^*) \right| < 1$$

is the stability condition for a period-N cycle. For the logistic map

$$x_{n+1} = a x_n (1 - x_n),$$

let a_n denote the parameter value at which the n-th period-doubling bifurcation occurs. Thus,

$$a_1 = 3, \quad \text{Period-1} \to \text{Period-2}, \quad a_2 \approx 3.44949, \quad \text{Period-2} \to \text{Period-4},$$
$$a_3 \approx 3.54409, \quad \text{Period-4} \to \text{Period-8}, \quad a_4 \approx 3.56441, \quad \text{Period-8} \to \text{Period-16}.$$

The successive bifurcation points become increasingly close

$$a_2 - a_1 > a_3 - a_2 > a_4 - a_3 > \cdots$$

The ratio of two successive intervals approaches a universal constant:

$$\delta = \lim_{n \to \infty} \frac{a_n - a_{n-1}}{a_{n+1} - a_n} = 4.6692016 \cdots$$

This number is called the **Feigenbaum constant**. The bifurcation points themselves accumulate at

$$a_\infty \approx 3.56995.$$

Thus,

$$a_n \longrightarrow a_\infty \qquad \text{as } n \to \infty.$$

The Feigenbaum constant is universal for a large class of nonlinear maps with a quadratic maximum.

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Behaviour Along the Route to Chaos Now for the logistic map as the parameter a increases, the stable orbit repeatedly loses stability through period-doubling bifurcations: $(1, 2, 4, 8, 16, 32, \cdots)$. The successive bifurcation points become closer and closer and accumulate at

$$a_\infty \approx 3.56995.$$

Beyond this accumulation point, chaotic dynamics appear. Thus, the sequence of repeated period doublings provides an important route from regular periodic behaviour to chaos, called the Feigenbaum route to chaos.

Sensitivity to Initial Conditions In a regime with an attracting periodic orbit, two trajectories starting from different initial conditions may initially be different. However, after the transient stage, both trajectories approach the same attracting periodic orbit. Thus, the long-term behaviour is predictable despite small differences in the initial conditions. In a chaotic regime, this behaviour changes dramatically. Consider two nearby initial conditions

$$x_0^{(1)} = x_0, \qquad x_0^{(2)} = x_0 + \varepsilon, \qquad |\varepsilon| \ll 1.$$

Their initial separation is

$$|\delta_0| = |x_0^{(2)} - x_0^{(1)}| = |\varepsilon|.$$

In a chaotic system, the separation of nearby trajectories can grow approximately as

$$|\delta_n| \approx |\delta_0| e^{\lambda n},$$

where $\lambda > 0$ is the Lyapunov exponent. Hence, even an extremely small initial difference can eventually produce substantially different trajectories. This phenomenon is called Sensitive dependence on initial conditions. It is one of the characteristic signatures of chaos.

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3.8 Onset of Chaos: Sensitive Dependence on Initial Conditions and Lyapunov Exponent

As explained the previous sections, here after the period-doubling cascade, the logistic map enters the chaotic regime for

$$a > a_c \approx 3.57.$$

In this region, the iterates

$$\{x_n\} = \{x_0, x_1, x_2, \ldots\}$$

do not converge to a fixed point or a periodic orbit. Instead, the trajectory remains bounded but becomes aperiodic. Such motion is called chaotic motion.

Sensitive Dependence on Initial Conditions The most important characteristic of chaos is the sensitive dependence on initial conditions. Two trajectories starting from nearly identical initial states eventually separate exponentially with time. Let the initial conditions be

$$x_0, \qquad x_0' = x_0 + \delta_0,$$

where

$$|\delta_0| \ll 1.$$

After one iteration,

$$\delta_1 = x_1' - x_1 = f(x_0 + \delta_0) - f(x_0).$$

Using the Taylor series expansion,

$$f(x_0 + \delta_0) = f(x_0) + f'(x_0)\delta_0 + \mathcal{O}(\delta_0^2),$$

and neglecting higher-order terms,

$$\delta_1 = f'(x_0)\delta_0.$$

Similarly,

$$\delta_2 = f'(x_1)\delta_1 = f'(x_1)f'(x_0)\delta_0.$$

Proceeding inductively,

$$\delta_n = \delta_0 \prod_{i=0}^{n-1} f'(x_i).$$

Thus, the separation between nearby trajectories depends on the product of the derivatives along the orbit.

3.9 Lyapunov Exponent

To quantify the average rate of separation of nearby trajectories, the Lyapunov exponent is defined as

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \ln \left| \frac{\delta_n}{\delta_0} \right|$$

Using the previous result,

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|.$$

Hence,

$$\delta_n = \delta_0 e^{\lambda n}.$$

Therefore,

$$\lambda \begin{cases} < 0, & \text{Nearby trajectories converge (stable motion),} \\ = 0, & \text{Neutral stability,} \\ > 0, & \text{Nearby trajectories diverge exponentially (chaos).} \end{cases}$$

Lyapunov Exponent for the Logistic Map given by

$$x_{n+1} = a x_n (1 - x_n),$$

the derivative is

$$f'(x) = a(1 - 2x).$$

Hence,

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |a(1 - 2x_i)|.$$

The Lyapunov exponent characterizes the dynamics as follows:

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$\lambda < 0$	Stable fixed point or periodic orbit
$\lambda = 0$	Bifurcation point (critical state)
$\lambda > 0$	Chaotic motion

For the logistic map,

$$\lambda < 0 \quad (0 < a < a_c),$$

while

$$\lambda > 0 \quad (a > a_c \approx 3.57),$$

for most parameter values.

Butterfly Effect The sensitive dependence on initial conditions was first observed by E. N. Lorenz while studying atmospheric dynamics. A minute perturbation in the initial state grows exponentially with time, making long-term prediction impossible. This phenomenon is known as the Butterfly Effect. It is commonly summarized as:

> *A butterfly flapping its wings today may ultimately influence the formation of a tornado weeks later.*

Thus, although chaotic systems are completely deterministic, accurate long-term prediction becomes practically impossible because tiny errors in the initial conditions grow exponentially.

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3.10 Bifurcation Diagram

A **bifurcation diagram** is a graphical representation of the long-term (asymptotic) behaviour of a dynamical system as a control parameter is varied. For the logistic map

$$x_{n+1} = ax_n(1 - x_n), \qquad 0 \le x_n \le 1,\ 0 \le a \le 4,$$

the bifurcation diagram shows all attracting values of x_n corresponding to each value of the parameter a.

3.11 Strange Attractor in the Hénon Map

The logistic map is a one-dimensional nonlinear discrete dynamical system. Although it exhibits complicated behaviour, it is noninvertible because different points can have the same image. To study more general chaotic behaviour, particularly chaotic behaviour in invertible dynamical systems, it is useful to consider higher-dimensional maps. One of the most important two-dimensional nonlinear maps is the Hénon map, introduced by Michel Hénon in 1976. The Hénon map provides a simple mathematical model in which nonlinear stretching, folding, dissipation, period-doubling bifurcations, and strange attractors can be studied explicitly.

The Hénon Map It is defined by

$$x_{n+1} = 1 - ax_n^2 + y_n,$$
$$y_{n+1} = bx_n,$$

where

$$a > 0, \qquad b > 0.$$

We introduce the state vector

$$X_n = \begin{pmatrix} x_n \\ y_n \end{pmatrix}.$$

Then the Hénon map can be written compactly as

$$X_{n+1} = F(X_n),$$

where

$$F(x, y) = \begin{pmatrix} 1 - ax^2 + y \\ bx \end{pmatrix}.$$

Thus,

$$X_{n+1} = \begin{pmatrix} 1 - ax_n^2 + y_n \\ bx_n \end{pmatrix}.$$

Since the state contains two variables,

$$X_n = (x_n, y_n)^T,$$

the Hénon map is a two-dimensional discrete dynamical system. Unlike a one-dimensional map, the evolution of x_n alone does not determine the complete state because

$$x_{n+1} = 1 - ax_n^2 + y_n$$

depends on both x_n and y_n.

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Invertibility of the Hénon Map The second equation is

$$y_{n+1} = bx_n.$$

If $b \neq 0$, we can solve this equation for x_n:

$$x_n = \frac{y_{n+1}}{b}.$$

Therefore,

$$x_n = \frac{y_{n+1}}{b}.$$

Now consider the first equation:

$$x_{n+1} = 1 - ax_n^2 + y_n.$$

Solving for y_n,

$$y_n = x_{n+1} - 1 + ax_n^2.$$

Substituting

$$x_n = \frac{y_{n+1}}{b},$$

we obtain

$$y_n = x_{n+1} - 1 + a\left(\frac{y_{n+1}}{b}\right)^2.$$

Hence the previous state can be reconstructed from the current state:

$$x_n = \frac{y_{n+1}}{b},$$
$$y_n = x_{n+1} - 1 + \frac{a}{b^2} y_{n+1}^2.$$

Therefore, for

$$b \neq 0,$$

the Hénon map is invertible. This is one of the important differences between the Hénon map and the one-dimensional logistic map.

At $b = 0$ When

$$b = 0,$$

the second equation becomes

$$y_{n+1} = 0.$$

Therefore, after one iteration,

$$y_1 = 0,$$

and consequently

$$y_n = 0, \qquad n \geq 1.$$

The first equation then becomes

$$x_{n+1} = 1 - a x_n^2.$$

Hence, for $b = 0$, the Hénon map reduces after one iteration to the one-dimensional quadratic map

$$x_{n+1} = 1 - a x_n^2.$$

This quadratic map is related to the logistic map by an affine change of variables and an appropriate transformation of the parameter. Thus, the Hénon map can be viewed as a two-dimensional extension of the one-dimensional quadratic dynamics.

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Conservative and Dissipative Maps Consider a general N-dimensional discrete dynamical system

$$X_{n+1} = F(X_n),$$

where

$$X_n = \begin{pmatrix} X_{1,n} \\ X_{2,n} \\ \vdots \\ X_{N,n} \end{pmatrix}.$$

The Jacobian matrix of the map is $DF(X) = \left[\frac{\partial F_i}{\partial X_j}\right]$. The determinant of the Jacobian matrix is $\det DF(X)$. The determinant measures the local change in phase-space volume under one iteration. For a sufficiently small region of phase-space with volume ΔV_n, we have approximately

$$\Delta V_{n+1} \approx |\det DF(X_n)|\, \Delta V_n.$$

Therefore, the quantity

$$|\det DF(X_n)|$$

is the local volume-expansion or volume-contraction factor. If

$$|\det DF(X)| = 1,$$

then phase-space volume is locally preserved. Such a map is called volume preserving or conservative. If

$$|\det DF(X)| < 1,$$

then phase-space volume contracts. Such a map is called dissipative. If

$$|\det DF(X)| > 1,$$

then phase-space volume expands. For the two-dimensional Hénon map, "volume" means two-dimensional area.

Now, the Jacobian matrix is defined by

$$DF(x,y) = \begin{pmatrix} \dfrac{\partial}{\partial x}(1-ax^2+y) & \dfrac{\partial}{\partial y}(1-ax^2+y) \\ \dfrac{\partial}{\partial x}(bx) & \dfrac{\partial}{\partial y}(bx) \end{pmatrix}.$$

Therefore,

$$DF(x,y) = \begin{pmatrix} -2ax & 1 \\ b & 0 \end{pmatrix}.$$

At the point (x_n, y_n),

$$M_n = DF(x_n, y_n) = \begin{pmatrix} -2ax_n & 1 \\ b & 0 \end{pmatrix}.$$

Therefore,

$$\det(M_n) = (-2ax_n)(0) - (1)(b).$$

Hence,

$$\det(M_n) = -b. \implies |\det(M_n)| = b.$$

Therefore, if

$$0 < b < 1,$$

then

$$b^k \to 0 \qquad \text{as} \qquad k \to \infty.$$

Hence areas contract exponentially. Therefore,

$$0 < b < 1 \quad \Rightarrow \quad \text{Hénon map is dissipative.}$$

Similarly,

$$\begin{aligned} b < 1 \quad &\Rightarrow \quad \text{area contracting,} \\ b = 1 \quad &\Rightarrow \quad \text{area preserving,} \\ b > 1 \quad &\Rightarrow \quad \text{area expanding.} \end{aligned}$$

The classical strange attractor of the Hénon map is studied in the dissipative regime $0 < b < 1$. Let (x^*, y^*) denote an equilibrium point. By definition,

$$x_{n+1} = x_n = x^* \quad \text{and} \quad y_{n+1} = y_n = y^*.$$

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Substituting these conditions into the Hénon map gives

$$x^* = 1 - a(x^*)^2 + y^* \quad \text{and} \quad y^* = bx^*.$$

We first use the second equation:

$$y^* = bx^*.$$

Substitute this into the first equilibrium equation we obtain

$$a(x^*)^2 + (1-b)x^* - 1 = 0.$$

Therefore,

$$x^*_{\pm} = \frac{b - 1 \pm \sqrt{(1-b)^2 + 4a}}{2a}.$$

The corresponding y-coordinates are obtained from

$$y^* = bx^*.$$

Therefore,

$$y^*_{\pm} = bx^*_{\pm}.$$

Thus the two equilibrium points are

$$E_{\pm} = \left(x^*_{\pm},\, bx^*_{\pm}\right).$$

Now, assume

$$a > 0, \qquad b > 0.$$

Consider

$$x^*_{+} = \frac{b - 1 + \sqrt{(1-b)^2 + 4a}}{2a}.$$

Since

$$4a > 0,$$

we have

$$\sqrt{(1-b)^2 + 4a} > |1 - b|.$$

Therefore,

$$\sqrt{(1-b)^2 + 4a} > 1 - b$$

when $b < 1$. Hence,

$$b - 1 + \sqrt{(1-b)^2 + 4a} > 0.$$

Since

$$2a > 0,$$

we obtain

$$x^*_{+} > 0.$$

Similarly,

$$x^*_{-} = \frac{b - 1 - \sqrt{(1-b)^2 + 4a}}{2a}$$

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has a negative numerator and positive denominator. Hence, $x^*_- < 0$. Thus the two equilibrium points lie on opposite sides of the y-axis. Now, to determine the stability of an equilibrium point, we study the behaviour of a small perturbation around it. Let

$$X^* = \begin{pmatrix} x^* \\ y^* \end{pmatrix}$$

be an equilibrium point. Consider a nearby state

$$X_n = X^* + \delta X_n,$$

where

$$\delta X_n = \begin{pmatrix} \delta x_n \\ \delta y_n \end{pmatrix}$$

is a small perturbation. The map is

$$X_{n+1} = F(X_n).$$

Therefore,

$$X_{n+1} = F(X^* + \delta X_n).$$

Using the first-order Taylor expansion,

$$F(X^* + \delta X_n) \approx F(X^*) + DF(X^*)\delta X_n.$$

Since X^* is an equilibrium,

$$F(X^*) = X^*.$$

Therefore,

$$X_{n+1} \approx X^* + DF(X^*)\delta X_n.$$

Subtracting X^* from both sides gives

$$X_{n+1} - X^* \approx DF(X^*)\delta X_n.$$

But

$$X_{n+1} - X^* = \delta X_{n+1}.$$

Hence,

$$\delta X_{n+1} = M^* \delta X_n,$$

where

$$M^* = DF(X^*).$$

For the Hénon map,

$$M^* = \begin{pmatrix} -2ax^* & 1 \\ b & 0 \end{pmatrix}.$$

To determine the stability of the linearized system, we calculate the eigenvalues of M^*. For a nonzero eigenvector to exist,

$$\det(M^* - \lambda I) = 0.$$

Thus,

$$M^* - \lambda I = \begin{pmatrix} -2ax^* - \lambda & 1 \\ b & -\lambda \end{pmatrix}.$$

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we obtain

$$(-2ax^* - \lambda)(-\lambda) - (1)(b) = 0.$$

Therefore,

$$\lambda^2 + 2ax^*\lambda - b = 0.$$

This is the characteristic equation of the linearized Hénon map. Therefore,

$$\lambda_\pm = -ax^* \pm \sqrt{a^2(x^*)^2 + b}.$$

Discrete-Time Stability Condition The linearized perturbation equation is

$$\delta X_{n+1} = M^* \delta X_n.$$

Applying the equation repeatedly gives

$$\delta X_n = (M^*)^n \delta X_0.$$

Suppose the eigenvalues of M^* are

$$\lambda_1, \lambda_2.$$

If

$$|\lambda_1| < 1 \quad \text{and} \quad |\lambda_2| < 1,$$

then repeated multiplication by the matrix causes perturbations to decay:

$$\delta X_n \to 0.$$

Therefore, the equilibrium is locally asymptotically stable. Now, the characteristic polynomial is

$$\lambda^2 + 2ax^*\lambda - b = 0.$$

Write this in the general form

$$\lambda^2 + p\lambda + q = 0,$$

where

$$p = 2ax^*, \qquad q = -b.$$

For a second-order discrete-time system, both roots of

$$\lambda^2 + p\lambda + q = 0$$

lie strictly inside the unit circle if and only if the stability conditions are satisfied:

$$1 + p + q > 0,$$

$$1 - p + q > 0,$$

and

$$1 - q > 0.$$

We now substitute

$$p = 2ax^*, \qquad q = -b.$$

The first condition becomes

$$1 + 2ax^* - b > 0.$$

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Therefore,

$$1 - b + 2ax^* > 0.$$

The second condition becomes

$$1 - 2ax^* - b > 0.$$

Therefore,

$$1 - b - 2ax^* > 0.$$

The third condition becomes

$$1 - (-b) > 0,$$

so

$$1 + b > 0.$$

Since

$$b > 0,$$

this condition is automatically satisfied. Thus the two relevant conditions are

$$1 - b + 2ax^* > 0$$

and

$$1 - b - 2ax^* > 0.$$

These two inequalities can be combined into

$$|2ax^*| < 1 - b.$$

Notice that this also requires

$$1 - b > 0.$$

Hence the stability analysis is naturally in the dissipative regime

$$0 < b < 1.$$

Now, consider the positive equilibrium

$$x_+^* = \frac{b - 1 + \sqrt{(1-b)^2 + 4a}}{2a}.$$

Multiply both sides by $2a$:

$$2ax_+^* = b - 1 + \sqrt{(1-b)^2 + 4a}.$$

Since

$$b - 1 = -(1 - b),$$

we obtain

$$2ax_+^* = -(1 - b) + \sqrt{(1-b)^2 + 4a}.$$

Thus,

$$2ax_+^* = \sqrt{(1-b)^2 + 4a} - (1 - b).$$

For the positive equilibrium,

$$x_+^* > 0,$$

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and hence

$$2ax_+^* > 0.$$

The stability condition

$$|2ax_+^*| < 1 - b$$

therefore becomes

$$2ax_+^* < 1 - b.$$

Substituting the expression for $2ax_+^*$,

$$\sqrt{(1-b)^2 + 4a} - (1-b) < 1 - b.$$

Add $(1-b)$ to both sides:

$$\sqrt{(1-b)^2 + 4a} < 2(1-b).$$

Because we are considering

$$0 < b < 1,$$

we have

$$1 - b > 0.$$

Therefore both sides are positive, and we can safely square the inequality:

$$(1-b)^2 + 4a < 4(1-b)^2.$$

Subtract $(1-b)^2$ from both sides:

$$4a < 4(1-b)^2 - (1-b)^2.$$

Therefore,

$$4a < 3(1-b)^2.$$

Dividing by 4 gives

$$a < \frac{3}{4}(1-b)^2.$$

Hence the positive equilibrium is locally asymptotically stable provided

$$0 < b < 1, \qquad a < \frac{3(1-b)^2}{4}.$$

Now consider the second equilibrium

$$x_-^* = \frac{b - 1 - \sqrt{(1-b)^2 + 4a}}{2a}.$$

Multiplying by $2a$ gives

$$2ax_-^* = b - 1 - \sqrt{(1-b)^2 + 4a}.$$

Using

$$b - 1 = -(1-b),$$

we obtain

$$2ax_-^* = -(1-b) - \sqrt{(1-b)^2 + 4a}.$$

Therefore,

$$2ax_-^* < 0.$$

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Its absolute value is

$$|2ax_-^*| = (1-b) + \sqrt{(1-b)^2 + 4a}.$$

Since

$$a > 0,$$

we have

$$\sqrt{(1-b)^2 + 4a} > 1 - b.$$

Consequently,

$$|2ax_-^*| > (1-b) + (1-b).$$

Thus,

$$|2ax_-^*| > 1 - b.$$

But stability requires

$$|2ax_-^*| < 1 - b.$$

The two inequalities are incompatible. Therefore,

$$E_- \text{ is always unstable.}$$

Thus, the boundary between stability and instability of the positive equilibrium occurs when

$$a = \frac{3(1-b)^2}{4}.$$

At this boundary,

$$4a = 3(1-b)^2.$$

Therefore,

$$(1-b)^2 + 4a = (1-b)^2 + 3(1-b)^2.$$

Hence,

$$(1-b)^2 + 4a = 4(1-b)^2.$$

Taking the positive square root,

$$\sqrt{(1-b)^2 + 4a} = 2(1-b),$$

because

$$0 < b < 1.$$

For the positive equilibrium,

$$2ax_+^* = \sqrt{(1-b)^2 + 4a} - (1-b).$$

Therefore, at the boundary,

$$2ax_+^* = 2(1-b) - (1-b).$$

Thus,

$$2ax_+^* = 1 - b.$$

The characteristic equation is

$$\lambda^2 + 2ax_+^*\lambda - b = 0.$$

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Substituting

$$2ax_+^* = 1 - b,$$

we obtain

$$\lambda^2 + (1-b)\lambda - b = 0.$$

This polynomial factors as

$$(\lambda - b)(\lambda + 1) = 0.$$

Therefore,

$$\lambda_1 = b, \qquad \lambda_2 = -1.$$

Since

$$|\lambda_2| = 1,$$

the equilibrium is exactly at the boundary of stability. The eigenvalue

$$\lambda = -1$$

is characteristic of a flip or period-doubling bifurcation. Thus, as the parameter a increases through the stability boundary, the fixed point can lose stability through a period-doubling bifurcation.

Classical Hénon Parameters The classical Hénon map is commonly studied using the parameters $a = 1.4, \quad b = 0.3$. First consider the determinant $|\det M| = b$. Therefore, $|\det M| = 0.3 < 1$. Hence the map is strongly dissipative. Hénon map is dissipative for $a = 1.4,\ b = 0.3$. Now consider the fixed-point stability condition:

$$a < \frac{3(1-b)^2}{4}.$$

Substituting $b = 0.3$, we obtain

$$a < \frac{1.47}{4}.$$

Hence,

$$a < 0.3675.$$

However, the classical value is $a = 1.4$. Consequently, the positive equilibrium is unstable for the classical Hénon parameters. At the same time, $|\det M| = 0.3 < 1$, so the system is dissipative. Thus the classical Hénon map simultaneously exhibits instability, nonlinearity, and dissipation.

3.12 Period Doubling Phenomenon

Fix $b = 0.3$. The map is dissipative. As the parameter a increases, the following sequence of bifurcations occurs. Period-1 to Period-2 to Period-4 then Period-8 and continues. The important bifurcation values are approximately $a_1 \approx 0.3675$, where the fixed point loses stability. Next, $a_2 \approx 0.905$, where the period-2 orbit loses stability. Successive bifurcations accumulate at $a_c \approx 1.06$, which marks the onset of chaos. Thus the Hénon map follows the same Feigenbaum period-doubling route to chaos observed in the logistic map.

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Chaotic Attractor For parameter values such as

$$a = 1.4, \qquad b = 0.3,$$

the trajectories no longer settle onto a periodic orbit. Instead, (x_n, y_n) wander forever inside a bounded complicated geometric object called the chaotic attractor. Its important properties are bounded, non-periodic, deterministic, sensitive to initial conditions, and attracts nearby trajectories.

3.13 Self-Similar Structure

The Hénon attractor possesses an geometric property called **self-similarity**. If a small region of the attractor is enlarged,

- several thin layers become visible,
- enlarging one layer again reveals similar folded layers,
- repeated magnification reproduces essentially the same structure.

Thus structure inside structure inside structure appears indefinitely. Only the scale changes; the geometric pattern remains similar. This property is called Self-similarity. The self-similar geometry results from two repeated dynamical operations.

1. Stretching Nearby trajectories separate exponentially because of nonlinear instability. Stretching increases the distance between neighbouring trajectories.

2. Folding Since trajectories remain bounded, they cannot diverge forever. The stretched trajectories fold back into the bounded region. Hence, every iteration performs Stretching, Folding, Stretching, Folding. Repeated stretching and folding create finer and finer layers, producing the fractal geometry of the attractor. This process is analogous to repeatedly stretching and folding a sheet of paper or shuffling a deck of cards.

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3.14 Route to Chaos

The Hénon map reaches chaos through the same sequence observed in many dissipative nonlinear systems.Starts with Stable Fixed Point then to Period-2 then to Period-4 then to Period-8 and continues to Chaos. This mechanism is called the Period-Doubling Route to Chaos or Feigenbaum Scenario. Other important routes to chaos include

- Quasiperiodic Route (Ruelle-Takens-Newhouse),
- Intermittency Route (Pomeau-Manneville),
- Period-adding and Farey-sequence routes.

4. Poincaré Cross Section and Orbits in Conservative Systems

Unlike dissipative systems, conservative (Hamiltonian) systems preserve phase-space volume and do not possess attractors. Their dynamics is therefore studied using Poincaré Surface of Section (SOS), which converts continuous motion into a discrete sequence of points while preserving the essential dynamics.

4.1 Poincaré Cross Section (Surface of Section)

Consider an autonomous dynamical system

$$\dot{X} = F(X), \qquad X \in \mathbb{R}^m,$$

and let

$$X(t) = \phi_t(X_0)$$

denote the trajectory starting from X_0, where ϕ_t is the flow generated by the system. Since a continuous trajectory may be difficult to visualize and analyse directly, Poincaré introduced a method of replacing the continuous flow by a discrete sequence of intersections with a lower-dimensional surface.

Definition Let

$$\Sigma = \{X \in \mathbb{R}^m : h(X) = 0\}$$

be an $(m-1)$-dimensional surface. For Σ to be a valid local cross section, the trajectory must cross it transversally. Thus,

$$\nabla h(X) \cdot F(X) \neq 0, \qquad X \in \Sigma.$$

Indeed, along a trajectory,

$$\frac{d}{dt} h(X(t)) = \nabla h(X(t)) \cdot \dot{X}(t) = \nabla h(X(t)) \cdot F(X(t)).$$

Hence,

$$\nabla h(X) \cdot F(X) \neq 0$$

implies that $h(X(t))$ changes as the trajectory passes through Σ, so the trajectory crosses the surface rather than remaining tangent to it.

For a three-dimensional system,

$$X = (x_1, x_2, x_3),$$

a simple choice is

$$\Sigma = \{(x_1, x_2, x_3) : x_3 = c\}.$$

Here,

$$h(X) = x_3 - c, \qquad \nabla h = (0, 0, 1),$$

and therefore the transversality condition becomes

$$F_3(X) \neq 0.$$

To avoid counting crossings in both directions, one often imposes an orientation condition such as

$$F_3(X) > 0.$$

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Construction of the Poincaré Map Suppose a trajectory starting from $P_n \in \Sigma$ returns to Σ after a return time

$$\tau(P_n) > 0.$$

The next intersection is therefore

$$P_{n+1} = \phi_{\tau(P_n)}(P_n).$$

Define the Poincaré map

$$T : \Sigma \to \Sigma$$

by

$$T(P) = \phi_{\tau(P)}(P).$$

Hence,

$$P_{n+1} = T(P_n),$$

and repeated intersections satisfy

$$P_n = T^n(P_0).$$

Thus, the continuous-time system

$$\dot{X} = F(X)$$

is converted into the discrete dynamical system

$$P_{n+1} = T(P_n).$$

The return time is generally state dependent: $\tau = \tau(P)$, so successive intersections need not occur at equally spaced times.

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Dimension Reduction The original flow evolves in an m-dimensional phase space, $X(t) \in \mathbb{R}^m$, whereas the Poincaré map acts only on the $(m-1)$-dimensional section $T : \Sigma \to \Sigma$. Therefore, m-dimensional continuous flow $\longrightarrow$ $(m-1)$-dimensional discrete map. For example, a three-dimensional flow (x_1, x_2, x_3) can be studied through a two-dimensional map on Σ. This dimensional reduction is one of the main advantages of the Poincaré section.

Periodic Orbits and Fixed Points of the Poincaré Map An important theoretical property is that a periodic orbit of the continuous system corresponds to a fixed point, or more generally a periodic point, of the Poincaré map. Suppose $X^*(t)$ is a periodic orbit with period T_p:

$$X^*(t + T_p) = X^*(t).$$

Let the orbit intersect the section at

$$P^* \in \Sigma.$$

After one complete period, the trajectory returns to the same point:

$$\phi_{T_p}(P^*) = P^*.$$

By the definition of the Poincaré map,

$$T(P^*) = P^*.$$

Hence, Periodic orbit of the flow*is*fixed point of the Poincaré map. More generally, if the orbit returns to its original intersection after k crossings, then

$$T^k(P^*) = P^*,$$

so it corresponds to a period-k point of the Poincaré map. This result is important because the stability theory developed for discrete maps can now be applied to periodic orbits of continuous dynamical systems. We demonstrate this in the next part.

Stability Through the Poincaré Map Let P^* be a fixed point of the Poincaré map:

$$T(P^*) = P^*.$$

Introduce a small perturbation

$$P_n = P^* + \delta P_n.$$

Using the first-order Taylor expansion,

$$T(P^* + \delta P_n) = T(P^*) + DT(P^*)\delta P_n + \mathcal{O}(\|\delta P_n\|^2).$$

Since

$$T(P^*) = P^*,$$

we obtain, to first order,

$$\delta P_{n+1} = DT(P^*)\delta P_n.$$

After n returns,

$$\delta P_n \approx [DT(P^*)]^n \delta P_0.$$

Therefore, the periodic orbit is locally asymptotically stable if all eigenvalues μ_i of $DT(P^*)$ satisfy

$$|\mu_i| < 1.$$

These eigenvalues are the characteristic multipliers of the return map. Thus, the stability of a periodic orbit of a continuous-time system can be investigated by studying the eigenvalues of a discrete Poincaré map.

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Remark The geometry of the intersection points provides useful information about the long-term dynamics as seen in the previous chapters. That is Periodic orbit has finite number of points, Quasiperiodic motion has closed curve or collection of curves, and Chaotic motion has irregular fractal-like set of points. Thus, a complicated continuous trajectory can often be classified more easily by examining its Poincaré section.

Stroboscopic Section for Periodically Forced Systems Consider a periodically forced system

$$\dot{X} = F(X,t), \qquad F(X,t+T) = F(X,t).$$

For a harmonic forcing term

$$A\cos(\omega t),$$

the forcing repeats whenever

$$\omega(t+T) = \omega t + 2\pi.$$

Therefore,

$$\omega T = 2\pi,$$

and hence

$$T = \frac{2\pi}{\omega}.$$

Instead of detecting intersections with a geometric surface, the state is sampled once every forcing period:

$$t_n = t_0 + nT, \qquad n = 0, 1, 2, \ldots$$

and

$$P_n = X(t_0 + nT).$$

Consequently,

$$P_{n+1} = T_s(P_n),$$

where T_s denotes the stroboscopic map. Because every sample is taken at the same phase of the external forcing,

$$\omega t_n = \omega t_0 + 2\pi n,$$

we have

$$\cos(\omega t_n) = \cos(\omega t_0).$$

Thus, the explicit periodic forcing is observed at the same phase at every sample, allowing the periodically forced continuous system to be represented by a discrete map.

Geometric and Stroboscopic Sections For an autonomous system, intersections are determined geometrically:

$$h(X) = 0, \qquad \nabla h(X) \cdot F(X) > 0.$$

The corresponding return times

$$\tau_1, \tau_2, \tau_3, \ldots$$

are generally unequal. For a periodically forced system, intersections are determined temporally:

$$t_n = t_0 + nT.$$

Hence, the sampling interval is constant and equal to the forcing period.

Conservative Hamiltonian Systems Consider a Hamiltonian system with N degrees of freedom,

$$H = H(q_1, \ldots, q_N, p_1, \ldots, p_N),$$

where q_i are generalized coordinates and p_i are the corresponding generalized momenta. Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \qquad i = 1, \ldots, N.$$

If the Hamiltonian has no explicit time dependence, then

$$\frac{dH}{dt} = \sum_{i=1}^{N} \left(\frac{\partial H}{\partial q_i}\dot{q}_i + \frac{\partial H}{\partial p_i}\dot{p}_i \right).$$

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Using Hamilton's equations,

$$\frac{dH}{dt} = \sum_{i=1}^{N} \left(\frac{\partial H}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i}\frac{\partial H}{\partial q_i} \right) = 0.$$

Therefore,

$$H = \text{constant}.$$

Thus, trajectories remain confined to constant-energy surfaces in phase space.

Example: Consider

$$H(q,p) = \frac{p^2}{2m} + \frac{1}{2}kq^2.$$

Hamilton's equations are

$$\dot{q} = \frac{p}{m}, \qquad \dot{p} = -kq.$$

Since

$$\frac{dH}{dt} = 0,$$

the trajectory satisfies

$$\frac{p^2}{2m} + \frac{1}{2}kq^2 = E,$$

which is an ellipse in the (q,p)-phase plane. Hence the oscillator moves repeatedly along the same constant-energy curve.

Conservation of Phase-Space Volume The phase-space velocity is

$$V = (\dot{q}_1, \ldots, \dot{q}_N, \dot{p}_1, \ldots, \dot{p}_N).$$

Its divergence is

$$\nabla \cdot V = \sum_{i=1}^{N} \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right).$$

Using Hamilton's equations,

$$\nabla \cdot V = \sum_{i=1}^{N} \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0.$$

Hence, $\nabla \cdot V = 0$. (already detailed explanation available in the last chapter) Therefore phase-space volume is preserved under Hamiltonian evolution. This is Liouville's theorem.

Example For the harmonic oscillator,

$$\dot{q} = \frac{p}{m}, \qquad \dot{p} = -kq.$$

Therefore,

$$\frac{\partial \dot{q}}{\partial q} = 0, \qquad \frac{\partial \dot{p}}{\partial p} = 0,$$

and hence

$$\nabla \cdot V = 0.$$

A small region of initial conditions may deform as it evolves, but its phase-space area remains unchanged. Consequently, conservative Hamiltonian systems do not possess ordinary asymptotically attracting fixed points or attractors. That means contraction in all phase-space directions would contradict volume preservation.

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Integrable Motion and Invariant Tori A Hamiltonian system with N degrees of freedom is called completely integrable if it possesses N independent conserved quantities in involution. Under suitable conditions, the system can be transformed to action-angle variables

$$(J_1, \ldots, J_N, \theta_1, \ldots, \theta_N)$$

such that

$$H = H_0(J_1, \ldots, J_N).$$

Hamilton's equations become

$$\dot{J}_i = -\frac{\partial H_0}{\partial \theta_i} = 0$$

and

$$\dot{\theta}_i = \frac{\partial H_0}{\partial J_i} = \omega_i(J).$$

Therefore,

$$J_i(t) = J_i(0)$$

and

$$\theta_i(t) = \theta_i(0) + \omega_i t.$$

Since every action J_i remains constant, the trajectory remains on an invariant N-dimensional torus. For two degrees of freedom,

$$(J_1, J_2) = \text{constant},$$

and the motion takes place on a two-dimensional torus.

4.2 Orbits in Conservative Systems

In a conservative system there is no dissipation of phase-space volume. For an autonomous Hamiltonian system with N degrees of freedom,

$$H = H(q_1, \ldots, q_N, p_1, \ldots, p_N),$$

the equations of motion are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \qquad i = 1, \ldots, N.$$

For an autonomous Hamiltonian,

$$\frac{dH}{dt} = 0,$$

and therefore every trajectory remains on a constant-energy surface

$$H(q, p) = E.$$

Thus the orbit of a conservative system does not approach an attracting equilibrium or attracting periodic orbit. Instead, the trajectory continues to move on the dynamically accessible part of the constant-energy surface. For a one-degree-of-freedom mechanical Hamiltonian,

$$H(q, p) = \frac{p^2}{2m} + V(q),$$

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the constant-energy condition is

$$\frac{p^2}{2m} + V(q) = E.$$

Solving for p,

$$p = \pm\sqrt{2m[E - V(q)]}.$$

Thus the orbit in the (q, p)-phase plane is obtained directly from the energy equation. Motion is possible only when

$$E - V(q) \geq 0.$$

The points at which

$$E = V(q)$$

satisfy

$$p = 0$$

and are called turning points. Suppose $q = q_c$ is a local minimum of the potential. Then

$$V'(q_c) = 0, \qquad V''(q_c) > 0.$$

The corresponding equilibrium

$$(q, p) = (q_c, 0)$$

is a center. For energies slightly greater than $V(q_c)$, the trajectory forms a closed curve surrounding the center. Hence the motion is periodic. If $q = q_s$ is a local maximum of the potential, then

$$V'(q_s) = 0, \qquad V''(q_s) < 0.$$

The corresponding equilibrium

$$(q, p) = (q_s, 0)$$

is a saddle point. The energy level passing through the saddle,

$$E_s = V(q_s),$$

has a special orbit called a separatrix. The separatrix separates regions having qualitatively different types of motion. Thus, for a one-degree-of-freedom autonomous conservative system, the phase portrait typically contains centers, periodic orbits, saddles and separatrices. An important observation is that an autonomous Hamiltonian system with only one degree of freedom cannot exhibit chaos. Its phase space is two-dimensional, and conservation of energy confines every trajectory to a one-dimensional curve

$$H(q, p) = E.$$

The trajectory therefore has insufficient freedom to wander chaotically through a finite region of phase space. For two degrees of freedom,

$$H = H(q_1, q_2, p_1, p_2),$$

the phase space is four-dimensional. Conservation of energy reduces the motion to the three-dimensional energy surface

$$H = E.$$

A Poincaré section reduces this further to a two-dimensional map. The resulting section can contain invariant curves, resonance islands, hyperbolic points and chaotic regions. Thus, one autonomous degree of freedom$\Longrightarrow$no chaos, whereas two or more autonomous degrees of freedom$\Longrightarrow$chaos is possible. A periodically forced one-degree-of-freedom system also may exhibit chaos because the explicit time dependence introduces an additional phase variable.

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4.3 Regular and Irregular Trajectories

Trajectories in conservative systems may be broadly classified as regular or irregular. Regular motion includes periodic and quasiperiodic motion. Irregular deterministic motion is identified with chaotic motion.

Periodic Trajectories A trajectory is periodic if there exists a finite period $T_p > 0$ such that

$$X(t + T_p) = X(t)$$

for all t. In action-angle variables,

$$J_i(t) = J_i(0), \qquad \theta_i(t) = \theta_i(0) + \omega_i t.$$

For two degrees of freedom, periodic motion occurs when the frequencies are rationally related. Suppose

$$\frac{\omega_2}{\omega_1} = \frac{l}{k},$$

where k and l are relatively prime integers. Then

$$k\omega_2 = l\omega_1, \implies l\omega_1 - k\omega_2 = 0.$$

A finite time T_p exists such that

$$\omega_1 T_p = 2\pi k, \qquad \omega_2 T_p = 2\pi l.$$

Therefore,

$$T_p = \frac{2\pi k}{\omega_1} = \frac{2\pi l}{\omega_2}.$$

After this time both angular variables return to their initial values modulo 2π. Hence the trajectory closes. Thus,

$$\frac{\omega_2}{\omega_1} \in \mathbb{Q} \Longrightarrow \text{periodic motion.}$$

On a Poincaré section, a periodic orbit produces a finite number of repeating points. A period-one orbit gives one fixed point, a period-two orbit gives two repeating points, and more generally a period-k orbit gives k repeating points. Thus periodic orbit means finite repeating points on the Poincaré section.

Quasiperiodic Trajectories Suppose

$$\frac{\omega_2}{\omega_1} \notin \mathbb{Q}.$$

If the orbit were periodic, there would exist integers k and l and a finite time T_p satisfying

$$\omega_1 T_p = 2\pi k, \qquad \omega_2 T_p = 2\pi l.$$

Dividing these equations gives

$$\frac{\omega_2}{\omega_1} = \frac{l}{k},$$

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which is rational and contradicts the assumption. Therefore no finite return time exists. The trajectory never closes exactly but winds around the invariant torus. Hence,

$$\frac{\omega_2}{\omega_1} \notin \mathbb{Q} \Longrightarrow \text{quasiperiodic motion.}$$

On a Poincaré section, successive intersections of a quasiperiodic trajectory lie densely on a smooth invariant curve, that is quasiperiodic orbit means smooth invariant curve on the Poincaré section. A quasiperiodic trajectory can appear complicated when viewed as a time series, but it is still regular motion. It remains confined to an invariant torus and does not show exponential sensitivity to nearby initial conditions.

Effect of a Small Perturbation Consider a nearly integrable Hamiltonian

$$H(J,\theta) = H_0(J) + \varepsilon H_1(J,\theta), \qquad 0 < |\varepsilon| \ll 1.$$

When

$$\varepsilon = 0,$$

the actions are constant:

$$\dot{J}_i = 0,$$

and the system consists entirely of invariant tori. When

$$\varepsilon \neq 0,$$

the equations for the actions become

$$\dot{J}_i = -\varepsilon \frac{\partial H_1}{\partial \theta_i}.$$

Therefore the actions are no longer exactly constant. For sufficiently weak perturbation, many nonresonant invariant tori survive in slightly deformed form. Resonant tori, however, are affected much more strongly. A resonance occurs when

$$m_1\omega_1 + m_2\omega_2 + \cdots + m_N\omega_N = 0,$$

where the m_i are integers not all equal to zero. For two degrees of freedom,

$$m_1\omega_1 + m_2\omega_2 = 0,$$

so

$$\frac{\omega_2}{\omega_1} = -\frac{m_1}{m_2}.$$

Thus resonance corresponds to a rational frequency ratio.

Irregular or Chaotic Trajectories A chaotic trajectory is deterministic but aperiodic and sensitive to initial conditions. It is not confined to a smooth invariant curve on the Poincaré section. Let two nearby initial conditions be separated initially by

$$\|\delta X(0)\|.$$

For a chaotic trajectory, the separation generally behaves locally as

$$\|\delta X(t)\| \approx \|\delta X(0)\| e^{\lambda_{\max} t},$$

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where

$$\lambda_{\max} > 0$$

is the largest Lyapunov exponent. Thus, $\lambda_{\max} > 0$ means sensitive dependence on initial conditions. For regular periodic or quasiperiodic motion, $\lambda_{\max} = 0$ in the long-time limit. The positive Lyapunov exponent does not mean that trajectories escape to infinity. In a bounded conservative system they may remain confined to an allowed energy region while separating exponentially from nearby trajectories.

Hyperbolic Points Chaotic motion in conservative systems is closely connected with hyperbolic periodic points. Let P^* be a fixed point of a two-dimensional Poincaré map

$$P_{n+1} = T(P_n).$$

Linearizing near P^*,

$$\delta P_{n+1} = DT(P^*)\delta P_n.$$

For an area-preserving map,

$$\det DT(P^*) = 1.$$

If P^* is hyperbolic, its eigenvalues are real and satisfy

$$|\lambda_s| < 1, \qquad |\lambda_u| > 1,$$

with

$$\lambda_s \lambda_u = 1.$$

The eigenvector associated with λ_s defines the local contracting direction, while the eigenvector associated with λ_u defines the local expanding direction. The stable manifold is

$$W^s(P^*) = \{P : T^n(P) \to P^* \text{ as } n \to \infty\},$$

and the unstable manifold is

$$W^u(P^*) = \left\{P : T^{-n}(P) \to P^* \text{ as } n \to \infty\right\}.$$

Thus, W^s : approach to the hyperbolic point and W^u : departure from the hyperbolic point.

Regular and Chaotic Motion on the Poincaré Section The different kinds of motion can therefore be distinguished geometrically as follows

Periodic motion	finite repeating points,
Quasiperiodic motion	smooth invariant curve,
Resonant regular motion	island chains,
Chaotic motion	irregular scattered set of points.

The scattered points of a chaotic conservative trajectory must not be interpreted as an attractor. Phase-space volume is preserved, so neighbouring trajectories are not collectively contracting towards the chaotic set as in dissipative systems. Consequently, a nonintegrable conservative system commonly exhibits a mixed phase space.

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4.4 Canonical Perturbation Theory: Overlapping Resonances and Chaos

Consider a nearly integrable Hamiltonian

$$H(J,\theta) = H_0(J) + \varepsilon H_1(J,\theta), \qquad 0 < |\varepsilon| \ll 1,$$

where

$$J = (J_1, \ldots, J_N), \qquad \theta = (\theta_1, \ldots, \theta_N).$$

For the unperturbed system,

$$H = H_0(J),$$

Hamilton's equations are

$$\dot{J}_i = 0, \qquad \dot{\theta}_i = \omega_i(J),$$

where

$$\omega_i(J) = \frac{\partial H_0}{\partial J_i}.$$

Hence,

$$J_i(t) = J_i(0),$$

and

$$\theta_i(t) = \theta_i(0) + \omega_i t.$$

The unperturbed motion therefore takes place on invariant tori. When the perturbation is introduced,

$$\dot{J}_i = -\varepsilon \frac{\partial H_1}{\partial \theta_i},$$

and hence the actions slowly vary. Canonical perturbation theory attempts to construct new action-angle variables in which the small angle-dependent perturbation is removed, at least approximately.

Near-Identity Canonical Transformation Introduce new action-angle variables

$$(J', \theta')$$

through the type-2 generating function

$$F_2(\theta, J') = \theta \cdot J' + \varepsilon S_1(\theta, J').$$

The transformation equations are

$$J_i = \frac{\partial F_2}{\partial \theta_i}$$

and

$$\theta'_i = \frac{\partial F_2}{\partial J'_i}.$$

Therefore,

$$J_i = J'_i + \varepsilon \frac{\partial S_1}{\partial \theta_i},$$

and

$$\theta'_i = \theta_i + \varepsilon \frac{\partial S_1}{\partial J'_i}.$$

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Since ε is small, this transformation differs only slightly from the identity transformation. The transformed Hamiltonian is

$$K(J', \theta') = H(J, \theta).$$

Substituting

$$J = J' + \varepsilon \frac{\partial S_1}{\partial \theta}$$

and expanding H_0 to first order,

$$H_0(J) = H_0(J') + \varepsilon \frac{\partial H_0}{\partial J'} \cdot \frac{\partial S_1}{\partial \theta} + \mathcal{O}(\varepsilon^2).$$

Since

$$\frac{\partial H_0}{\partial J'} = \omega(J'),$$

we obtain

$$H_0(J) = H_0(J') + \varepsilon \omega(J') \cdot \frac{\partial S_1}{\partial \theta} + \mathcal{O}(\varepsilon^2).$$

Also,

$$\varepsilon H_1(J, \theta) = \varepsilon H_1(J', \theta) + \mathcal{O}(\varepsilon^2).$$

Therefore,

$$K = H_0(J') + \varepsilon \left[H_1(J', \theta) + \omega(J') \cdot \frac{\partial S_1}{\partial \theta} \right] + \mathcal{O}(\varepsilon^2).$$

We choose S_1 so that the angle-dependent part of the first-order perturbation disappears. Let

$$\langle H_1 \rangle$$

denote the average of H_1 over all angle variables. We require

$$H_1 - \langle H_1 \rangle + \omega \cdot \frac{\partial S_1}{\partial \theta} = 0.$$

Hence

$$\omega \cdot \frac{\partial S_1}{\partial \theta} = -(H_1 - \langle H_1 \rangle).$$

If this equation can be solved, then

$$K(J', \theta') = H_0(J') + \varepsilon \langle H_1 \rangle + \mathcal{O}(\varepsilon^2),$$

which is independent of the angles to first order. The new actions are therefore approximately constant.

Fourier Expansion and the Origin of Resonance Since H_1 is periodic in the angle variables, write it as a Fourier series:

$$H_1(J, \theta) = \langle H_1 \rangle + \sum_{m \neq 0} H_m(J) e^{i\, m \cdot \theta},$$

where

$$m = (m_1, \ldots, m_N) \in \mathbb{Z}^N$$

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and

$$m \cdot \theta = m_1\theta_1 + \cdots + m_N\theta_N.$$

Similarly, write

$$S_1(J,\theta) = \sum_{m\neq 0} S_m(J)e^{i\,m\cdot\theta}.$$

Now,

$$\frac{\partial S_1}{\partial \theta} = \sum_{m\neq 0} i\,m\,S_m e^{i\,m\cdot\theta}.$$

Therefore,

$$\omega \cdot \frac{\partial S_1}{\partial \theta} = \sum_{m\neq 0} i(m\cdot\omega) S_m e^{i\,m\cdot\theta}.$$

Comparing Fourier coefficients in

$$\omega \cdot \frac{\partial S_1}{\partial \theta} = -(H_1 - \langle H_1\rangle),$$

we obtain

$$i(m\cdot\omega)S_m = -H_m.$$

Hence

$$S_m = \frac{iH_m}{m\cdot\omega}.$$

This expression explains why resonances are important. If

$$m\cdot\omega$$

is not close to zero, then S_m remains small and the corresponding perturbation term can be approximately removed by the canonical transformation. However, if

$$m\cdot\omega \approx 0,$$

the denominator becomes small and S_m becomes large. The assumption of a small near-identity transformation then fails. Exactly at resonance,

$$m\cdot\omega = 0,$$

and the resonant Fourier term cannot be eliminated by ordinary canonical perturbation theory. For two degrees of freedom,

$$m_1\omega_1 + m_2\omega_2 = 0,$$

or

$$\frac{\omega_2}{\omega_1} = -\frac{m_1}{m_2}.$$

Thus the difficulty occurs precisely at rational frequency ratios.

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Example. For the one-action Hamiltonian

$$H(J,\theta) = \frac{1}{2}J^2 + \varepsilon a\cos\theta,$$

construct a first-order near-identity canonical transformation, determine the transformed Hamiltonian, and identify where the construction becomes invalid.
solution Write $H_0 = J^2/2$ and $H_1 = a\cos\theta$. The unperturbed frequency is $\omega(J) = J$, while the angle average is $\langle H_1\rangle = (a/2\pi)\displaystyle\int_0^{2\pi}\cos\theta\,d\theta = 0$. Let $x = J'$ and take $F_2(\theta, x) = \theta x + \varepsilon S_1(\theta, x)$. The homological equation becomes

$$x\frac{\partial S_1}{\partial\theta} = -a\cos\theta.$$

For $x \neq 0$ a periodic solution with zero angular mean is

$$S_1(\theta, x) = -\frac{a}{x}\sin\theta.$$

The generating equations yield

$$J = x - \varepsilon\frac{a}{x}\cos\theta, \qquad \theta' = \theta + \varepsilon\frac{a}{x^2}\sin\theta.$$

Substitute the action correction into the Hamiltonian. Since $J^2/2 = x^2/2 - \varepsilon a\cos\theta + O(\varepsilon^2)$, the term $+\varepsilon a\cos\theta$ cancels, giving

$$K(x,\theta') = \frac{1}{2}x^2 + O(\varepsilon^2).$$

The action and angle corrections contain x^{-1} and x^{-2}, respectively. Thus the transformation is not uniformly near identity near $x = 0$. This is the resonance $\omega = 0$: the formal first-order cancellation cannot be used there.

Problem A reduced model of two interacting vibration modes is

$$H = \omega_1 J_1 + \omega_2 J_2 + \varepsilon\lambda J_1 J_2[1 + \cos(\theta_1 - \theta_2)].$$

Find an exact conserved combination of actions, the generating function, the first-order normal form and the corrected frequencies. Explain the resonance condition.
Solution Set $\phi = \theta_1 - \theta_2$. Hamilton's action equations give

$$\dot{J}_1 = \varepsilon\lambda J_1 J_2\sin\phi, \qquad \dot{J}_2 = -\varepsilon\lambda J_1 J_2\sin\phi.$$

Adding them proves the exact conservation law $J_1 + J_2 = \text{constant}$. The unperturbed frequencies are (ω_1, ω_2) and the angular average is $\langle H_1\rangle = \lambda J_1 J_2$. Let $x = J_1'$, $y = J_2'$ and $D = \omega_1 - \omega_2$. With $F_2 = \theta_1 x + \theta_2 y + \varepsilon S_1$, the homological equation reads

$$\omega_1 S_{1,\theta_1} + \omega_2 S_{1,\theta_2} = -\lambda xy\cos\phi.$$

Trying $S_1 = A(x,y)\sin\phi$ gives $A(\omega_1 - \omega_2) = -\lambda xy$; hence, when $D \neq 0$,

$$S_1 = -\frac{\lambda xy}{D}\sin(\theta_1 - \theta_2).$$

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The canonical transformation follows by differentiation:

$$J_1 = x - \varepsilon\frac{\lambda xy}{D}\cos\phi, \quad J_2 = y + \varepsilon\frac{\lambda xy}{D}\cos\phi,$$
$$\theta_1' = \theta_1 - \varepsilon\frac{\lambda y}{D}\sin\phi, \quad \theta_2' = \theta_2 - \varepsilon\frac{\lambda x}{D}\sin\phi.$$

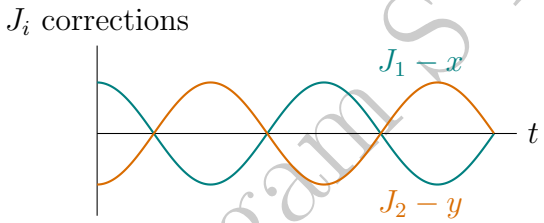
The corrections to J_1 and J_2 cancel in their sum, as required by the exact invariant. In the transformed Hamiltonian the oscillatory cosine is eliminated to first order:

$$K = \omega_1 x + \omega_2 y + \varepsilon\lambda xy + O(\varepsilon^2).$$

Therefore $\dot{x}, \dot{y} = O(\varepsilon^2)$ and the frequency corrections are

$$\Omega_1 = \omega_1 + \varepsilon\lambda y, \qquad \Omega_2 = \omega_2 + \varepsilon\lambda x.$$

The divisor D vanishes at $\omega_1 = \omega_2$. There the phase difference ceases to rotate at leading order and the coupling cannot be averaged away by this generator. For $\omega_1 = 2$, $\omega_2 = 3$, $\varepsilon = .05$, $\lambda = 1$, $x = 1$, $y = 2$, one obtains $J_1 = 1 + .1\cos\phi$, $J_2 = 2 - .1\cos\phi$, $\Omega_1 = 2.10$ and $\Omega_2 = 3.05$.



Opposite oscillatory action corrections preserve the total action.

Problem For

$$H = \frac{1}{2}J_1^2 + \frac{1}{4}J_2^2 + \varepsilon a J_1 J_2 \cos\theta_1 \cos\theta_2,$$

construct the first-order generating function, determine the transformation and locate both formal resonance lines. *solution* The unperturbed frequencies are $\omega_1 = J_1$ and $\omega_2 = J_2/2$. The trigonometric identity

$$\cos\theta_1\cos\theta_2 = \tfrac{1}{2}\cos(\theta_1 + \theta_2) + \tfrac{1}{2}\cos(\theta_1 - \theta_2)$$

shows that two Fourier modes are present. Their angular means vanish. Write $x = J_1'$, $y = J_2'$, $\phi_+ = \theta_1 + \theta_2$ and $\phi_- = \theta_1 - \theta_2$. The homological equation is

$$xS_{1,\theta_1} + \frac{y}{2}S_{1,\theta_2} = -\frac{axy}{2}(\cos\phi_+ + \cos\phi_-).$$

Solving separately for the two modes yields

$$S_1 = -\frac{axy}{2x+y}\sin\phi_+ - \frac{axy}{2x-y}\sin\phi_-.$$

Let $C_+ = axy/(2x+y)$ and $C_- = axy/(2x-y)$. The action equations from $F_2 = \theta_1 x + \theta_2 y + \varepsilon S_1$ are

$$J_1 = x - \varepsilon C_+\cos\phi_+ - \varepsilon C_-\cos\phi_-,$$
$$J_2 = y - \varepsilon C_+\cos\phi_+ + \varepsilon C_-\cos\phi_-.$$

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The coefficient derivatives are

$$C_{+,x} = \frac{ay^2}{(2x+y)^2}, \quad C_{+,y} = \frac{2ax^2}{(2x+y)^2}, \quad C_{-,x} = -\frac{ay^2}{(2x-y)^2}, \quad C_{-,y} = \frac{2ax^2}{(2x-y)^2}.$$

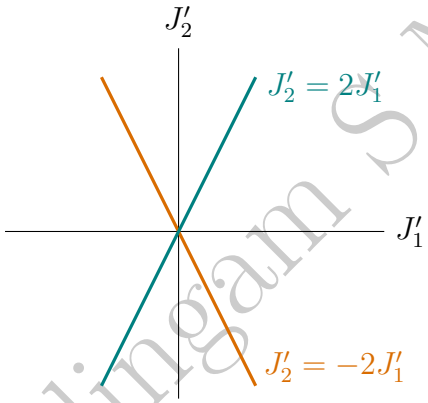
Consequently,

$$\theta_1' = \theta_1 - \varepsilon\frac{ay^2}{(2x+y)^2}\sin\phi_+ + \varepsilon\frac{ay^2}{(2x-y)^2}\sin\phi_-,$$
$$\theta_2' = \theta_2 - \varepsilon\frac{2ax^2}{(2x+y)^2}\sin\phi_+ - \varepsilon\frac{2ax^2}{(2x-y)^2}\sin\phi_-.$$

The first-order perturbation averages to zero, and hence

$$K = \tfrac{1}{2}x^2 + \tfrac{1}{4}y^2 + O(\varepsilon^2).$$

The denominators vanish on $2x+y=0$ and $2x-y=0$. In the positive-action quadrant, only the line $y=2x$ enters the interior. The other lies outside that quadrant, apart from the origin where the perturbing Fourier amplitude also vanishes.



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Problem Consider

$$H(J,\theta) = \frac{1}{2}(J-1)^2 + \varepsilon[\alpha J^2 + \beta J\cos(2\theta)].$$

Determine the first-order canonical transformation and frequency correction. Explain the breakdown near $J=1$.

solution The unperturbed frequency is $\omega(J) = J-1$. The angular mean is $\langle H_1\rangle = \alpha J^2$, so with $x = J'$ the homological equation becomes

$$(x-1)S_{1,\theta} = -\beta x\cos(2\theta).$$

For $x \neq 1$, integration gives

$$S_1 = -\frac{\beta x}{2(x-1)}\sin(2\theta).$$

Differentiation of $F_2 = \theta x + \varepsilon S_1$ gives

$$J = x - \varepsilon\frac{\beta x}{x-1}\cos(2\theta), \qquad \theta' = \theta + \varepsilon\frac{\beta}{2(x-1)^2}\sin(2\theta).$$

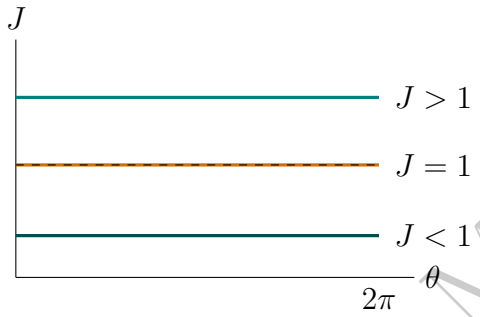
Since the angle-dependent term cancels, the normal form is

$$K = \frac{1}{2}(x-1)^2 + \varepsilon\alpha x^2 + O(\varepsilon^2).$$

Therefore $\dot{x} = O(\varepsilon^2)$ and the frequency is

$$\Omega(x) = x - 1 + 2\varepsilon\alpha x + O(\varepsilon^2).$$

At $x = 1$, the unperturbed frequency is zero. The corrections contain $1/(x-1)$ and $1/(x-1)^2$, so neither remains small near that point. For $\varepsilon = .02$, $\alpha = \beta = 1$ and $x = 2$, the action correction amplitude is .04, the angle correction amplitude is .01 and $\Omega \approx 1.08$. At $x = 1.01$ the corresponding amplitudes become 2.02 and 100: the proposed near-identity transformation has lost validity even though the original Hamiltonian is smooth.



Unperturbed action circles on the phase cylinder, the angular frequency vanishes at $J = 1$.

Problem Let

$$H = \frac{1}{2}J_1^2 + J_2^2 + \frac{3}{2}J_3^2 + \varepsilon\mu J_1 J_2 J_3 \cos(\theta_1 + \theta_2 - \theta_3).$$

Find two independent exact action invariants, the first-order canonical transformation. *solution* Define $\phi = \theta_1 + \theta_2 - \theta_3$. The unperturbed frequency vector is

$$\boldsymbol{\omega} = (J_1, 2J_2, 3J_3), \qquad \dot{\phi}\big|_{\varepsilon=0} = J_1 + 2J_2 - 3J_3.$$

Differentiating the full Hamiltonian with respect to the three angles gives

$$\dot{J}_1 = G, \qquad \dot{J}_2 = G, \qquad \dot{J}_3 = -G, \qquad G = \varepsilon\mu J_1 J_2 J_3 \sin\phi.$$

Consequently $\frac{d}{dt}(J_1 - J_2) = 0$ and $\frac{d}{dt}(J_1 + J_3) = 0$. These are independent invariants because their gradients $(1, -1, 0)$ and $(1, 0, 1)$ are linearly independent. Set $x = J_1'$, $y = J_2'$, $z = J_3'$ and $D = x + 2y - 3z$. The perturbation has zero angular average. Therefore

$$xS_{1,\theta_1} + 2yS_{1,\theta_2} + 3zS_{1,\theta_3} = -\mu xyz\cos\phi.$$

The sine ansatz gives

$$S_1 = -\frac{\mu xyz}{D}\sin\phi, \qquad D \neq 0.$$

With $C = \mu xyz/D$, the old actions satisfy

$$J_1 = x - \varepsilon C\cos\phi, \quad J_2 = y - \varepsilon C\cos\phi, \quad J_3 = z + \varepsilon C\cos\phi.$$

Their combinations obey $J_1 - J_2 = x - y$ and $J_1 + J_3 = x + z$, matching the exact invariants. Differentiating C with respect to the new actions gives

$$C_x = \frac{\mu yz(2y - 3z)}{D^2}, \quad C_y = \frac{\mu xz(x - 3z)}{D^2}, \quad C_z = \frac{\mu xy(x + 2y)}{D^2}.$$

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The new angles are therefore

$$\theta_1' = \theta_1 - \varepsilon C_x \sin\phi, \quad \theta_2' = \theta_2 - \varepsilon C_y \sin\phi, \quad \theta_3' = \theta_3 - \varepsilon C_z \sin\phi.$$

The angle-dependent term disappears at first order, yielding

$$K = \frac{1}{2}x^2 + y^2 + \frac{3}{2}z^2 + O(\varepsilon^2).$$

4.5 Periodically Driven Undamped Duffing Oscillator

The periodically driven undamped Duffing oscillator is an important example in which regular and chaotic motion can be studied in a continuous-time nonlinear system. The general equation is

$$\ddot{x} + \alpha x + \beta x^3 = f\cos(\Omega t),$$

where α is linear stiffness parameter, β is nonlinear stiffness parameter, f is forcing amplitude, and Ω is forcing frequency. There is no damping term proportional to $\dot{x}$. Hence the system is called an undamped Duffing oscillator. Introducing

$$y = \dot{x},$$

the equation becomes

$$\dot{x} = y,$$
$$\dot{y} = -\alpha x - \beta x^3 + f\cos(\Omega t).$$

The system is explicitly time dependent because of the periodic forcing.

Unforced Duffing Oscillator First consider

$$f = 0.$$

Then

$$\ddot{x} + \alpha x + \beta x^3 = 0.$$

Multiplying by $\dot{x}$,

$$\dot{x}\ddot{x} + \alpha x\dot{x} + \beta x^3\dot{x} = 0.$$

Using

$$\dot{x}\ddot{x} = \frac{d}{dt}\left(\frac{1}{2}\dot{x}^2\right),$$

$$x\dot{x} = \frac{d}{dt}\left(\frac{1}{2}x^2\right),$$

and

$$x^3\dot{x} = \frac{d}{dt}\left(\frac{1}{4}x^4\right),$$

we obtain

$$\frac{d}{dt}\left(\frac{1}{2}\dot{x}^2 + \frac{\alpha}{2}x^2 + \frac{\beta}{4}x^4\right) = 0.$$

Therefore,

$$H_0(x,y) = \frac{1}{2}y^2 + \frac{\alpha}{2}x^2 + \frac{\beta}{4}x^4 = E.$$

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The effective potential is

$$V(x) = \frac{\alpha}{2}x^2 + \frac{\beta}{4}x^4.$$

Thus the unforced Duffing oscillator is a one-degree-of-freedom autonomous Hamiltonian system and cannot exhibit chaos.

Double-Well Duffing Oscillator This one is a particularly important form obtained by taking

$$\alpha = -1, \qquad \beta = 1.$$

The forced equation becomes

$$\ddot{x} - x + x^3 = f\cos(\Omega t).$$

For the unforced case,

$$\ddot{x} - x + x^3 = 0,$$

and the Hamiltonian is

$$H_0(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

The potential is

$$V(x) = -\frac{1}{2}x^2 + \frac{1}{4}x^4.$$

The equilibrium points satisfy

$$y = 0$$

and

$$x - x^3 = 0.$$

Therefore,

$$x(1 - x^2) = 0,$$

which gives

$$x^* = 0, \qquad x^* = \pm 1.$$

Hence the three equilibria are

$$(0, 0), \qquad (1, 0), \qquad (-1, 0).$$

The first-order system is

$$\dot{x} = y,$$

$$\dot{y} = x - x^3.$$

Its Jacobian is

$$J(x, y) = \begin{bmatrix} 0 & 1 \\ 1 - 3x^2 & 0 \end{bmatrix}.$$

At

$$(0, 0),$$

we obtain

$$J(0, 0) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

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The characteristic equation is

$$\det \begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = 0,$$

so

$$\lambda^2 - 1 = 0.$$

Therefore,

$$\lambda = \pm 1.$$

Hence $(0, 0)$ is a saddle point. At

$$(1, 0)$$

or

$$(-1, 0),$$

we have

$$1 - 3x^2 = 1 - 3 = -2.$$

Thus

$$J = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}.$$

The characteristic equation is

$$\lambda^2 + 2 = 0,$$

and therefore

$$\lambda = \pm i\sqrt{2}.$$

Hence $(\pm 1, 0)$ are centers. The potential values are

$$V(0) = 0$$

and

$$V(\pm 1) = -\frac{1}{4}.$$

Therefore the saddle energy is

$$E_s = 0.$$

For

$$-\frac{1}{4} < E < 0,$$

there are two separate families of closed periodic orbits, one surrounding $x = 1$ and the other surrounding $x = -1$. At

$$E = 0,$$

the energy curve passes through the saddle and forms the separatrix. For

$$E > 0,$$

the oscillator has sufficient energy to pass from one well to the other and the trajectory forms a larger periodic orbit surrounding both wells. Thus,

$-\frac{1}{4} < E < 0$	oscillation inside one well,
$E = 0$	separatrix through the saddle,
$E > 0$	motion across both wells.

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Equation of the Separatrix At the separatrix,

$$E = 0.$$

Hence

$$\frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4 = 0.$$

Multiplying by 2,

$$y^2 - x^2 + \frac{1}{2}x^4 = 0.$$

Therefore,

$$y^2 = x^2 - \frac{1}{2}x^4.$$

Factoring,

$$y^2 = x^2\left(1 - \frac{x^2}{2}\right).$$

Thus,

$$y = \pm x\sqrt{1 - \frac{x^2}{2}}.$$

The separatrix forms two homoclinic loops attached to the saddle point $(0, 0)$. One explicit homoclinic solution is

$$x_h(t) = \sqrt{2}\,\mathrm{sech}(t - t_0).$$

Differentiating,

$$y_h(t) = \dot{x}_h(t) = -\sqrt{2}\,\mathrm{sech}(t - t_0)\tanh(t - t_0).$$

As

$$t \to \pm\infty,$$

we have

$$x_h(t) \to 0, \qquad y_h(t) \to 0.$$

Hence the orbit leaves the neighbourhood of the saddle and eventually returns to the same saddle, which confirms its homoclinic character.

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Effect of Periodic Forcing Now consider

$$\ddot{x} - x + x^3 = f\cos(\Omega t).$$

Define the unperturbed mechanical energy

$$E_0 = \frac{1}{2}\dot{x}^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

Differentiating,

$$\frac{dE_0}{dt} = \dot{x}\ddot{x} - x\dot{x} + x^3\dot{x}.$$

Factoring $\dot{x}$,

$$\frac{dE_0}{dt} = \dot{x}(\ddot{x} - x + x^3).$$

Using the equation of motion,

$$\ddot{x} - x + x^3 = f\cos(\Omega t),$$

we obtain

$$\frac{dE_0}{dt} = f\dot{x}\cos(\Omega t).$$

Therefore the mechanical energy E_0 is no longer constant. The absence of damping does not mean that the mechanical energy of the oscillator is constant because the external periodic force continuously exchanges energy with the oscillator. The time-dependent Hamiltonian is

$$H(x, p, t) = \frac{1}{2}p^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4 - fx\cos(\Omega t),$$

where

$$p = \dot{x}.$$

Indeed,

$$\dot{x} = \frac{\partial H}{\partial p} = p,$$

and

$$\dot{p} = -\frac{\partial H}{\partial x} = x - x^3 + f\cos(\Omega t),$$

which gives

$$\ddot{x} - x + x^3 = f\cos(\Omega t).$$

Thus the periodically driven undamped Duffing oscillator is a time-periodic Hamiltonian system.

Poincaré Section for the Driven Duffing Oscillator The forcing period is

$$T = \frac{2\pi}{\Omega}.$$

A stroboscopic Poincaré section is obtained by sampling

$$(x(t), y(t))$$

at

$$t_n = t_0 + nT, \qquad n = 0, 1, 2, \ldots.$$

Thus,

$$P_n = (x(t_n), y(t_n)).$$

Every point is recorded at the same phase of the external forcing because

$$\Omega t_n = \Omega t_0 + 2\pi n.$$

The continuous nonautonomous system is therefore represented by a two-dimensional Poincaré map

$$P_{n+1} = T_s(P_n).$$

Different motions produce different structures on this section response periodic with forcing gives one point, subharmonic periodic response gives finite set of points, quasiperiodic response gives smooth closed curve, and chaotic response gives irregular scattered set.

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4.6 The Standard Map

The standard map, also called the Chirikov–Taylor map, is one of the most important two-dimensional nonlinear maps used in the study of Hamiltonian chaos. It is an area-preserving map and may be obtained as the discrete-time description of a periodically kicked rotor. The state variables are θ_n=angular position at the nth iteration, and I_n =angular momentum at the nth iteration, and K denotes the strength of the periodic kick. The standard map is

$$\begin{aligned} I_{n+1} &= I_n + K \sin\theta_n, \\ \theta_{n+1} &= \theta_n + I_{n+1} \pmod{2\pi}. \end{aligned}$$

When the dynamics is considered on the two-dimensional torus, I may also be reduced modulo 2π. Thus one sometimes writes

$$\begin{aligned} I_{n+1} &= I_n + K \sin\theta_n \pmod{2\pi}, \\ \theta_{n+1} &= \theta_n + I_{n+1} \pmod{2\pi}. \end{aligned}$$

The parameter K controls the strength of the nonlinearity. For small values of K, much of the motion is regular, whereas increasing K produces resonance islands, destruction of invariant curves and increasingly chaotic motion. Consider the Hamiltonian (derivation available in class notes)

$$H(\theta, I, t) = \frac{I^2}{2} + K\cos\theta \sum_{m=-\infty}^{\infty} \delta(t-m),$$

where $\delta(\cdot)$ denotes the Dirac delta function. Hamilton's equations are

$$\begin{aligned} \dot{\theta} &= \frac{\partial H}{\partial I} = I, \\ \dot{I} &= -\frac{\partial H}{\partial \theta} = K\sin\theta \sum_{m=-\infty}^{\infty} \delta(t-m). \end{aligned}$$

Between two successive kicks the delta function vanishes. Hence

$$\dot{I} = 0,$$

so that the angular momentum remains constant during the free motion. At the instant $t = n$, the system receives an impulsive kick. Integrating across a small interval containing $t = n$,

$$\int_{n^-}^{n^+} \dot{I}\, dt = K\sin\theta_n \int_{n^-}^{n^+} \delta(t-n)\, dt.$$

Using

$$\int_{n^-}^{n^+} \delta(t-n)\, dt = 1,$$

we obtain

$$I(n^+) - I(n^-) = K\sin\theta_n.$$

If I_n denotes the angular momentum immediately before the nth kick and I_{n+1} denotes the angular momentum immediately after the kick, then

$$I_{n+1} - I_n = K\sin\theta_n,$$

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and therefore

$$I_{n+1} = I_n + K \sin\theta_n.$$

After the kick, the rotor evolves freely until the next kick. During this interval the angular momentum remains constant and is equal to I_{n+1}. Hence

$$\dot{\theta} = I_{n+1}.$$

Integrating over one unit of time,

$$\begin{aligned}
\int_{\theta_n}^{\theta_{n+1}} d\theta &= \int_{n}^{n+1} I_{n+1}\, dt, \\
\theta_{n+1} - \theta_n &= I_{n+1}(n+1-n), \\
\theta_{n+1} - \theta_n &= I_{n+1}.
\end{aligned}$$

Therefore,

$$\theta_{n+1} = \theta_n + I_{n+1}.$$

Since θ is an angular variable, angles differing by integral multiples of 2π represent the same angular position. Hence

$$\theta_{n+1} = \theta_n + I_{n+1} \pmod{2\pi}.$$

Thus the periodically kicked rotor gives the standard map

$$\begin{aligned}
I_{n+1} &= I_n + K \sin\theta_n, \\
\theta_{n+1} &= \theta_n + I_{n+1} \pmod{2\pi}.
\end{aligned}$$

Substituting the first equation into the second in the standard map,

$$\theta_{n+1} = \theta_n + I_n + K \sin\theta_n.$$

The Jacobian matrix of the transformation is

$$J = \frac{\partial(I_{n+1}, \theta_{n+1})}{\partial(I_n, \theta_n)} = \begin{bmatrix} \dfrac{\partial I_{n+1}}{\partial I_n} & \dfrac{\partial I_{n+1}}{\partial \theta_n} \\ \dfrac{\partial \theta_{n+1}}{\partial I_n} & \dfrac{\partial \theta_{n+1}}{\partial \theta_n} \end{bmatrix}.$$

From

$$I_{n+1} = I_n + K \sin\theta_n,$$

we obtain

$$\frac{\partial I_{n+1}}{\partial I_n} = 1, \qquad \frac{\partial I_{n+1}}{\partial \theta_n} = K\cos\theta_n.$$

From

$$\theta_{n+1} = \theta_n + I_n + K \sin\theta_n,$$

we obtain

$$\frac{\partial \theta_{n+1}}{\partial I_n} = 1, \qquad \frac{\partial \theta_{n+1}}{\partial \theta_n} = 1 + K\cos\theta_n.$$

Therefore,

$$J = \begin{bmatrix} 1 & K\cos\theta_n \\ 1 & 1 + K\cos\theta_n \end{bmatrix}.$$

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The determinant of the Jacobian is

$$\begin{aligned}\det J &= \begin{vmatrix} 1 & K\cos\theta_n \\ 1 & 1 + K\cos\theta_n \end{vmatrix} \\ &= 1(1 + K\cos\theta_n) - 1(K\cos\theta_n) \\ &= 1 + K\cos\theta_n - K\cos\theta_n \\ &= 1.\end{aligned}$$

Hence

$$\det J = 1.$$

Suppose that a small region in phase space has area A_n before one iteration of the map. After one iteration its area is

$$A_{n+1} = |\det J| A_n.$$

Since

$$|\det J| = 1,$$

we obtain

$$A_{n+1} = A_n.$$

Thus the standard map preserves phase-space area and is therefore an area-preserving conservative map. For a dissipative map one usually has

$$|\det J| < 1,$$

and phase-space areas contract under repeated iteration. Such contraction allows trajectories to approach attractors. For the standard map,

$$|\det J| = 1,$$

so phase-space area does not contract. Consequently, the standard map does not possess dissipative attractors.

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Dynamics for $K = 0$ When

$$K = 0,$$

the standard map reduces to

$$\begin{aligned} I_{n+1} &= I_n, \\ \theta_{n+1} &= \theta_n + I_{n+1} \pmod{2\pi}. \end{aligned}$$

Since

$$I_{n+1} = I_n,$$

we have

$$I_1 = I_0, \qquad I_2 = I_1 = I_0, \qquad I_3 = I_2 = I_0, \qquad \ldots$$

and therefore

$$I_n = I_0.$$

The angular equation becomes

$$\theta_{n+1} = \theta_n + I_0 \pmod{2\pi}.$$

Starting from θ_0,

$$\theta_1 = \theta_0 + I_0,$$
$$\theta_2 = \theta_1 + I_0 = \theta_0 + 2I_0,$$
$$\theta_3 = \theta_2 + I_0 = \theta_0 + 3I_0.$$

Continuing in this way,

$$\theta_n = \theta_0 + nI_0 \pmod{2\pi}.$$

Hence, when $K = 0$, the angular momentum remains constant and every trajectory lies on an invariant curve

$$I = I_0.$$

The nature of the motion depends on whether

$$\frac{I_0}{2\pi}$$

is rational or irrational.

Rational Rotation Number Suppose

$$\frac{I_0}{2\pi} = \frac{N}{M},$$

where N and M are relatively prime integers. Then

$$I_0 = \frac{2\pi N}{M}.$$

After M iterations,

$$\begin{aligned}\theta_M &= \theta_0 + MI_0 \\ &= \theta_0 + M\left(\frac{2\pi N}{M}\right) \\ &= \theta_0 + 2\pi N.\end{aligned}$$

Since angular positions differing by an integer multiple of 2π are identical,

$$\theta_M = \theta_0 \pmod{2\pi}.$$

Also,

$$I_M = I_0.$$

Therefore,

$$(\theta_M, I_M) = (\theta_0, I_0).$$

Hence the orbit returns to its initial point after M iterations. When N and M are relatively prime, the period is M. Thus

$$\frac{I_0}{2\pi} \in \mathbb{Q} \quad \Longrightarrow \quad \text{periodic motion.}$$

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Irrational Rotation Number Suppose

$$\frac{I_0}{2\pi} \notin \mathbb{Q}.$$

Assume that the orbit is periodic. Then there must exist a positive integer M such that

$$\theta_M = \theta_0 \pmod{2\pi}.$$

Using

$$\theta_M = \theta_0 + MI_0,$$

periodicity requires

$$MI_0 = 2\pi N$$

for some integer N. Hence

$$\frac{I_0}{2\pi} = \frac{N}{M}.$$

This would imply that $I_0/(2\pi)$ is rational, which contradicts the assumption. Therefore the orbit can never repeat exactly. The sequence

$$\theta_n = \theta_0 + nI_0 \pmod{2\pi}$$

becomes dense on the circle. Hence

$$\frac{I_0}{2\pi} \notin \mathbb{Q} \quad \Longrightarrow \quad \text{quasiperiodic motion.}$$

Thus, for $K = 0$, the dynamics is completely regular. Rational rotation numbers generate periodic trajectories, whereas irrational rotation numbers generate quasiperiodic trajectories.

Period-One Fixed Points A period-one fixed point is a point that returns to itself after one application of the map. Let

$$(\theta^*, I^*)$$

denote such a fixed point. Then

$$I_{n+1} = I_n = I^*$$

and

$$\theta_{n+1} = \theta_n = \theta^* \pmod{2\pi}.$$

From the first equation of the standard map,

$$I_{n+1} = I_n + K \sin\theta_n,$$

the fixed-point condition gives

$$I^* = I^* + K \sin\theta^*.$$

Subtracting I^* from both sides,

$$K \sin\theta^* = 0.$$

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For $K \neq 0$,

$$\sin\theta^* = 0.$$

Therefore,

$$\theta^* = m\pi, \qquad m \in \mathbb{Z}.$$

Restricting the angle to

$$0 \leq \theta < 2\pi,$$

we obtain

$$\theta^* = 0, \qquad \theta^* = \pi.$$

The second fixed-point equation is

$$\theta^* = \theta^* + I^* \pmod{2\pi}.$$

Subtracting θ^*,

$$I^* = 0 \pmod{2\pi}.$$

Taking the principal value,

$$I^* = 0.$$

Hence the two basic period-one fixed points are

$$(0, 0), \qquad (\pi, 0).$$

To investigate the stability of a fixed point, consider a small perturbation from (θ^*, I^*):

$$I_n = I^* + \delta I_n, \qquad \theta_n = \theta^* + \delta\theta_n.$$

For sufficiently small perturbations, the nonlinear map may be approximated by its linearization:

$$\begin{pmatrix} \delta I_{n+1} \\ \delta\theta_{n+1} \end{pmatrix} = J^* \begin{pmatrix} \delta I_n \\ \delta\theta_n \end{pmatrix},$$

where J^* is the Jacobian evaluated at the fixed point:

$$J^* = \begin{bmatrix} 1 & K\cos\theta^* \\ 1 & 1 + K\cos\theta^* \end{bmatrix}.$$

The behaviour of small perturbations is determined by the eigenvalues of J^*. These eigenvalues satisfy

$$\det(J^* - \lambda I) = 0.$$

Therefore,

$$\det \begin{bmatrix} 1-\lambda & K\cos\theta^* \\ 1 & 1 + K\cos\theta^* - \lambda \end{bmatrix} = 0.$$

Thus

$$\lambda^2 - (2 + K\cos\theta^*)\,\lambda + 1 = 0.$$

The trace of J^* is

$$\mathrm{Tr}(J^*) = 1 + (1 + K\cos\theta^*) = 2 + K\cos\theta^*,$$

and

$$\det(J^*) = 1.$$

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Hence the characteristic equation becomes

$$\lambda^2 - \mathrm{Tr}(J^*)\lambda + 1 = 0.$$

Using the quadratic formula,

$$\lambda_{1,2} = \frac{\mathrm{Tr}(J^*) \pm \sqrt{[\mathrm{Tr}(J^*)]^2 - 4}}{2}.$$

Since the constant term of the characteristic polynomial is 1,

$$\lambda_1 \lambda_2 = 1.$$

This reciprocal relation between the two eigenvalues is a direct consequence of the area-preserving nature of the standard map.

Stability criterion Let

$$T = \mathrm{Tr}(J^*).$$

Then

$$\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4}}{2}.$$

Consider first

$$|T| < 2.$$

Then

$$T^2 - 4 < 0.$$

Therefore the eigenvalues are complex:

$$\lambda_{1,2} = \frac{T}{2} \pm i\frac{\sqrt{4 - T^2}}{2}.$$

Their modulus is

$$\begin{aligned} |\lambda|^2 &= \left(\frac{T}{2}\right)^2 + \left(\frac{\sqrt{4 - T^2}}{2}\right)^2 \\ &= \frac{T^2}{4} + \frac{4 - T^2}{4} \\ &= 1. \end{aligned}$$

Therefore,

$$|\lambda_1| = |\lambda_2| = 1.$$

The eigenvalues lie on the unit circle in the complex plane. Such a fixed point is called an **elliptic fixed point**. Nearby trajectories generally rotate around the fixed point and form closed invariant curves. Hence

$$|\,\mathrm{Tr}(J^*)| < 2 \quad \Longrightarrow \quad \text{elliptic fixed point.}$$

Now consider

$$|T| > 2.$$

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Then

$$T^2 - 4 > 0,$$

so both eigenvalues are real. Since

$$\lambda_1 \lambda_2 = 1,$$

the eigenvalues are reciprocal. One therefore has magnitude greater than one and the other has magnitude less than one:

$$|\lambda_1| > 1, \qquad |\lambda_2| < 1.$$

A perturbation grows along one eigendirection and decreases along the other. Such a fixed point is called a **hyperbolic fixed point**. Hence

$$|\operatorname{Tr}(J^*)| > 2 \quad \Longrightarrow \quad \text{hyperbolic fixed point.}$$

Finally, when

$$|\operatorname{Tr}(J^*)| = 2,$$

the eigenvalues are repeated:

$$\lambda = \pm 1.$$

This represents the boundary between elliptic and hyperbolic behaviour. Such a point is called **parabolic** or **marginal**. Thus

$$\begin{aligned} |\operatorname{Tr}(J^*)| < 2 &\quad \Longrightarrow \quad \text{elliptic,} \\ |\operatorname{Tr}(J^*)| > 2 &\quad \Longrightarrow \quad \text{hyperbolic,} \\ |\operatorname{Tr}(J^*)| = 2 &\quad \Longrightarrow \quad \text{parabolic.} \end{aligned}$$

Stability of the Fixed Point $(0, 0)$ For the fixed point

$$(\theta^*, I^*) = (0, 0),$$

we have

$$\cos\theta^* = \cos 0 = 1.$$

Therefore the Jacobian becomes

$$J^* = \begin{bmatrix} 1 & K \\ 1 & 1+K \end{bmatrix}.$$

Its trace is

$$\operatorname{Tr}(J^*) = 1 + (1+K) = 2 + K.$$

For

$$K > 0,$$

we have

$$2 + K > 2.$$

Therefore,

$$|\operatorname{Tr}(J^*)| > 2.$$

Hence $(0, 0)$ is hyperbolic and unstable for $K > 0$. The eigenvalues can also be calculated explicitly. The characteristic equation is

$$\lambda^2 - (2+K)\lambda + 1 = 0.$$

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Hence

$$\lambda_{1,2} = \frac{2 + K \pm \sqrt{(2+K)^2 - 4}}{2}.$$

Now,

$$\begin{aligned}(2+K)^2 - 4 &= 4 + 4K + K^2 - 4 \\ &= K^2 + 4K \\ &= K(K+4).\end{aligned}$$

Therefore,

$$\lambda_{1,2} = \frac{2 + K \pm \sqrt{K(K+4)}}{2}.$$

For $K > 0$, these eigenvalues are real and reciprocal. One eigenvalue lies outside the unit circle and the other lies inside the unit circle, confirming that $(0, 0)$ is a hyperbolic fixed point.

Stability of the Fixed Point $(\pi, 0)$ For the fixed point

$$(\theta^*, I^*) = (\pi, 0),$$

we have

$$\cos\theta^* = \cos\pi = -1.$$

Therefore,

$$J^* = \begin{bmatrix} 1 & -K \\ 1 & 1-K \end{bmatrix}.$$

Its trace is

$$\operatorname{Tr}(J^*) = 1 + (1 - K) = 2 - K.$$

For the fixed point to be elliptic, we require

$$|\operatorname{Tr}(J^*)| < 2.$$

Thus

$$|2 - K| < 2.$$

Writing this as a double inequality,

$$-2 < 2 - K < 2.$$

Subtracting 2 throughout,

$$-4 < -K < 0.$$

Multiplying by -1 reverses both inequalities:

$$0 < K < 4.$$

Hence $(\pi, 0)$ is elliptic for $0 < K < 4$. The corresponding characteristic equation is

$$\lambda^2 - (2 - K)\lambda + 1 = 0.$$

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Therefore,

$$\lambda_{1,2} = \frac{2 - K \pm \sqrt{(2-K)^2 - 4}}{2}.$$

The discriminant is

$$\begin{aligned}(2-K)^2 - 4 &= 4 - 4K + K^2 - 4 \\ &= K^2 - 4K \\ &= K(K-4).\end{aligned}$$

Hence

$$\lambda_{1,2} = \frac{2 - K \pm \sqrt{K(K-4)}}{2}.$$

For

$$0 < K < 4,$$

we have

$$K(K-4) < 0,$$

and therefore the eigenvalues form a complex-conjugate pair. Since

$$\lambda_1 \lambda_2 = 1,$$

both eigenvalues have modulus one. We may therefore write

$$\lambda_{1,2} = e^{\pm i\omega}.$$

The sum of the eigenvalues is

$$\lambda_1 + \lambda_2 = \mathrm{Tr}(J^*) = 2 - K.$$

Thus

$$e^{i\omega} + e^{-i\omega} = 2 - K.$$

Using

$$e^{i\omega} + e^{-i\omega} = 2\cos\omega,$$

we obtain

$$2\cos\omega = 2 - K,$$

and hence

$$\cos\omega = 1 - \frac{K}{2}.$$

Thus small perturbations around the fixed point $(\pi, 0)$ undergo oscillatory motion when

$$0 < K < 4.$$

At

$$K = 4,$$

the trace is

$$\mathrm{Tr}(J^*) = 2 - 4 = -2.$$

The characteristic equation becomes

$$\lambda^2 + 2\lambda + 1 = 0,$$

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or

$$(\lambda + 1)^2 = 0.$$

Hence

$$\lambda_1 = \lambda_2 = -1.$$

Therefore $K = 4$ is the stability boundary of the fixed point $(\pi, 0)$. For

$$K > 4,$$

we have

$$2 - K < -2,$$

so

$$|2 - K| > 2.$$

Therefore the fixed point becomes hyperbolic. Thus

$$\begin{aligned} 0 < K < 4 \quad & (\pi, 0) \text{ is elliptic,} \\ K = 4 \quad & (\pi, 0) \text{ is parabolic,} \\ K > 4 \quad & (\pi, 0) \text{ is hyperbolic.} \end{aligned}$$

Invariant Curves Consider a continuously differentiable, invertible map

$$\mathbf{z}_{n+1} = F(\mathbf{z}_n), \qquad \mathbf{z}_n = (q_n, p_n) \in \mathbb{R}^2.$$

A set $C \subseteq \mathbb{R}^2$ is said to be forward invariant under F if $F(C) \subseteq C$, while it is invariant if $F(C) = C$. In other words, whenever a point belongs to an invariant set, its image under the map also belongs to that set. For an invariant curve, the image of a point need not coincide with the original point. It is only required to remain on the same curve. This distinguishes invariant curves from fixed points, which satisfy $F(\mathbf{z}^*) = \mathbf{z}^*$, and periodic orbits, for which $F^r(\mathbf{z}_0) = \mathbf{z}_0$ for some positive integer r. An invariant curve may contain a periodic orbit, but it may also support a nonperiodic trajectory.
Suppose a curve C is represented implicitly by

$$C = \{(q, p) : g(q, p) = 0\}.$$

To verify its forward invariance, it is sufficient to establish that

$$g(q, p) = 0 \quad \Longrightarrow \quad g(F(q, p)) = 0.$$

If the map is written as $F(q, p) = (Q(q, p), P(q, p))$, this condition becomes

$$g(q, p) = 0 \quad \Longrightarrow \quad g(Q(q, p), P(q, p)) = 0.$$

A particularly useful situation occurs when the curve can be expressed as $p = f(q)$. A point on the curve has coordinates $(q, f(q))$. After one iteration, its new coordinates are

$$q' = Q(q, f(q)), \qquad p' = P(q, f(q)).$$

For the new point to remain on the same curve, it must satisfy $p' = f(q')$. Consequently, the unknown function f must satisfy

$$P(q, f(q)) = f(Q(q, f(q))).$$

This is called the invariance equation. It expresses the geometric requirement that the map sends points of a candidate curve back onto that curve. Finding a function satisfying this equation can be considerably more difficult than calculating individual trajectories of the original map.

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Example 1. A circular platform rotates uniformly about its centre, and a camera records the position of a marked point at equally spaced time intervals. If the platform rotates through an angle α between consecutive observations, show that the marked point moves on an invariant circle. Determine the conditions under which its observed motion is periodic or nonperiodic.

Solution. Let (x_n, y_n) denote the position of the marked point at the nth observation. A rotation through an angle α transforms its coordinates according to

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x_n \\ y_n \end{pmatrix}.$$

Thus,

$$x_{n+1} = x_n\cos\alpha - y_n\sin\alpha, \qquad y_{n+1} = x_n\sin\alpha + y_n\cos\alpha.$$

Squaring and adding these equations,

$$\begin{aligned} x_{n+1}^2 + y_{n+1}^2 &= (x_n\cos\alpha - y_n\sin\alpha)^2 \\ &\quad + (x_n\sin\alpha + y_n\cos\alpha)^2 \\ &= x_n^2(\cos^2\alpha + \sin^2\alpha) \\ &\quad + y_n^2(\sin^2\alpha + \cos^2\alpha) \\ &\quad + 2x_ny_n(-\cos\alpha\sin\alpha + \sin\alpha\cos\alpha). \end{aligned}$$

Since $\cos^2\alpha + \sin^2\alpha = 1$ and the mixed terms cancel,

$$x_{n+1}^2 + y_{n+1}^2 = x_n^2 + y_n^2.$$

Therefore, the function

$$I(x, y) = x^2 + y^2$$

is conserved under the map, since $I(F(x, y)) = I(x, y)$. If the marked point is initially at a distance R from the centre, then

$$x_0^2 + y_0^2 = R^2$$

and hence

$$x_n^2 + y_n^2 = R^2, \qquad n = 0, 1, 2, \ldots.$$

Every circle

$$C_R = \{(x, y) : x^2 + y^2 = R^2\}$$

is therefore invariant. Different initial radii give different invariant circles, and a marked point cannot move from one such circle to another under the given rotation. To examine the motion along an invariant circle, introduce polar coordinates

$$x_n = R\cos\theta_n, \qquad y_n = R\sin\theta_n.$$

Substitution into the rotation equations and application of the angle-addition formulae give

$$x_{n+1} = R\cos(\theta_n + \alpha), \qquad y_{n+1} = R\sin(\theta_n + \alpha).$$

Consequently,

$$\theta_{n+1} = \theta_n + \alpha \pmod{2\pi}$$

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and, after n observations,

$$\theta_n = \theta_0 + n\alpha \pmod{2\pi}.$$

The observed position returns to its initial value after r iterations if

$$r\alpha = 2\pi k$$

for some integers $r \geq 1$ and k. Thus, a return is possible precisely when

$$\frac{\alpha}{2\pi} = \frac{k}{r} \in \mathbb{Q}.$$

For instance, if $\alpha = \pi/2$, the marked point occupies four distinct positions before returning to the starting point. If instead

$$\alpha = 2\pi\frac{\sqrt{5}-1}{2},$$

the ratio $\alpha/(2\pi)$ is irrational. No positive integer r then satisfies $r\alpha = 2\pi k$, so the sampled trajectory never repeats exactly. Nevertheless, every observed point lies on the same invariant circle, and the sequence of positions becomes dense on that circle. The latter is an example of irrational rotation, a simple model of regular nonperiodic motion. The continuously rotating platform itself remains periodic; the nonperiodicity concerns the sequence observed at the fixed sampling times.

4.7 Numerical Analysis: Regular and Chaotic Motions

Numerical iteration of the map reveals several different dynamical regimes.

(i) Small Nonlinearity ($K \ll 1$) Most invariant curves survive. The phase space consists mainly of Elliptic fixed points, Closed invariant curves, Quasiperiodic trajectories. Motion is predominantly regular.

(ii) Intermediate Values of K Increasing K distorts invariant curves. Resonances appear. Regular and chaotic trajectories coexist.

(iii) Large Values of K For sufficiently large K, most invariant curves disappear. The remaining phase space is occupied by large connected chaotic regions.

Period-Doubling The standard map also exhibits Hamiltonian period-doubling cascades. As K increases, period-q, 2q, 4q, 8q until chaotic motion develops. Unlike dissipative systems, these periodic orbits are surrounded by invariant curves rather than being attractors.

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5. Characterization of Regular and Chaotic Motions

5.1 Lyapunov Exponents

In nonlinear dynamical systems, trajectories may exhibit periodic, quasiperiodic, or chaotic behaviour depending on the system parameters and initial conditions. One of the most important quantitative measures used to distinguish these motions is the Lyapunov exponent. It measures the average exponential rate at which two initially nearby trajectories either diverge or converge in phase space.

The existence of a positive Lyapunov exponent is regarded as one of the strongest signatures of deterministic chaos because it quantifies the phenomenon of sensitive dependence on initial conditions.

Definition Consider an n-dimensional autonomous dynamical system

$$\dot{\mathbf{X}} = \mathbf{F}(\mathbf{X}),$$

where

$$\mathbf{X}(t) = (x_1(t), x_2(t), \ldots, x_n(t))^T.$$

Suppose two trajectories start from two nearby initial conditions

$$\mathbf{X}_0, \qquad \mathbf{X}_0 + \delta\mathbf{X}_0.$$

At time t, the separation between them becomes

$$\delta\mathbf{X}(t).$$

The Euclidean distance between the trajectories is

$$d(\mathbf{X}_0, t) = \|\delta\mathbf{X}(t)\| = \sqrt{(\delta x_1)^2 + (\delta x_2)^2 + \cdots + (\delta x_n)^2}.$$

The evolution of the infinitesimal perturbation is governed by the linearized (variational) equation

$$\delta\dot{\mathbf{X}} = J(\mathbf{X}(t))\,\delta\mathbf{X}$$

where

$$J(\mathbf{X}) = \frac{\partial \mathbf{F}}{\partial \mathbf{X}}$$

is the Jacobian matrix evaluated along the reference trajectory. Assume that the perturbation evolves exponentially,

$$\delta\mathbf{X}(t) \approx \delta\mathbf{X}_0 e^{\lambda t}.$$

Taking norms,

$$d(t) = d(0)e^{\lambda t}.$$

Taking logarithm,

$$\ln d(t) = \ln d(0) + \lambda t.$$

Hence,

$$\lambda = \lim_{t\to\infty} \frac{1}{t} \ln\left(\frac{d(t)}{d(0)}\right)$$

This quantity is called the Lyapunov exponent. It represents the average exponential rate of divergence (or convergence) of nearby trajectories.

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Multiple Lyapunov Exponents Since an n-dimensional system has n independent directions in phase space, there exist n independent Lyapunov exponents,

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n.$$

Each exponent corresponds to stretching or contraction along one principal direction of the phase space. The perturbation along the i^{th} eigen-direction evolves as

$$d_i(t) \approx d_i(0)e^{\lambda_i t}, \qquad i = 1, \ldots, n.$$

Thus,

- $\lambda_i > 0$: exponential stretching,
- $\lambda_i < 0$: exponential contraction,
- $\lambda_i = 0$: neutral direction.

Usually, only the largest Lyapunov exponent

$$\lambda_{\max} = \lambda_1(\text{assume})$$

is sufficient to determine whether the system is chaotic.

Case I : $\lambda_{\max} < 0$ If

$$d(t) = d(0)e^{\lambda_{\max} t}, \quad \lambda_{\max} < 0,$$

then

$$d(t) \to 0 \quad (t \to \infty).$$

Hence nearby trajectories converge towards one another. This behaviour corresponds to a stable equilibrium point, stable periodic orbit, and asymptotically stable attractor. There is no sensitive dependence on initial conditions.

Case II : $\lambda_{\max} = 0$ If

$$\lambda_{\max} = 0,$$

then

$$d(t) = d(0),$$

and the distance between neighbouring trajectories neither grows nor decays. This occurs in periodic motion (one zero exponent), and quasiperiodic motion (two or more zero exponents). Case III : $\lambda_{\max} > 0$ If

$$\lambda_{\max} > 0,$$

then

$$d(t) = d(0)e^{\lambda_{\max} t},$$

which grows exponentially with time. Therefore, Positive Lyapunov exponent means Sensitive dependence on initial conditions and consequently,

$$\lambda_{\max} > 0 \Longrightarrow \text{Deterministic Chaos.}$$

Even an extremely small uncertainty in the initial condition increases exponentially,

$$\delta(t) = \delta_0 e^{\lambda_{\max} t},$$

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making long-term prediction impossible.

Lyapunov exponents measure the local stretching and contraction of phase-space trajectories. They provide quantitative information about the predictability of the system, stability of trajectories, complexity of attractors, and degree of chaos. A larger positive Lyapunov exponent indicates faster divergence of trajectories and therefore a shorter prediction horizon.

Numerical Computation of Lyapunov Exponents For most nonlinear dynamical systems, analytical evaluation of Lyapunov exponents is difficult. Hence, they are usually computed numerically by linearizing the system along a reference trajectory. In this section, numerical computation is discussed for both one-dimensional maps and continuous-time dynamical systems.

Lyapunov Exponent for One-Dimensional Maps Consider the one-dimensional discrete map

$$x_{n+1} = f(x_n).$$

Suppose two trajectories start from nearby initial conditions

$$x_0, \qquad x_0 + \delta x_0,$$

where δx_0 is infinitesimally small. After one iteration,

$$x_1 = f(x_0),$$

and the perturbation becomes

$$\delta x_1 = f(x_0 + \delta x_0) - f(x_0).$$

Using the first-order Taylor expansion,

$$f(x_0 + \delta x_0) = f(x_0) + f'(x_0)\delta x_0,$$

we obtain

$$\delta x_1 = f'(x_0)\delta x_0.$$

Similarly,

$$\delta x_2 = f'(x_1)\delta x_1 = f'(x_1)f'(x_0)\delta x_0.$$

Proceeding recursively,

$$\delta x_n = \delta x_0 \prod_{i=0}^{n-1} f'(x_i).$$

Thus, the perturbation evolves as the product of local stretching factors. Taking absolute values,

$$|\delta x_n| = |\delta x_0| \prod_{i=0}^{n-1} |f'(x_i)|.$$

Taking logarithms,

$$\ln |\delta x_n| = \ln |\delta x_0| + \sum_{i=0}^{n-1} \ln |f'(x_i)|.$$

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Dividing by n,

$$\frac{1}{n}\ln\frac{|\delta x_n|}{|\delta x_0|} = \frac{1}{n}\sum_{i=0}^{n-1}\ln|f'(x_i)|.$$

Taking the limit as $n \to \infty$,

$$\lambda = \lim_{n\to\infty}\frac{1}{n}\sum_{i=0}^{n-1}\ln|f'(x_i)|.$$

This is the standard numerical formula for the Lyapunov exponent of a one-dimensional map.

$$\begin{cases}\lambda < 0, & \text{stable fixed point or periodic orbit,}\\ \lambda = 0, & \text{bifurcation or neutral stability,}\\ \lambda > 0, & \text{chaotic motion.}\end{cases}$$

Example: Logistic Map The equation is

$$x_{n+1} = ax_n(1-x_n),$$

whose derivative is

$$f'(x) = a(1-2x).$$

Hence,

$$\lambda = \lim_{n\to\infty}\frac{1}{n}\sum_{i=0}^{n-1}\ln|a(1-2x_i)|.$$

The numerical computation proceeds as follows:

1. Choose the parameter a and initial condition x_0.
2. Iterate the map sufficiently many times to remove transient behaviour.
3. Continue iterating while evaluating
$$\ln|a(1-2x_i)|.$$
4. Compute the average of these logarithms.

The computed Lyapunov exponent satisfies

$$\lambda < 0$$

for periodic solutions,

$$\lambda = 0$$

at bifurcation points, and

$$\lambda > 0$$

in the chaotic region.

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Lyapunov Exponents for Continuous-Time Dynamical Systems Consider the nonlinear dynamical system

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}),$$

where

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)^T.$$

Suppose

$$\mathbf{x}(t)$$

is the reference trajectory and

$$\delta\mathbf{x}(t)$$

is an infinitesimal perturbation. Linearizing about the reference trajectory gives the variational equation

$$\delta\dot{\mathbf{x}} = J(\mathbf{x})\delta\mathbf{x},$$

where

$$J(\mathbf{x}) = \frac{\partial \mathbf{F}}{\partial \mathbf{x}}$$

is the Jacobian matrix evaluated along the trajectory. The perturbation evolves approximately as

$$\delta\mathbf{x}(t) \approx e^{Jt}\delta\mathbf{x}(0),$$

and therefore

$$\|\delta\mathbf{x}(t)\| \approx \|\delta\mathbf{x}(0)\|e^{\lambda t}.$$

Hence,

$$\lambda = \lim_{t\to\infty} \frac{1}{t} \ln \frac{\|\delta\mathbf{x}(t)\|}{\|\delta\mathbf{x}(0)\|}.$$

Since an n-dimensional system has n independent perturbation directions, it possesses n Lyapunov exponents.

Suppose after integrating for a finite time interval T, the perturbation vectors become

$$\Delta X_1, \Delta X_2, \ldots, \Delta X_n.$$

These vectors are orthonormalized as

$$V_1 = \Delta X_1, \qquad U_1 = \frac{V_1}{\|V_1\|}.$$

For

$$i = 2, \ldots, n,$$

the orthogonal component is

$$V_i = \Delta X_i - \sum_{j=1}^{i-1} (\Delta X_i, U_j) U_j,$$

and the normalized vector is

$$U_i = \frac{V_i}{\|V_i\|}.$$

The norms

$$\|V_i\|$$

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represent the stretching factors during one integration interval. This process is repeated throughout the numerical integration. Suppose reorthonormalization is performed after every time interval T, and the procedure is repeated N times. Then the i^{th} Lyapunov exponent is

$$\lambda_i = \frac{1}{NT}\sum_{k=1}^{N} \ln \|V_i^{(k)}\|.$$

As

$$N \to \infty,$$

the values converge to the true Lyapunov spectrum.

System	Lyapunov Exponent Formula
1-D Map	$\lambda = \lim_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} \ln \lvert f'(x_i)\rvert$
Continuous System	$\lambda_i = \frac{1}{NT}\sum_{k=1}^{N} \ln \|V_i^{(k)}\|$

Example 1. Consider the doubling map

$$x_{n+1} = 2x_n \pmod 1, \qquad x_n \in [0,1).$$

Determine its Lyapunov exponent and discuss the nature of the motion.
Solution. The doubling map can be expressed as

$$f(x) = \begin{cases} 2x, & 0 \le x < \frac{1}{2}, \\ 2x-1, & \frac{1}{2} \le x < 1. \end{cases}$$

On each smooth branch, $f'(x) = 2$. Consequently, for any orbit avoiding the discontinuity of the interval representation, the Lyapunov exponent is

$$\lambda = \lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} \ln \lvert f'(x_n)\rvert = \lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} \ln 2.$$

It follows that

$$\lambda = \ln 2 \approx 0.6931.$$

Indeed, an infinitesimal perturbation δx_0 evolves locally according to

$$|\delta x_n| = 2^n |\delta x_0|,$$

so long as the separation is sufficiently small to be described by the linearized dynamics. Thus, nearby trajectories exhibit exponential separation. The positive Lyapunov exponent is consistent with the chaotic behaviour of the doubling map for typical initial conditions.

Example 2. Determine the Lyapunov exponent of the linear map

$$x_{n+1} = \frac{3}{4}x_n, \qquad x_0 \in \mathbb{R},$$

and examine the stability of its fixed point.

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Solution. Successive iterations yield

$$x_1 = \frac{3}{4}x_0, \qquad x_2 = \left(\frac{3}{4}\right)^2 x_0, \qquad x_n = \left(\frac{3}{4}\right)^n x_0.$$

Since $0 < 3/4 < 1$, every trajectory converges to the fixed point $x^* = 0$. Moreover, $f'(x) = 3/4$, and therefore

$$\lambda = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln \left|\frac{3}{4}\right| = \ln\left(\frac{3}{4}\right).$$

Hence,

$$\lambda \approx -0.28768.$$

The negative Lyapunov exponent shows that infinitesimal perturbations decay exponentially, confirming the asymptotic stability of the equilibrium.

Example 3. Consider the logistic map

$$x_{n+1} = r x_n(1 - x_n)$$

with $r = 3/2$. Determine the Lyapunov exponent associated with its nonzero fixed point.

Solution. The fixed points satisfy

$$x^* = \frac{3}{2}x^*(1 - x^*),$$

which gives $x^* = 0$ and $x^* = 1/3$. Since

$$f'(x) = \frac{3}{2}(1 - 2x),$$

the derivative at the nonzero fixed point is

$$f'\left(\frac{1}{3}\right) = \frac{3}{2}\left(1 - \frac{2}{3}\right) = \frac{1}{2}.$$

The Lyapunov exponent of this fixed-point orbit is therefore

$$\lambda = \ln\left|f'\left(\frac{1}{3}\right)\right| = \ln\left(\frac{1}{2}\right).$$

Thus,

$$\lambda = -\ln 2 \approx -0.69315.$$

Since $|f'(1/3)| < 1$, the nonzero fixed point is locally asymptotically stable. Small disturbances from the equilibrium decrease exponentially under iteration.

Example 4. A point on a uniformly rotating circular disk is observed at equal time intervals. Its angular position is governed by

$$\theta_{n+1} = \theta_n + \alpha \pmod{2\pi},$$

where $\alpha/(2\pi)$ is irrational. Determine the Lyapunov exponent and describe the resulting motion.

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Solution. Repeated application of the map gives

$$\theta_n = \theta_0 + n\alpha \pmod{2\pi}.$$

The derivative of the angular map is $f'(\theta) = 1$. Hence,

$$\lambda = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln |f'(\theta_n)| = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln 1 = 0.$$

Therefore,

$$\lambda = 0.$$

Since $\alpha/(2\pi)$ is irrational, there are no integers m and $n > 0$ satisfying $n\alpha = 2\pi m$. Consequently, the observed orbit never returns exactly to its initial angular position. Nevertheless, all iterates remain on the same circle and are dense on it. This is an example of regular nonperiodic motion with a zero Lyapunov exponent. The continuous rotation of the disk itself remains periodic; the nonperiodicity refers to the discretely sampled orbit.

Example 5. Calculate the Lyapunov exponent of the tent map

$$x_{n+1} = \begin{cases} 2x_n, & 0 \le x_n \le \frac{1}{2}, \\ 2(1 - x_n), & \frac{1}{2} < x_n \le 1. \end{cases}$$

Solution. The derivative is given by

$$f'(x) = \begin{cases} 2, & 0 < x < \frac{1}{2}, \\ -2, & \frac{1}{2} < x < 1. \end{cases}$$

Thus, $|f'(x)| = 2$ wherever the derivative exists. For an orbit avoiding the nondifferentiable point and its preimages,

$$\lambda = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln |f'(x_n)| = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \ln 2.$$

Therefore,

$$\lambda = \ln 2 \approx 0.69315.$$

Although the interval is bounded, the map expands infinitesimal perturbations on each smooth branch. The folding of the interval prevents trajectories from escaping, while repeated stretching produces sensitive dependence on initial conditions. The tent map consequently provides a simple example of deterministic chaos.

Example 6. Consider the logistic map

$$x_{n+1} = 4x_n(1 - x_n), \qquad 0 \le x_n \le 1.$$

Show that its Lyapunov exponent is $\ln 2$ for almost every initial condition.

Solution. Introduce the transformation

$$x_n = \sin^2(\pi\theta_n).$$

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Then

$$\begin{aligned} x_{n+1} &= 4\sin^2(\pi\theta_n)\left[1-\sin^2(\pi\theta_n)\right] \\ &= 4\sin^2(\pi\theta_n)\cos^2(\pi\theta_n) \\ &= \sin^2(2\pi\theta_n). \end{aligned}$$

Thus, the logistic map is related to the doubling map

$$\theta_{n+1} = 2\theta_n \pmod 1.$$

Let $h(\theta) = \sin^2(\pi\theta)$ and $f(x) = 4x(1-x)$. The identity

$$f(h(\theta)) = h(2\theta)$$

implies, after N iterations,

$$f^N(h(\theta)) = h(2^N\theta).$$

Differentiating with respect to θ,

$$(f^N)'(h(\theta))h'(\theta) = 2^N h'(2^N\theta).$$

At points for which the derivatives involved are nonzero,

$$|(f^N)'(x_0)| = 2^N \left|\frac{h'(2^N\theta_0)}{h'(\theta_0)}\right|.$$

Taking logarithms and dividing by N gives

$$\frac{1}{N}\ln|(f^N)'(x_0)| = \ln 2 + \frac{1}{N}\ln|h'(2^N\theta_0)| - \frac{1}{N}\ln|h'(\theta_0)|.$$

For almost every initial condition, the remaining terms tend to zero in the limit. Therefore,

$$\lambda = \ln 2 \approx 0.69315.$$

The positive Lyapunov exponent indicates exponential sensitivity to initial conditions for typical trajectories. In contrast to the preceding examples of stable and neutral motion, this map exhibits chaotic behaviour, although exceptional initial conditions may produce different asymptotic exponents.

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